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TEMPORAL RELATIONAL GRAPH CONVOLUTIONAL NETWORKS FOR FINANCIAL APPLICATIONS

Brindha Priyadarshini Jeyaraman

SINGAPORE MANAGEMENT UNIVERSITY
2025



Temporal Relational Graph Convolutional Networks for Financial Applications

By

Brindha Priyadarshini Jeyaraman

Submitted to School of Computing and Information Systems
in partial fulfillment of the requirements for the
Degree of Doctor of Engineering

Dissertation Committee:

Dai *Bing Tian* (Supervisor / Chair)
Associate Professor of School of Computing and Information Systems
Singapore Management University

Fang *Yuan*
Assistant Professor of School of Computing and Information Systems
Singapore Management University

Dr. Wang *Jiwei*
Associate Professor of School of Accountancy
Singapore Management University.

Singapore Management University

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DRAFT 2025-04-17 16:35

I hereby declare that this PhD dissertation is my original work and it has been written by me in its entirety. I have duly acknowledged all the sources of information which have been used in this dissertation.

This EngD dissertation has also not been submitted for any degree in any university previously.



Brindha Priyadarshini Jeyaraman

2025-04-17

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Abstract

The financial industry operates within a highly dynamic and interconnected ecosystem, presenting unique challenges for predictive modeling and decision-making. Accurately forecasting financial performance, assessing credit risk, detecting fraud, and ensuring compliance require methodologies that can capture complex temporal, relational, and contextual dependencies within financial data. This thesis investigates the use of **Temporal Relational Graph Convolutional Networks (TRGCNs)** combined with financial knowledge graphs (FKGs) to address these challenges and enable advanced analytics in the financial domain.

We introduce **FintechKG**, a financial knowledge graph constructed through a three-dimensional information extraction process, incorporating entities, temporal dimensions, and domain-specific financial relationships. A TRGCN-based framework is proposed to model both temporal and relational dependencies within FintechKG, enhanced by FinBERT embeddings and social media data integration for richer feature representations. The framework is evaluated using financial performance prediction as a case study, where a logistic regression model is applied to classify revenue trends, demonstrating the effectiveness of combining relational and textual embeddings.

Additionally, this thesis presents a **Temporal Credit Knowledge Graph (TCKG)** framework for corporate credit risk assessment. By employing temporal embeddings, relational reasoning, and generative AI techniques, we establish a comprehensive methodology for credit risk prediction. A comparative analysis of five predictive models—including logistic regression, Relational Graph Convolutional Networks (RGCNs), generative AI-based Gemini models, and Graph Recurrent Neural Networks (GRNNs)—reveals the superiority of the Gemini-based approach in terms of accuracy, robustness, and interpretability.

By integrating temporal learning, relational reasoning, and generative AI, this research advances the state of the art in financial analytics. It provides a systematic framework for constructing financial knowledge graphs and demonstrates their transformative potential in addressing critical challenges in the financial sector. This work not only highlights the efficacy of TRGCNs for financial applications but also paves the way for future research into integrating knowledge graphs and generative AI for enhanced decision-making in dynamic financial environments.

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CHAPTER 1

Introduction

Knowledge graphs (KGs) have become increasingly valuable in the finance industry, where they facilitate the organization, analysis, and interpretation of complex interrelated data. These graphs provide a structured representation of entities and their relationships, facilitating advanced analytics in areas such as credit risk assessment, systemic risk analysis, compliance monitoring, and fraud detection. By capturing interdependencies within financial data, KGs offer actionable insights that support data-driven decision-making. For instance, prior research has highlighted the utility of KGs in visualizing systemic liquidity risks through the representation of interbank relationships [1], as well as in enterprise risk management, where domain-specific KGs improve risk evaluation capabilities [2].

1.1 Applications of Knowledge Graphs in Finance

Knowledge graphs (KGs) have become increasingly significant in the financial sector due to their ability to represent and analyze complex relationships between financial entities. By integrating structured and unstructured data sources, KGs facilitate various financial applications, including credit risk assessment, fraud detection, regulatory compliance, and market trend analysis. These applications leverage the interconnected nature of financial entities, transactions, and events, providing deeper insights and improving decision-making processes.

1.1.1 Credit Risk Assessment

Credit risk assessment plays a crucial role in financial institutions, determining the creditworthiness of borrowers and mitigating potential defaults. Traditional credit scoring models rely on historical financial records, income levels, and credit history. However, these methods often fail to capture nuanced relationships and real-time financial indicators. Knowledge graphs provide a more comprehensive approach by integrating multi-source financial data, including company filings, economic indicators, and transactional history, into a structured network of interconnected financial entities.

Mitra et al. [3] highlight how financial institutions utilize KGs to enhance credit risk evaluation by linking diverse financial indicators and borrower profiles. These graphs enable institutions to uncover hidden dependencies between financial variables and assess credit risk more holistically. By applying graph-based machine learning techniques, such as Graph Neural Networks (GNNs) and Temporal Relational Graph Convolutional Networks (TRGCNs), financial institutions can dynamically update risk assessments based on evolving market conditions.

1.1.2 Fraud Detection and Anomaly Detection

Fraudulent financial activities, including money laundering, cyber fraud, and insider trading, pose significant risks to financial markets. Traditional fraud detection systems often rely on rule-based techniques and statistical anomaly detection, which struggle with the complexity and evolving nature of financial fraud schemes. Knowledge graphs enhance fraud detection by representing financial transactions as structured networks, allowing the identification of unusual patterns and relationships.

Cai et al. [4] demonstrate how KG-based anomaly detection systems uncover hidden patterns in transaction networks by analyzing relationships between financial entities. By constructing graph-based fraud detection models, institutions can detect suspicious activities such as circular trading, shell company linkages, and illicit fund transfers. Graph-based techniques, including community detection and link prediction, help identify potential fraudulent actors and their associations within financial networks.

Moreover, the integration of KGs with temporal anomaly detection methods enables real-time fraud detection by continuously monitoring transactional data streams. By leveraging unsupervised learning models on financial KGs, financial institutions can significantly reduce false positives while enhancing fraud detection efficiency.

1.1.3 Regulatory Compliance and Risk Management

Regulatory compliance is a critical aspect of financial operations, ensuring adherence to legal frameworks and mitigating systemic risks. Financial institutions are required to comply with regulations such as Anti-Money Laundering (AML), Know Your Customer (KYC), and Basel III, which demand structured reporting and transparent financial practices. Traditional compliance mechanisms rely on manual audits and rule-based systems, which are labor-intensive and prone to errors.

Padmanaban et al. [5] explore how KGs streamline regulatory compliance by structuring financial data for efficient reporting and audit processes. By leveraging entity resolution techniques and relationship extraction models, financial institutions can automate compliance checks and detect inconsistencies in regulatory filings.

Additionally, KGs facilitate automated risk assessment by linking financial transactions, company ownership structures, and regulatory filings into an interconnected graph. This structured representation enhances the identification of high-risk entities and improves the efficiency of regulatory audits. For instance, by using knowledge graph embeddings, financial institutions can identify potential compliance risks in corporate networks and detect inconsistencies in financial disclosures.

1.1.4 Market Trend Analysis and Financial Forecasting

Financial markets are influenced by a multitude of factors, including macroeconomic indicators, corporate earnings, geopolitical events, and investor sentiment. Identifying market trends and forecasting financial performance require the integration of diverse data sources, ranging from structured financial reports to unstructured news articles and social media data. Traditional market trend analysis methods often rely on econometric models, which have limitations in capturing the dynamic and interconnected nature of financial events.

Knowledge graphs offer a powerful solution by combining financial metrics, event data, and textual embeddings into a unified graph structure. Vasiliu et al. [6] discuss the Graph-Massivizer project, which employs synthetic time-series data generation to improve financial sustainability modeling. By constructing temporal knowledge graphs, financial analysts can capture evolving market trends and assess their impact on financial entities.

Similarly, Sarmah et al. [7] propose HybridRAG, an approach that integrates vector retrieval techniques with KGs to enhance financial information extraction. This method enables financial analysts to retrieve relevant insights from vast textual data repositories by leveraging graph-based relationships between entities and financial events.

Furthermore, KGs facilitate the integration of alternative data sources, such as social media sentiment analysis and alternative credit risk indicators, into traditional financial models. By incorporating entity linking and knowledge graph embeddings, financial institutions can enhance their predictive models for stock price forecasting, asset allocation, and portfolio optimization.

1.1.5 Sentiment Analysis and Financial News Understanding

Financial sentiment analysis plays a crucial role in understanding market reactions to news events, earnings reports, and policy announcements. Traditional sentiment analysis approaches rely on natural language processing (NLP) models that analyze individual textual components without considering entity relationships and contextual dependencies.

By integrating knowledge graphs with NLP techniques, financial institutions can enhance sentiment analysis by capturing entity co-references and contextual linkages. For instance, knowledge graphs can link company announcements, stock performance, and analyst sentiment into a structured framework, enabling more accurate sentiment-driven investment strategies.

Knowledge graphs also enable event detection by linking financial news articles to entities within the financial ecosystem. This structured representation allows for the extraction of causal relationships between events and their impact on market fluctuations. For example, in the case of a major merger announcement, knowledge graphs can trace the ripple effects on related stocks, supply chains, and industry sectors.

1.1.6 Risk Propagation and Counterparty Risk Analysis

Risk propagation is a fundamental concern in interconnected financial markets, where the failure of one institution can have cascading effects on the entire ecosystem. Traditional risk assessment models often rely on historical correlation analysis and stress testing, which may not fully capture hidden dependencies between financial entities.

By leveraging knowledge graphs, financial institutions can model counterparty risk relationships and simulate risk propagation scenarios. KGs enable the identification of systemic risk factors by representing interdependencies between financial institutions, debt obligations, and investment portfolios. This approach enhances financial stability analysis by providing a holistic view of market vulnerabilities and potential contagion effects.

1.1.7 Integration with Large Language Models (LLMs) for Financial Analysis

The integration of knowledge graphs with large language models (LLMs) such as GPT and BERT-based models has opened new frontiers in financial data analysis. Hybrid models combining LLMs with knowledge graphs enable advanced financial reasoning by leveraging both structured and unstructured data.

Recent advancements, such as the use of generative AI for financial document summarization, entity extraction, and automated financial reporting, highlight the potential of this integration. Knowledge graphs act as a structured backbone, enhancing the interpretability and contextual awareness of LLM-generated financial insights.

The application of knowledge graphs in finance is rapidly expanding, addressing challenges in credit risk assessment, fraud detection, regulatory compliance, and market trend analysis. By structuring financial data and integrating it with machine learning techniques, KGs provide a powerful tool for financial decision-making and risk management. The future of financial knowledge graphs lies in their integration with real-time data streams, large-scale LLMs, and enhanced anomaly detection systems, paving the way for more sophisticated and intelligent financial analytics.

1.2 Methods and Advances in Knowledge Graph Construction

Robust methodologies are crucial for constructing financial KGs. Multi-source data integration unifies structured and unstructured datasets, enabling the discovery of complex relationships [8].

Graph embeddings convert KGs into machine-readable formats, retaining relational and semantic information critical for analytics. Advanced machine learning models, such as Graph Neural Networks (GNNs), facilitate tasks like link prediction, anomaly detection, and temporal forecasting [9]. Integration with explainable AI (XAI) and large language models (LLMs) enhances the interpretation of unstructured data and improves analytical explainability [10].

1.3 Taxonomy of Financial Knowledge Graphs

Financial Knowledge Graphs (KGs) have evolved to meet the diverse needs of the financial industry, leading to the development of several distinct types of KGs. Each type serves specific purposes and applications, ranging from risk management to compliance and predictive analytics. This section provides a taxonomy of financial KGs, categorizing them into static, temporal, hybrid, and application-specific types, with a discussion on their characteristics, uses, and foundational references.

1.3.1 Static Knowledge Graphs

Static KGs focus on capturing stable, long-term relationships among financial entities, such as ownership structures, partnership ties, and regulatory classifications. These KGs are particularly useful for compliance reporting, where consistent and accurate representation of financial relationships is crucial. They also play a key role in systemic risk analysis, providing a macroscopic view of financial networks to identify potential points of systemic failure [11]. Static KGs are characterized by their stability and reliability, making them a foundational tool for strategic financial management and planning.

1.3.2 Temporal Knowledge Graphs

Temporal KGs incorporate time-dependent data to model dynamic relationships within the financial sector. These graphs are essential for applications that require understanding changes over time, such as financial trend forecasting, event-based risk modeling, and market dynamics analysis. By integrating temporal data, these KGs help analysts and decision-makers track the evolution of financial entities and their interrelations, aiding in the prediction of future trends and potential market shifts [12]. Temporal KGs are invaluable for capturing the transient nature of financial data and its implications on short-term financial decisions and long-term strategic planning.

1.3.3 Hybrid Knowledge Graphs

Hybrid KGs represent a combination of static and temporal KGs while integrating multimodal data sources, including textual, numerical, and multimedia information. These graphs are particu-

larly effective for complex analytical tasks such as fraud detection, where integrating diverse data types can uncover sophisticated fraudulent schemes [13]. They are also used in supply chain risk management, where understanding the interplay between different modalities of data is crucial for assessing and mitigating risks. Hybrid KGs provide a comprehensive analytical framework that leverages the strengths of both static and temporal data, making them versatile and powerful tools for advanced financial analytics.

1.3.4 Application-Specific Knowledge Graphs

Application-specific KGs are tailored to address particular use cases within the financial industry, embedding domain-specific knowledge and methodologies to enhance precision and relevance. Examples include KGs designed for anti-money laundering (AML), which incorporate specific indicators of financial malpractice, and enterprise risk management, which focus on the unique risk factors relevant to a particular organization or sector [3]. These KGs are developed with targeted functionalities to support specialized tasks, ensuring that the insights they provide are directly applicable and highly actionable.

Overall, the taxonomy of financial KGs illustrates the breadth and depth of applications these tools can support. From ensuring regulatory compliance to enabling dynamic market analysis and tackling specific financial crimes, the versatility of KGs is key to their growing importance in the financial sector.

1.4 Our Contributions

While prior research has explored applications of knowledge graphs (KGs) in finance, many surveys have been narrowly focused on specific tasks or techniques, often overlooking the broader integration of KGs with temporal, dynamic, and multimodal financial data [14], [11]. This study addresses this gap by presenting a comprehensive analysis of financial KGs, including their construction methodologies, integration with advanced machine learning models, and their applications in financial analytics. Our key contributions are outlined below:

1.4.1 A Comprehensive Taxonomy of Financial Knowledge Graphs

Existing research has categorized KGs in general-purpose domains, but a detailed taxonomy tailored to financial applications remains underdeveloped. This survey introduces a fine-grained taxonomy that classifies financial KGs into:

- **Static Knowledge Graphs:** Representing long-term, relatively unchanging financial relationships, such as corporate hierarchies, regulatory compliance structures, and historical economic indicators.

- **Temporal Knowledge Graphs:** Capturing dynamic relationships in financial markets, including stock price movements, economic policy changes, and real-time trade transactions [14].
- **Hybrid Knowledge Graphs:** Combining static and temporal components to model both stable and evolving financial relationships, improving predictive analytics and anomaly detection [11].
- **Application-Specific Knowledge Graphs:** Customized KGs tailored for specific financial tasks, such as fraud detection, credit risk assessment, and regulatory compliance [3], [4].

This classification provides a structured framework for researchers and practitioners to identify the most suitable KG type for their financial applications.

1.4.2 Systematic Review of Knowledge Graph Construction Methods

A significant challenge in financial KG adoption is the lack of standardized methodologies for their construction. We systematically review and categorize KG construction techniques, including:

- **Graph Embeddings:** Methods such as TransE, node2vec, and relational graph convolutional networks (RGCNs) for representing financial entities and their relationships in low-dimensional space [15].
- **Multi-Source Data Integration:** Approaches for unifying structured and unstructured financial data sources, such as earnings reports, regulatory filings, and real-time market data, into coherent graph structures [16].
- **Temporal Modeling in KGs:** Techniques like time-aware graph neural networks (GNNs) and recurrent relational networks (RRNs) that incorporate temporal dependencies into financial predictions [17], [6].
- **Large Language Models (LLMs) for Knowledge Extraction:** The use of transformer-based models, such as FinBERT and HybridRAG, for extracting financial entities and relationships from textual data [7].

This review highlights the state-of-the-art techniques used in constructing financial KGs and their implications for improving data-driven decision-making in finance.

1.4.3 Addressing Challenges in Financial Knowledge Graphs

Financial KGs face several challenges, including data privacy, scalability, and interpretability. Our study critically analyzes these challenges and proposes strategies to address them:

- **Data Privacy and Security:** Ensuring that sensitive financial data remains protected during KG construction, utilizing privacy-preserving graph learning techniques [14].
- **Scalability and Computational Efficiency:** Exploring efficient indexing and distributed graph processing techniques to handle large-scale financial data [11].

- **Interpretability and Explainability:** Integrating explainable AI (XAI) methods, such as SHAP and LIME, to enhance the transparency of financial predictions derived from KGs [16].

We highlight how these challenges are being mitigated through recent innovations in KG architectures, machine learning models, and regulatory compliance frameworks.

1.4.4 Exploring Opportunities in Generative AI and Temporal Modeling

The integration of generative AI and temporal knowledge graphs presents new opportunities for financial analysis. This survey explores:

- **Generative AI for Financial Data Augmentation:** Using techniques like Variational Autoencoders (VAEs) and GANs to synthesize financial transactions for stress-testing predictive models [17].
- **Temporal Graph Neural Networks (TGNNs) for Event Forecasting:** Leveraging models such as Temporal Graph Attention Networks (TGATs) and Temporal Relational Graph Convolutional Networks (TRGCNs) to predict market shifts and systemic risks [6].

These advancements enable financial institutions to create more robust risk management frameworks and enhance market intelligence.

1.4.5 Future Research Directions

This study identifies several promising directions for future research in financial KGs:

- **Multi-Modal Integration:** Expanding KGs to incorporate diverse data modalities, such as satellite imagery for economic forecasting and audio transcripts for investor sentiment analysis [13].
- **Real-Time Dynamic Knowledge Graph Updates:** Developing algorithms for continuously updating financial KGs based on real-time transaction data and regulatory changes [18].
- **KG-Enhanced Financial Virtual Assistants:** Integrating KGs with conversational AI systems to provide intelligent financial insights through natural language interfaces [15].

By outlining these directions, this survey serves as a foundation for further research in the evolving field of financial knowledge graphs.

1.5 Taxonomy of Knowledge Graph Types in Finance

The taxonomy of Knowledge Graph (KG) types in finance categorizes KGs into four primary categories: Static, Temporal, Hybrid, and Application-Specific, each designed to cater to distinct analytical requirements within the financial sector. This taxonomy provides a structured approach

to understanding how KGs can be utilized to model financial data efficiently, enhance predictive analytics, and improve decision-making processes. A visual representation of this taxonomy is illustrated in Figure 1.1.

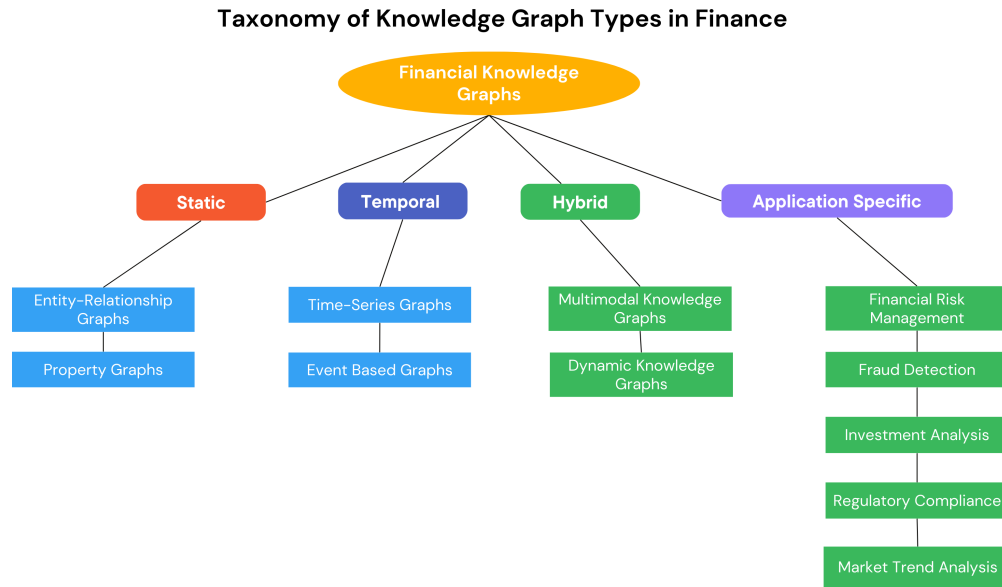


FIGURE 1.1 Taxonomy of Financial Knowledge Graphs

1.5.1 Static Knowledge Graphs

Static Knowledge Graphs represent financial relationships that remain relatively stable over time, making them highly valuable for tasks involving fundamental financial structures such as:

- Corporate ownership structures
- Regulatory compliance frameworks
- Industry classification hierarchies
- Long-term contractual relationships

These KGs maintain a fixed graph structure, typically designed for long-term financial analysis and risk management. Static knowledge graphs are widely used in entity-relationship modeling and property graphs, where financial entities (such as companies, investors, and financial instruments) are linked by well-defined relationships. Vasiliu et al. [6] demonstrated their application in modeling financial transactions and organizational hierarchies, providing a consistent framework for in-depth financial analysis and fraud detection.

1.5.2 Temporal Knowledge Graphs

While static KGs offer robust structural representations, financial systems are inherently dynamic and time-sensitive. Temporal Knowledge Graphs (TKGs) introduce a temporal dimension,

enabling the modeling of evolving financial relationships and patterns. These graphs are particularly useful in:

- Tracking changes in stock market movements
- Analyzing corporate earnings over time
- Modeling economic trends influenced by geopolitical events
- Predicting financial risk by incorporating sequential dependencies

Temporal KGs can be further classified into:

- **Time-Series Graphs:** Capturing continuous changes in financial metrics (e.g., stock prices, interest rates, and exchange rates).
- **Event-Based Graphs:** Modeling discrete financial events, such as mergers, acquisitions, and regulatory changes, which impact financial markets.

Ouyang et al. [10] illustrated the effectiveness of temporal KGs for financial predictions, integrating multimodal financial data (text, numerical, and image-based sources) to improve market trend forecasting. These time-aware KGs enable financial institutions to anticipate risks, identify opportunities, and improve portfolio management strategies.

1.5.3 Hybrid Knowledge Graphs

Hybrid Knowledge Graphs combine the advantages of both static and temporal KGs, offering a more comprehensive analytical perspective. These graphs integrate real-time data streams with historical financial knowledge, making them particularly effective for:

- Fraud detection in banking transactions
- Investment strategy optimization
- Predicting systemic financial risks

Two key subcategories of Hybrid KGs are:

- **Multimodal KGs:** Incorporating structured financial data with unstructured sources, such as earnings reports, news articles, and social media sentiment.
- **Dynamic KGs:** Automatically updating with real-time financial transactions, improving the accuracy and relevance of financial predictions.

Li et al. [13] demonstrated the application of multimodal KGs in phishing detection, leveraging multimodal learning techniques to improve fraud detection in financial transactions. This approach is also adaptable to other complex financial anomalies, such as insider trading and money laundering.

1.5.4 Application-Specific Knowledge Graphs

Application-Specific Knowledge Graphs are designed to solve specialized financial challenges by embedding domain-specific financial knowledge. These graphs enhance analytical precision in key financial tasks, including:

- Credit risk assessment
- Fraud detection and anomaly detection
- Market trend forecasting
- Regulatory compliance automation

Several studies have demonstrated the power of application-specific KGs in finance:

- **Credit Risk Assessment:** Mitra et al. [3] leveraged KGs to assess SME creditworthiness by integrating multi-source financial indicators. This method significantly improves the accuracy and transparency of credit scoring models.
- **Fraud Detection:** Cai and Xie [19] utilized KGs for detecting fraudulent transactions, enabling early-stage fraud prevention through pattern recognition and anomaly detection.
- **Market Trend Analysis:** Ouyang et al. [10] introduced modal-adaptive KGs, integrating economic policy impacts, sentiment analysis, and transaction patterns to enhance predictive market insights.
- **Regulatory Compliance:** Padmanaban et al. [5] demonstrated how KGs streamline regulatory compliance by automating financial reporting processes and ensuring adherence to evolving regulations.

Figure 1.2 provides an overview of different application-specific KGs tailored for various financial use cases.

1.5.5 Advantages of a Structured KG Taxonomy in Finance

The proposed taxonomy of financial knowledge graphs offers several benefits for research and industry applications:

- **Improved Interpretability:** By categorizing financial KGs, institutions can identify the most suitable graph model for specific use cases.
- **Enhanced Predictive Capabilities:** Temporal and hybrid KGs allow for real-time forecasting and financial risk modeling.
- **Scalability and Adaptability:** Application-specific KGs can be tailored to industry-specific challenges, providing more relevant and efficient financial insights.
- **Better Data Integration:** Multimodal KGs enable seamless integration of structured and unstructured financial data, improving analytical depth.

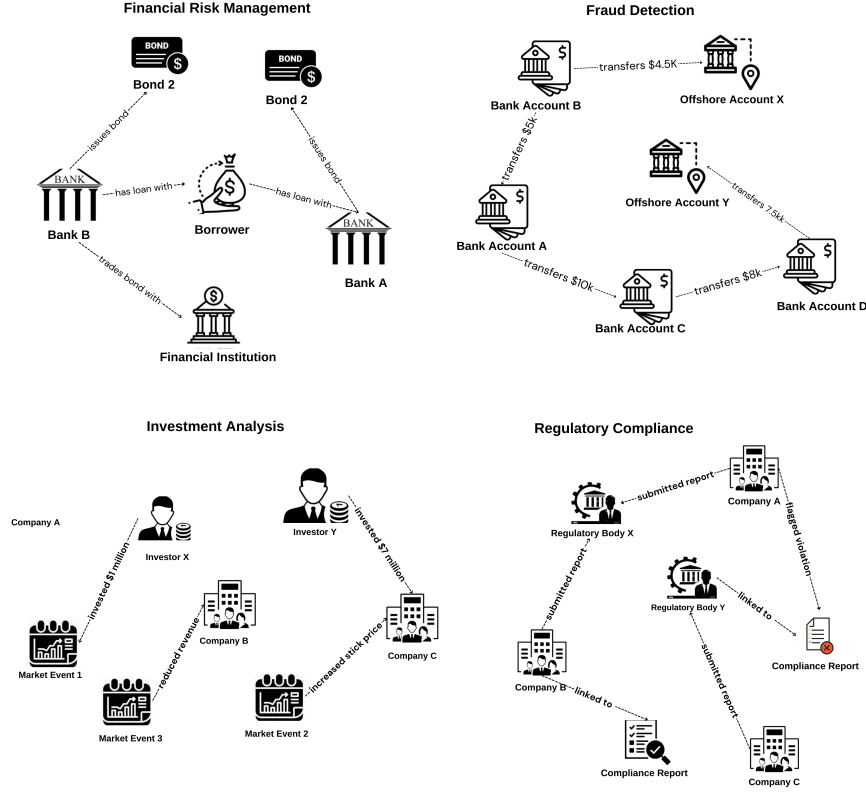


FIGURE 1.2 Application-Specific Knowledge Graphs Examples

This taxonomy provides a comprehensive framework for understanding and utilizing financial knowledge graphs. By distinguishing between static, temporal, hybrid, and application-specific KGs, this classification aids researchers and practitioners in selecting the most appropriate KG type for financial analysis. As financial markets continue to evolve, the adoption of dynamic and AI-enhanced KGs will play a pivotal role in shaping the future of financial intelligence and decision-making.

X₁Y₁Z₁.

CHAPTER 2

Methodologies of Financial Knowledge Graphs

The methodology for constructing and managing financial knowledge graphs (KGs) involves systematic processes, including data collection, integration, representation, and advanced analytical techniques. These methods enable the effective modeling of financial data, which is often complex, heterogeneous, and dynamic.

The Figure 2.1 shows the different types of knowledge graphs.

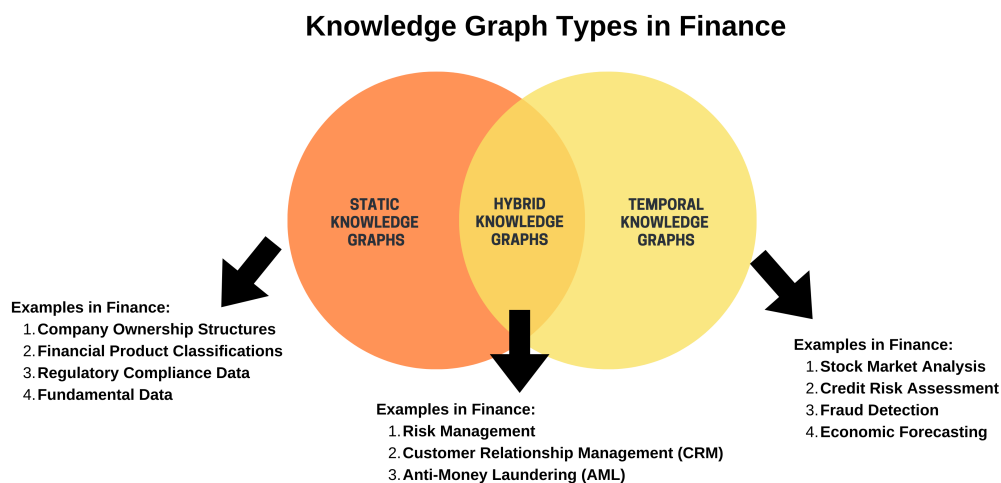


FIGURE 2.1 Types of Knowledge Graphs

2.1 Data Collection and Integration

Data collection and integration form the foundation for constructing high-quality financial knowledge graphs (KGs). In the financial domain, data originates from diverse sources, including corporate filings, regulatory documents, market transactions, and social media sentiment. The

integration of this heterogeneous data into cohesive graph structures enables cross-source analysis, pattern recognition, and complex relationship discovery.

This section explores key methodologies for data collection and integration, including automated extraction techniques, causality modeling, multimodal data fusion, and entity resolution. We also discuss challenges such as data consistency, scalability, and handling real-time updates in financial KGs.

2.1.1 Automated Data Extraction

Financial knowledge graphs require structured data extraction from various unstructured sources, such as annual reports, earnings calls, and SEC filings. Natural Language Processing (NLP) techniques and graph-based representations facilitate this transformation.

Mohsin et al. (2024) introduced an automated framework that extracts financial entities and relationships from structured documents using entity linking and dependency parsing techniques [8]. This method significantly enhances data accessibility and interpretability by transforming unstructured reports into structured graph representations.

Xu et al. (2024) extended this approach by incorporating causality detection frameworks that identify causative relationships between financial events [9]. Their model integrates temporal event sequences into financial KGs, enriching them with actionable insights relevant to financial decision-making.

2.1.2 Multimodal Data Integration

Multimodal integration enhances the utility of financial KGs by incorporating diverse data formats, including:

- **Textual Data:** Earnings press releases, annual reports, news articles, social media sentiment (e.g., Twitter sentiment embeddings).
- **Visual Data:** Stock price trend charts, financial infographics, and visual reports.
- **Numerical Data:** Market indices, asset prices, credit ratings, financial ratios.

Li et al. (2024) emphasized the importance of multimodal fusion in financial analytics by integrating disparate datasets for enterprise risk assessment [2]. Their model links structured and unstructured financial datasets into a unified KG framework, improving risk prediction accuracy.

2.1.3 Entity Resolution and Relationship Linking

Financial KGs require accurate entity resolution to merge redundant representations of the same financial entity. For example:

- Multiple representations of the same company (e.g., "Apple Inc.", "Apple", "AAPL").

- Variations in financial instruments' naming conventions (e.g., "SP500", "S&P 500").

Cai et al. (2024) developed a linking framework that synchronizes textual and visual elements within financial documents [20]. This method ensures that structured and unstructured data sources align, providing a cohesive knowledge graph representation.

Additionally, graph embeddings such as TransE, DistMult, and RotatE are widely used for entity disambiguation and relationship prediction in large-scale KGs. These embedding techniques improve the graph's ability to detect hidden patterns and relationships.

2.1.4 Scalability and Real-Time Data Handling

As financial markets evolve dynamically, KGs must handle real-time data ingestion and scalability challenges. Techniques for real-time KG construction include:

- Stream processing frameworks (e.g., Apache Kafka) for ingesting financial transactions.
- Incremental graph updates using real-time learning algorithms.
- Distributed graph databases (e.g., Neo4j, TigerGraph) for scalable KG storage.

Farghaly et al. (2024) proposed a dynamic KG construction model capable of handling evolving financial relationships [18]. Their approach enables adaptive graph updates to reflect market changes in real time.

Hofer et al. (2024) focused on enhancing KG quality and scalability by introducing hierarchical entity-linking mechanisms [21]. Their model supports large-scale KGs while preserving financial relationship accuracy.

2.1.5 Comparison of Static Knowledge Graph Types in Finance

Static knowledge graphs capture stable financial relationships, providing a fixed graph structure for long-term analysis, such as:

- Compliance monitoring (ensuring financial regulations are met).
- Systemic risk analysis (visualizing interconnections between financial entities).
- Corporate governance modeling (mapping ownership structures).

Chen and Zhang (2024) applied static KGs to visualize systemic risks in banking, identifying interdependencies among financial institutions [22].

Table 2.1 presents a comparative analysis of static knowledge graph types and their applications in finance.

2.1.6 Limitations of Static Knowledge Graphs

Despite their effectiveness, static KGs face limitations in handling evolving financial markets. They lack:

TABLE 2.1 Comparison of Static Knowledge Graph Types in Finance

Paper	Focus Area	Applications	Data Type	Challenges	Key Features
[22]	Systemic Risk in Banking	Visualizes interconnections in financial institutions	Financial relationships	Capturing systemic risk	Highlights connections and vulnerabilities
[8]	Information Extraction	Automates extraction from financial reports	Structured & unstructured data	Handling diverse formats	Enhances accessibility and interpretability
[20]	Text-Visualization Linkage	Links text to visualizations in reports	Text and visual elements	Synchronizing text with visuals	Enables integrated understanding of reports
[18]	KG Construction	Models static KGs for finance	Heterogeneous data	Data structuring & relationship definition	Initial structure for stability and scalability
[21]	KG Quality & Scalability	Enhances KG quality & scalability	Financial & economic data	Ensuring quality at scale	Supports complex inter-entity relationships

- **Real-time adaptability:** Inability to incorporate rapidly changing market events.
- **Dynamic relationship updates:** Static structures do not adjust to evolving company mergers, acquisitions, or new financial regulations.
- **Temporal sequence modeling:** Static graphs struggle to model the chronological progression of financial activities.

To address these challenges, hybrid and temporal KGs are employed, incorporating dynamic updates and predictive analytics.

Data collection and integration methodologies play a important role in constructing high-quality financial knowledge graphs. By combining automated extraction techniques, multimodal data fusion, entity resolution, and real-time data processing, financial KGs can achieve greater accuracy, scalability, and adaptability. Future research should explore more efficient causality modeling techniques and enhanced graph embedding approaches to further optimize knowledge extraction in financial markets.

2.2 Temporal Knowledge Graphs

Temporal Knowledge Graphs (TKGs) introduce a time-dependent dimension to financial knowledge representation, making them essential for applications such as stock trend analysis, economic forecasting, credit risk assessment, and event-driven market predictions. Unlike static KGs, which represent fixed relationships, TKGs continuously evolve by incorporating dynamic financial interactions, real-time transactions, and changing market conditions.

The importance of TKGs in finance lies in their ability to model evolving relationships, track historical trends, and enable predictive analytics. This is particularly relevant in stock price forecasting, credit risk prediction, regulatory compliance monitoring, and fraud detection, where financial data is inherently time-sensitive.

Researchers have proposed various graph-based learning techniques to enhance the predictive capability of TKGs, including Graph Convolutional Networks (GCNs), Long Short-Term Memory (LSTMs), Temporal Graph Attention Networks (TGATs), and Transformer-based models. These approaches enable the identification of sequential patterns, anomaly detection, and real-time decision-making.

2.2.1 Time-Series vs. Event-Based Temporal Knowledge Graphs

Temporal KGs can be categorized into two main types:

- **Time-Series Graphs:** These graphs focus on modeling changes in financial metrics over time, such as stock prices, economic indicators, and credit ratings.
- **Event-Based Graphs:** These graphs capture the relationships between discrete financial events, such as mergers, acquisitions, regulatory actions, and earnings reports.

Shi et al. (2024) applied hybrid GCN-LSTM models to capture temporal dependencies in stock price forecasting, effectively integrating graph structures with time-series data [12]. Their model demonstrated superior forecasting accuracy by leveraging both graph topology and temporal evolution of financial entities.

Additionally, Xia et al. (2024) employed Transformer-based models to enhance historical financial context integration, significantly improving the interpretability of TKGs [23]. This allows for better identification of market trends, regulatory changes, and investment risks.

Event-based TKGs focus on linking financial events to their outcomes. Xiong et al. (2024) introduced Temporal Rule Learning (TRL) to predict the impact of financial events on stock performance and corporate creditworthiness [24]. Their framework identifies causal relationships between events, providing deeper insights into financial forecasting.

2.2.2 Graph Neural Networks for Temporal Knowledge Graphs

The integration of Graph Neural Networks (GNNs) with temporal analysis has significantly enhanced the predictive power of financial KGs. Some notable techniques include:

- **Temporal Graph Attention Networks (TGATs):** These models leverage self-attention mechanisms to dynamically learn the influence of past events on financial entities.
- **Graph Convolutional Networks (GCNs):** Widely used for credit risk assessment, GCNs capture complex relationships in financial transactions.
- **Graph-based Reinforcement Learning (GRL):** GRL models optimize decision-making in financial trading and investment portfolios.

Kosasih et al. (2024) employed Graph Neural Networks (GNNs) for supply chain risk management, demonstrating their ability to detect disruptions and fraud in financial transactions [25]. Similarly, Liang et al. (2024) utilized KG-based reasoning techniques to analyze investment risks, integrating pattern recognition and temporal analysis [11].

2.2.3 Hybrid Temporal Knowledge Graphs

Hybrid models that combine multimodal and dynamic KGs have gained attention for their ability to handle unstructured, structured, and real-time financial data. These hybrid KGs integrate:

- **Large Language Models (LLMs):** Used for financial document understanding, entity extraction, and regulatory compliance.
- **Multimodal Graphs:** Combine text, images, and numerical data for comprehensive financial analysis.
- **Real-time Streaming Data:** Enables live updates for stock trading, market monitoring, and credit scoring.

Li et al. (2024) applied multimodal KGs for fraud detection, integrating real-time transaction analysis with financial reports and news sources [13]. Their framework enhanced anomaly detection by combining LLM-powered text embeddings with graph-based learning models.

In credit risk assessment, Mitra et al. (2024) demonstrated the effectiveness of dynamic knowledge graphs, integrating historical transaction data and external risk indicators [3]. Their approach improved loan default prediction accuracy.

2.2.4 Comparison of Hybrid Knowledge Graphs in Finance

Table 2.3 summarizes key applications of hybrid temporal KGs in finance, highlighting their methods and advantages.

TABLE 2.2 Comparison of Hybrid Knowledge Graphs in Finance

Reference	Multimodal KG	Dynamic KG	Applications	Methods
Mitra et al. (2024) [3]	×	✓	Credit Risk Assessment	Entity-Relationship Analysis
Li et al. (2024) [26]	✓	×	Enterprise Risk Management	Graph-Based Risk Modelling
Chen et al. (2024) [14]	×	✓	Privacy-Preserving Knowledge Graphs	Differential Privacy Techniques
Li et al. (2024) [13]	✓	✓	Phishing Detection	Large Language Models, Multi-modal Graphs
Shi et al. (2024) [27]	×	✓	Stock Price Prediction	GCN-LSTM Model Integration
Kosasih et al. (2024) [25]	×	✓	Supply Chain Risk Management	Graph Neural Networks
Liang et al. (2024) [11]	✓	✓	Knowledge Graph Reasoning	Pattern Recognition, Temporal Analysis
Zhang et al. (2024) [28]	✓	×	Recommender Systems	KG Embedding Techniques
Cadeddu et al. (2024) [29]	✓	×	Tourism Data Analysis	Language Models, KG Integration
Chen et al. (2024) [30]	×	✓	Knowledge Development Prediction	Hybrid Learning Models

Temporal Knowledge Graphs are gaining prominence in financial analytics by incorporating real-time event tracking, sequential pattern recognition, and predictive modeling. Their integration with graph-based learning techniques, multimodal data processing, and large language models enables:

- Improved credit risk assessment through historical and real-time transaction analysis.
- Enhanced fraud detection using dynamic entity relationship tracking.
- More accurate stock trend predictions leveraging GCN-LSTM hybrid models.
- Better regulatory compliance monitoring through automated knowledge extraction.

Future research should focus on scaling TKGs for high-frequency trading, improving explain-

ability with XAI methods, and enhancing real-time anomaly detection.

2.3 Hybrid Knowledge Graphs

Hybrid Knowledge Graphs (KGs) combine features of both static and temporal KGs, often integrating multimodal data sources and real-time updates. These KGs are particularly well-suited for handling complex financial applications, such as fraud detection, credit risk assessment, regulatory compliance, and supply chain risk management. The hybrid nature of these graphs enables them to model both stable relationships and evolving financial interactions, allowing for a more nuanced understanding of financial entities, transactions, and trends.

Unlike purely static KGs, which primarily capture long-term relationships, or temporal KGs, which focus on time-sensitive changes, hybrid KGs blend both approaches. They support real-time adaptation by continuously updating as new financial data, regulatory requirements, or market fluctuations emerge. This capability makes them invaluable in scenarios requiring real-time decision-making, risk assessment, and automated anomaly detection.

2.3.1 Multimodal Integration in Hybrid Knowledge Graphs

Hybrid KGs often integrate multiple data modalities, including:

- **Textual Data:** Financial news, regulatory filings, and earnings reports.
- **Numerical Data:** Stock prices, balance sheets, and financial ratios.
- **Graph-based Data:** Corporate structures, investment relationships, and supply chain linkages.
- **Visual and Multimedia Data:** Charts, scanned documents, and sentiment analysis from social media.

Liang et al. (2024) demonstrated the effectiveness of hybrid KGs in adaptive decision-making, employing pattern recognition and temporal analysis to model evolving relationships [11]. Their approach showcased how combining multimodal data sources enhances financial forecasting accuracy and anomaly detection.

Li et al. (2024) further explored multimodal hybrid KGs in phishing detection, showing that the integration of textual and multimedia data (such as images, emails, and web content) significantly improved fraud detection accuracy [13]. This research highlights the potential of hybrid KGs in identifying complex fraud patterns, which may not be evident in single-modality analysis.

Kosasih et al. (2024) employed dynamic Graph Neural Networks (GNNs) to monitor supply chain risks and enable proactive decision-making [25]. Their model continuously updates relationships between entities based on real-time transactions and market changes, allowing for immediate detection of financial and operational risks.

2.3.2 Types of Hybrid Knowledge Graphs in Finance

Hybrid KGs can be categorized into the following types:

- **Multimodal Knowledge Graphs:** These KGs integrate multiple data formats (text, images, numerical data, and structured databases) to provide a comprehensive financial analysis.
- **Dynamic Knowledge Graphs:** These KGs continuously update based on real-time financial transactions and market conditions.
- **Temporal Hybrid Knowledge Graphs:** These graphs capture both historical financial trends and real-time updates, enabling predictive modeling.

Example Applications:

- **Regulatory Compliance and Risk Management:** Hybrid KGs help financial institutions track evolving regulations, integrating real-time updates from regulatory bodies.
- **Fraud Detection and Anti-Money Laundering (AML):** By integrating transaction networks, textual reports, and visual evidence, hybrid KGs improve fraud detection accuracy.
- **Investment Analysis and Market Trend Prediction:** These graphs analyze historical performance and live market data to forecast stock movements.
- **Credit Scoring and Loan Risk Assessment:** Hybrid KGs assess borrower creditworthiness by integrating historical financial statements, alternative credit data, and macroeconomic trends.

2.3.3 Comparison of Hybrid Knowledge Graphs in Finance

Table 2.3 provides an overview of key applications of hybrid KGs in finance, highlighting their use cases, methodologies, and advantages.

2.3.4 Future Directions for Hybrid Knowledge Graphs in Finance

Despite their advancements, hybrid KGs face several challenges that future research can address:

- **Scalability:** Efficiently managing the computational complexity of large-scale hybrid financial knowledge graphs.
- **Explainability:** Enhancing interpretability in automated financial decision-making.
- **Security and Privacy:** Ensuring data protection in knowledge graph-based financial systems.
- **Real-time Adaptation:** Further improving real-time event tracking and dynamic updates in fast-changing financial markets.

Emerging Graph Transformer models and self-supervised learning techniques have the potential to significantly enhance hybrid KG performance, particularly in applications such as real-time fraud detection, predictive risk management, and automated trading strategies.

TABLE 2.3 Comparison of Hybrid Knowledge Graphs in Finance

Paper	Focus Area	Applications	Methods	Key Features
Li & Passino (2024) [31]	Financial Trend Detection	Extracting temporal KGs	Large Language Models	Real-time trend detection pipeline
Xu et al. (2024) [32]	Financial Causality	Causality analysis	Text Mining	Links financial events temporally
Xiong et al. (2024) [24]	Temporal Rule Learning	Event prediction	Temporal Rules	Sequential financial pattern analysis
Vasiliu et al. (2024) [33]	Synthetic Data Generation	Financial time-series modeling	Knowledge Graphs	Generates synthetic financial data
Xia et al. (2024) [23]	Temporal Forecasting	Temporal KG forecasting	Chain-of-History Reasoning	Integrates historical context
Lin et al. (2024) [34]	Temporal Reasoning	Credit risk, investment	Temporal Embeddings	Multi-hop reasoning on financial data
Farghaly et al. (2024) [18]	Dynamic KG Construction	Structural and temporal integration	Temporal Modeling	Integrates structural and temporal KG features
Bai et al. (2024) [35]	Cross-lingual Alignment	Multi-language financial datasets	Embedding-based Alignment	Expands TKGs to cross-lingual data
Shi et al. (2024) [12]	Stock Prediction	Stock movement forecasting	GCN-LSTM	Combines graph structure with temporal data
Dong et al. (2024) [36]	Temporal Path Reasoning	Time-sensitive prediction	Inductive Path Neural Network	Extracts time-sensitive prediction paths

Hybrid Knowledge Graphs provide a powerful framework for combining structured and unstructured financial data, enabling improved predictive analytics, fraud detection, and risk management. The integration of multimodal data sources and real-time processing makes them indispensable for financial decision-making. Future research should focus on enhancing scalability, security, and real-time adaptability to fully unlock their potential in financial applications.

2.4 Machine Learning Models for Knowledge Graphs

Machine learning (ML) models play a pivotal role in constructing, analyzing, and utilizing financial Knowledge Graphs (KGs). These models help in discovering hidden relationships, predict-

ing future financial trends, detecting fraudulent activities, and improving risk assessment methodologies. The primary ML techniques used for KGs include Relational Graph Convolutional Networks (RGCNs), Transformers, Large Language Models (LLMs), and hybrid learning approaches.

2.4.1 Graph Neural Networks (GNNs) for Financial Knowledge Graphs

Graph Neural Networks (GNNs) have revolutionized financial KGs by enabling deep learning over structured graph data. Among them, Relational Graph Convolutional Networks (RGCNs) have emerged as one of the most effective models for capturing multi-relational dependencies.

2.4.1.1 Relational Graph Convolutional Networks (RGCNs)

RGCNs, introduced by Schlichtkrull et al. (2018), extend traditional Graph Convolutional Networks (GCNs) to handle heterogeneous graphs where entities and relations belong to different types [37]. These networks are particularly useful in financial applications that involve complex interconnections among different financial entities.

Applications of RGCNs in Finance:

- **Credit Risk Assessment:** RGCNs analyze borrower-lender relationships, integrating transaction histories and financial statements to evaluate creditworthiness.
- **Systemic Risk Modeling:** By examining relationships between financial institutions, RGCNs identify potential contagion risks within the financial ecosystem.
- **Entity Resolution in KGs:** RGCNs improve the accuracy of entity linking and disambiguation, which is essential in automated financial reporting.

Tao et al. (2022) leveraged RGCNs for financial performance prediction, integrating graph-based and time-series data to enhance forecasting accuracy [38]. Similarly, Wang et al. (2024) introduced multi-relational embeddings within RGCNs for temporal knowledge graph completion, improving financial event prediction [39].

2.4.2 Transformer Models for Financial Knowledge Graphs

Transformers have gained widespread adoption in large-scale KG processing, especially in handling dynamic and evolving financial data.

2.4.2.1 Graph Transformers

Graph Transformers have proven highly effective in financial analytics, particularly in fraud detection, stock trend analysis, and financial text processing.

Key Contributions in Financial KGs:

- Tang and Liu (2024) applied Transformers in a distributed knowledge distillation framework to detect financial fraud, identifying anomalies in transaction patterns across multiple institutions [40].
- Li and Passino (2024) introduced the FindKG framework, which utilizes heterogeneous graph Transformers to dynamically construct KGs for market trend detection [31].
- Zhao et al. (2024) combined Graph Transformers with multimodal embeddings to predict macroeconomic trends from financial reports and historical data [17].

Transformer models are particularly effective in learning long-range dependencies within financial graphs, making them well-suited for financial event prediction, credit scoring, and financial knowledge extraction.

2.4.3 Large Language Models (LLMs) in Financial Knowledge Graphs

Recent advancements in LLMs such as Gemma, GPT-4, and Llama have significantly improved the automated construction, augmentation, and reasoning over financial knowledge graphs.

LLMs have been leveraged in financial KGs for:

- Financial Document Understanding: LLMs extract entities, relationships, and event sequences from earnings reports, regulatory filings, and news articles.
- Automated KG Construction: Models like HybridRAG integrate vector retrieval techniques with LLMs, improving knowledge extraction accuracy [7].
- Explainable AI in Finance: Shaikh et al. (2024) developed XKG, an explainable financial knowledge graph system using LLMs to enhance interpretability [16].

Integration of LLMs with Graph Models:

- Graph-Augmented LLMs: LLMs enhance financial knowledge graphs by learning from structured and unstructured data.
- KG-Guided LLMs: KGs improve LLM reasoning capabilities by constraining responses based on structured knowledge.

Pan et al. (2024) proposed a unified framework integrating LLMs with KG embeddings, improving performance in financial question answering and automated risk analysis [15].

2.4.4 Hybrid Learning Approaches

Hybrid learning approaches combine multiple ML models, such as GNNs, Transformers, and LLMs, to enhance financial KG performance.

Examples of Hybrid Models:

- Ouyang et al. (2024) introduced modal-adaptive GNNs that integrate financial news, policy documents, and time-series data into a unified KG for macroeconomic forecasting [10].

- Liao et al. (2024) applied generative LLMs to enrich financial KGs with synthetic data, improving predictive analytics in credit risk modeling [41].
- Lin et al. (2024) developed T-Flex, a hybrid KG model that incorporates temporal embeddings and LLM-generated features for multi-hop reasoning on financial data [34].

2.4.5 Challenges and Future Directions

Despite the advancements in machine learning for financial knowledge graphs, several challenges remain:

1. Scalability and Efficiency

- Training large-scale financial KGs with millions of nodes and billions of edges remains computationally expensive.
- Optimization techniques such as self-supervised learning and graph pruning can help improve model efficiency.

2. Explainability and Interpretability

- Financial decisions must be interpretable and auditable.
- The integration of XAI techniques with KG-based machine learning models can improve transparency.

3. Data Privacy and Security

- Financial KGs often involve sensitive data, necessitating privacy-preserving techniques such as differential privacy and federated learning.
- Chen et al. (2024) proposed privacy-preserving knowledge graphs for secure financial analytics [14].

4. Real-Time Adaptability

- Many financial applications require real-time updates in response to market fluctuations.
- Research in streaming GNNs and real-time graph embeddings is crucial for the next generation of financial KGs.

Machine learning models have transformed the landscape of financial knowledge graphs, enabling applications ranging from credit risk assessment and fraud detection to macroeconomic forecasting and regulatory compliance. The integration of RGCNs, Transformers, LLMs, and hybrid models has significantly improved the capabilities of financial KGs. Future research should focus on scalability, explainability, and privacy-preserving techniques to further enhance trustworthy AI-driven financial decision-making.

2.5 Enhancing Knowledge Graphs with Large Language Models

The integration of Large Language Models (LLMs) with Knowledge Graphs (KGs) has significantly expanded their utility by enhancing semantic depth, contextual understanding, and adaptability. LLMs improve the representation, reasoning, and querying capabilities of financial KGs, enabling more accurate and explainable financial decision-making.

2.5.1 Semantic Representation and Contextual Enhancement

Pan et al. (2024) highlighted the reasoning capabilities of LLM-enhanced KGs, demonstrating their ability to accurately represent complex financial relationships and perform entity linking across diverse datasets [15]. By incorporating LLM-generated embeddings, financial KGs can infer missing relationships, recognize ambiguous entities, and enhance overall knowledge retrieval.

Transformers and LLMs such as FinBERT, Open-finLLMs, and BloombergGPT have been particularly effective in financial applications where sentiment analysis, compliance monitoring, and fraud detection are required [42]. These models are fine-tuned on domain-specific datasets, ensuring higher accuracy in financial text interpretation compared to general-purpose LLMs.

2.5.2 Retrieval-Augmented Transformers (RATs) for Knowledge Graphs

Retrieval-Augmented Transformers (RATs) have been proposed as a method to dynamically integrate external knowledge into LLMs, improving their ability to handle real-time financial trend detection. Zhao et al. (2024) demonstrated the application of RATs in financial KGs, where real-time news, stock market updates, and regulatory changes were incorporated dynamically to improve forecasting accuracy [43].

Benefits of Retrieval-Augmented LLMs for Financial KGs:

- **Dynamic Data Integration:** Real-time market trends, earnings reports, and financial news can be continuously updated in the KG.
- **Improved Query Response:** LLMs enhance financial knowledge retrieval by reducing hallucinations and improving fact-based responses.
- **Adaptability in Financial Analysis:** RAT-based models allow financial institutions to adapt to regulatory changes and market shifts rapidly.

2.5.3 Trustworthiness and Explainability in LLM-Driven Financial KGs

One of the major challenges of integrating LLMs with KGs is ensuring trustworthiness and explainability. **Zafar et al. (2024)** addressed these concerns by developing frameworks for building transparent and explainable LLM-driven KGs, ensuring that financial decisions are auditable and interpretable [44].

Key Approaches for Explainable LLM-Driven KGs:

- **Natural Language Explanations:** Providing users with human-readable justifications for KG-based decisions.
- **Constraint-Based Querying:** Using logical constraints to ensure that LLM-generated knowledge aligns with established financial regulations.
- **Knowledge Provenance Tracking:** Ensuring that all extracted information from LLMs is traceable to original financial sources.

As financial institutions increasingly rely on AI-driven knowledge graphs, it is critical to develop governance mechanisms for LLM-powered KGs that ensure reliability, auditability, and regulatory compliance.

2.6 Graph Embedding and Machine Learning

Graph embedding techniques play a crucial role in converting KG structures into machine-readable vector representations, enabling advanced analytics such as link prediction, clustering, and financial anomaly detection. Embeddings preserve relational and temporal information, supporting real-time KG updates and improved interpretability.

2.6.1 Graph Embedding Techniques in Financial Knowledge Graphs

Graph embedding transforms structured financial knowledge graphs into numerical vector spaces, enabling efficient computation and pattern discovery. Shaikh et al. (2024) proposed explainable KG embeddings that allow natural language queries to retrieve interpretable financial insights [16].

Types of Graph Embeddings in Finance:

- **Node Embeddings:** Capturing the role of financial entities (e.g., companies, transactions) within a KG.
- **Edge Embeddings:** Representing relationships (e.g., financial transactions, risk connections) between entities.
- **Temporal Embeddings:** Modeling how financial data evolves over time, improving time-sensitive predictions.

2.6.2 Graph Neural Networks (GNNs) for Financial Knowledge Graphs

Graph Neural Networks (GNNs), particularly Relational Graph Convolutional Networks (RGCNs) and Graph Attention Networks (GATs), have been widely adopted for analyzing complex financial relationships.

Zhu and Wu (2024) demonstrated the utility of KG-GNNs in peer-to-peer lending, identifying credit risk factors through relational analysis [45]. Similarly, Cai and Xie (2024) leveraged GNN-based knowledge graphs for fraud detection, using a two-layer graph structure to enhance precision [19].

Key Financial Applications of GNNs:

- **Fraud Detection:** Analyzing transaction networks to detect suspicious financial patterns.
- **Credit Risk Modeling:** Predicting loan default probabilities by analyzing borrower relationships and historical payments.
- **Stock Market Analysis:** Identifying interdependencies between corporate financial reports and stock price fluctuations.

2.6.3 Advancements in Graph-Based Machine Learning for Finance

Recent advancements in graph-based machine learning models have led to the development of multi-modal and cross-domain KG applications in finance.

Cutting-Edge ML Models for Financial KGs:

- **Temporal Graph Neural Networks (TGNNs):** Used for predicting stock trends by analyzing historical financial time-series data.
- **Hybrid Graph Transformers:** Combining graph-based embeddings with transformer-based models for market risk predictions.
- **Multi-Relational Graph Embeddings:** Encoding diverse financial relationships (e.g., lender-borrower, firm-investor) for improved forecasting.

Xu et al. (2024) proposed a hybrid approach that integrates causal reasoning with KG embeddings, improving the interpretability of causal financial event detection [9].

2.6.4 Challenges and Future Directions in Graph-Based ML for Financial KGs

Despite significant advancements, several challenges remain in developing graph-based ML models for finance:

1. Scalability and Computational Complexity

- Handling large-scale financial networks (millions of entities, billions of transactions).
- Developing efficient graph sampling techniques to reduce computational costs.

2. Interpretability and Trustworthiness

- Ensuring explainable financial predictions in high-stakes decision-making.
- Incorporating human-in-the-loop validation for improved accuracy.

3. Real-Time Adaptability

- Enabling real-time updates in dynamic financial markets.
- Improving streaming GNN architectures for faster inference.

4. Privacy and Security in Financial KGs

- Implementing differential privacy techniques to protect sensitive financial data.
- Using federated learning for secure distributed KG training.

The integration of LLMs, Transformers, and GNNs has significantly enhanced financial KGs, enabling real-time analytics, fraud detection, and credit risk modeling. However, further research is needed to improve scalability, interpretability, and privacy while ensuring that financial KGs remain reliable and compliant with evolving regulations.

2.7 Graph Embedding and Machine Learning

Graph embedding techniques play a crucial role in transforming Knowledge Graph (KG) structures into numerical vector representations, enabling advanced analytics such as link prediction, clustering, and anomaly detection. These embeddings retain relational and temporal information, making them particularly effective for financial applications requiring dynamic graph analysis.

2.7.1 Graph Embeddings for Financial Knowledge Graphs

Graph embedding converts structured financial knowledge into dense numerical representations, facilitating machine learning applications such as fraud detection, credit scoring, and financial risk modeling. Shaikh et al. (2024) proposed explainable KG embeddings that allow natural language queries to retrieve interpretable financial insights [16].

Types of Graph Embeddings Used in Finance:

- **Node Embeddings:** Capture the roles of financial entities (e.g., companies, transactions, investment instruments) in a KG.
- **Edge Embeddings:** Represent relationships (e.g., transactions, contractual agreements) between entities.
- **Temporal Embeddings:** Preserve time-sensitive financial interactions, allowing for event-based forecasting.
- **Heterogeneous Embeddings:** Integrate multiple data modalities, such as textual financial reports, stock prices, and regulatory filings.

Graph embedding models such as DeepWalk, Node2Vec, and TransE have been widely adopted to generate low-dimensional vector representations of financial entities. However, these classical methods lack contextual understanding, which has led to the integration of Graph Neural Networks (GNNs) and Transformer-based embeddings for more sophisticated learning paradigms.

2.7.2 Graph Neural Networks (GNNs) for Financial Knowledge Graphs

Graph Neural Networks (GNNs) are a class of machine learning models specifically designed to handle graph-structured data. In financial knowledge graphs, GNNs enable tasks such as fraud detection, risk assessment, and predictive analytics.

Types of GNNs Used in Financial Applications:

- Relational Graph Convolutional Networks (RGCNs): Capture multi-relational financial entities and their dependencies [37].
- Graph Attention Networks (GATs): Use attention mechanisms to weigh important connections in financial transactions.
- Temporal Graph Neural Networks (TGNNs): Incorporate time-sensitive data for applications such as stock trend forecasting and market anomaly detection.
- Heterogeneous Graph Neural Networks (HGNNs): Model multi-type nodes and edges in diverse financial datasets.

Zhu and Wu (2024) demonstrated the application of KG-GNNs in peer-to-peer lending, where relational analysis of financial transactions helped identify credit risk factors [45]. Similarly, Cai and Xie (2024) leveraged GNN-based knowledge graphs for fraud detection, using a two-layer graph structure to improve precision and explainability [19].

2.7.3 Financial Applications of Graph-Based Machine Learning

Graph-based machine learning models have revolutionized the financial industry by enabling automated reasoning, real-time anomaly detection, and predictive analytics. The following sections discuss some key applications.

2.7.3.1 Fraud Detection

Fraud detection in financial networks benefits significantly from GNN-based anomaly detection models. Financial fraud often involves complex transaction patterns, hidden relationships, and synthetic identities, which can be identified using graph embeddings and link prediction techniques.

Cai and Xie (2024) employed multi-relational knowledge graphs with GNNs to detect fraud patterns, successfully uncovering suspicious transaction networks in large-scale datasets [19].

2.7.3.2 Credit Risk Assessment

Credit risk modeling relies on predicting borrower default probabilities based on their financial history, network interactions, and macroeconomic indicators. RGCNs and HGNNs are commonly used to model credit risk, incorporating structured and unstructured financial data.

Zhu and Wu (2024) introduced a GNN-based credit scoring framework, which outperformed traditional credit assessment models by leveraging multi-relational knowledge graphs [45].

2.7.3.3 Stock Market Prediction

The stock market is highly influenced by company relationships, investor sentiment, and economic events. Temporal GNNs and hybrid embedding models have been developed to predict stock trends based on financial knowledge graphs.

Shi et al. (2024) used a GCN-LSTM hybrid model to capture both graph-structured dependencies and sequential time-series trends, improving stock price forecasting accuracy [27].

2.7.3.4 Systemic Risk Analysis

Systemic financial risks arise from interconnected financial entities such as banks, hedge funds, and lending institutions. Knowledge graphs allow regulators to visualize financial networks, simulate systemic risk scenarios, and preempt financial crises.

Li et al. (2024) developed a risk propagation model using heterogeneous GNNs, allowing central banks to assess systemic vulnerabilities and respond proactively [26].

2.7.4 Challenges in Graph-Based Machine Learning for Finance

Despite significant advancements, several challenges remain in the development of graph-based ML models for financial knowledge graphs:

1. Scalability and Computational Complexity

- Financial knowledge graphs often contain millions of entities and billions of edges, making scalability a critical concern.
- Efficient graph sampling techniques and distributed training methods are needed for real-time processing.

2. Interpretability and Explainability

- Financial decision-making requires high interpretability, as model outputs impact risk assessments, investments, and regulatory compliance.
- Explainable GNNs and embedding techniques, such as attention-based models and concept-based graph reasoning, can enhance trustworthiness.

3. Handling Dynamic and Evolving Financial Data

- Financial relationships change over time, requiring adaptive learning models that update dynamically.
- Temporal GNNs and streaming graph learning algorithms are key to handling real-time updates in financial KGs.

4. Privacy and Security in Financial KGs

- Financial datasets are highly sensitive, necessitating differential privacy techniques and federated learning for secure graph model training.

2.7.5 Future Directions in Graph-Based Financial Knowledge Graphs

Future research will focus on addressing existing limitations and expanding the capabilities of financial knowledge graphs. Key areas of development include:

1. Integration of Large Language Models (LLMs) with GNNs
 - Combining pre-trained financial LLMs (e.g., FinBERT, BloombergGPT) with graph neural networks can enhance financial text understanding.
2. Hybrid Graph Learning Architectures
 - Hybrid architectures, such as Transformer-GNN hybrids, can improve multi-modal financial risk prediction.
3. Real-Time Financial Graph Streaming
 - Advances in online graph learning algorithms will allow real-time updates for financial KGs.
4. Explainable AI (XAI) for Financial Graphs
 - Developing post-hoc explainability techniques for graph models will improve regulatory compliance and trust in AI-driven financial decisions.

Summary

This chapter explores the methodologies involved in constructing, managing, and analyzing financial knowledge graphs (KGs). Financial KGs integrate heterogeneous data sources such as corporate filings, stock transactions, financial news, and regulatory documents to enable applications like credit risk assessment, fraud detection, and systemic risk modeling. The chapter outlines key techniques for data collection, including automated data extraction, multimodal data fusion, and entity resolution. It also discusses the challenges of scalability and real-time updates, highlighting technologies such as Apache Kafka for streaming data and distributed graph databases like Neo4j and TigerGraph.

Furthermore, the chapter categorizes financial KGs into static and temporal structures. While static KGs capture long-term relationships, temporal KGs incorporate dynamic financial interactions, supporting event-driven forecasting and anomaly detection. Machine learning models such as Relational Graph Convolutional Networks (RGCNs), Graph Attention Networks (GATs), and Large Language Models (LLMs) significantly enhance the predictive capabilities of KGs. The integration of explainable AI (XAI) and hybrid learning architectures ensures interpretability, trustworthiness, and regulatory compliance in financial decision-making.

Additionally, the chapter examines hybrid knowledge graphs that merge static, temporal, and multimodal data sources to provide comprehensive financial insights. Techniques such as retrieval-augmented transformers (RATs), self-supervised learning, and graph embeddings like TransE and

Node2Vec further improve knowledge extraction and analysis. Future research should focus on scalability, explainability, and privacy-preserving mechanisms to enhance the reliability and adaptability of financial KGs in real-world applications.

X₁ $\mathbb{R}^d$.

CHAPTER 3

Applications, Challenges, and Future Directions

In this section, we examine key applications of financial knowledge graphs (KGs), illustrating their role in diverse areas. KGs enhance credit risk assessment by integrating multiple financial indicators to evaluate creditworthiness; support fraud detection by analyzing graph-based relationships to spot anomalies; and aid market trend analysis by connecting and interpreting market data to forecast economic shifts. Additionally, KGs are critical in systemic risk management, identifying potential contagion across financial entities, and in regulatory compliance, where they streamline data processes to meet complex reporting requirements.

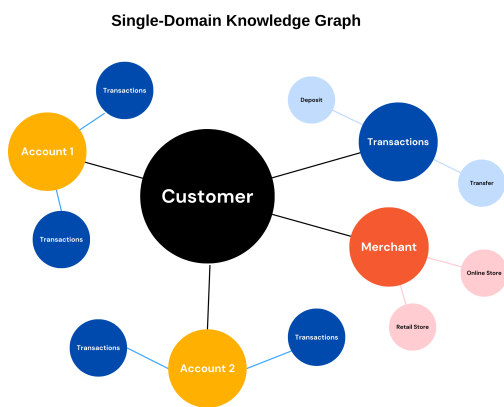


FIGURE 3.1 Single Domain Financial Knowledge Graphs

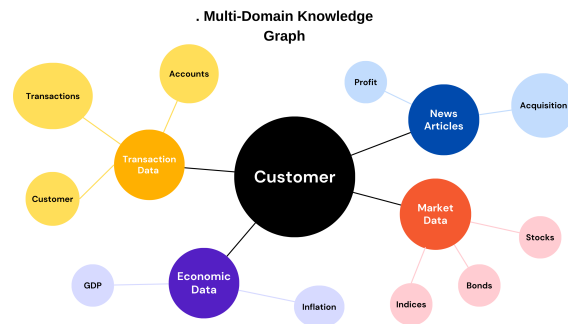


FIGURE 3.2 Multi-domain Financial Knowledge Graphs

Financial KG research distinguishes between single-domain and multi-domain graphs. A single-domain KG, as shown in Figure 3.1, focuses on specific areas like credit risk, integrating detailed financial histories and transaction data to provide deep insights into creditworthiness within lending frameworks [3]. In contrast, multi-domain KGs, illustrated in Figure 3.2, span various financial domains—combining data on market trends, corporate risks, and regulatory factors. This approach offers a holistic view for enterprise risk management, enabling analyses across financial areas, such as the impact of market changes on credit risk and compliance [2, 46].

3.1 Financial Risk Assessment

Financial risk assessment is a critical component of the financial industry, where organizations evaluate the potential risks associated with credit, liquidity, and systemic market fluctuations. Financial Knowledge Graphs (KGs) provide a structured and scalable approach to risk assessment by integrating diverse financial data sources and leveraging graph-based machine learning models to detect hidden patterns, assess vulnerabilities, and predict financial outcomes. This section explores the role of KGs in credit risk evaluation, liquidity risk management, and systemic risk assessment.

3.1.1 Credit Risk Assessment Using Knowledge Graphs

Credit risk assessment involves evaluating the likelihood that a borrower will default on their financial obligations. Traditional methods rely on credit scores and financial statements, but they often fail to capture complex interdependencies and hidden relationships in financial transactions. Knowledge Graphs (KGs) overcome these limitations by integrating multiple data sources, including banking transactions, corporate records, and alternative credit indicators.

Mitra et al. (2024) developed a KG-based framework for credit risk assessment in Micro, Small, and Medium Enterprises (MSMEs), which improves lending decisions by identifying risk factors beyond traditional credit scoring models [3]. By incorporating transaction histories, loan repayments, and supplier relationships, their model enhances creditworthiness evaluation and reduces loan default rates.

Song et al. (2024) introduced a multi-level KG-based approach to analyze enterprise credit risk [47]. Their model aggregates financial performance metrics, debt obligations, and inter-company trade data to predict financial distress and bankruptcy probabilities more accurately than conventional methods.

To ensure data privacy and security, Wang et al. (2024) proposed a federated learning framework for privacy-preserving credit scoring [48]. Their method allows financial institutions to collaboratively build credit risk models using distributed KGs while preserving user confidentiality. Federated KG learning is particularly useful in regulatory-sensitive environments where financial data cannot be shared across institutions.

3.1.2 Liquidity Risk Management

Liquidity risk refers to the inability of a financial institution to meet short-term obligations due to cash flow constraints. Knowledge Graphs provide a real-time, interconnected view of liquidity dynamics by analyzing relationships between banks, trading firms, and capital markets.

Chen and Zhang (2024) demonstrated how liquidity risk can be assessed using systemic KGs, where interbank lending, asset correlations, and market stress factors are represented as a structured

network [22]. Their approach enables financial institutions to simulate liquidity crises and take proactive measures to mitigate potential risks.

Wei et al. (2024) applied graph neural networks (GNNs) to liquidity risk forecasting, predicting sudden liquidity shortages based on market transaction networks [49]. By integrating real-time financial data streams, their model identifies at-risk institutions before liquidity crises emerge, offering early warning indicators for regulators and policymakers.

3.1.3 Systemic Risk Analysis and Financial Stability

Systemic risk refers to the potential collapse of an entire financial system due to the failure of interconnected institutions. Traditional models often fail to capture the cascading effects of financial distress, whereas Knowledge Graphs provide a more holistic view of systemic vulnerabilities.

Gambacorta et al. (2024) incorporated non-traditional data sources, such as supply chain disruptions and geopolitical risks, into financial KGs to improve systemic risk predictions [50]. Their model enhances financial stability assessments by identifying weak links within global financial networks.

The interconnected nature of financial markets makes them susceptible to contagion effects, where financial distress in one institution spreads to others. Financial KGs model these interdependencies through dynamic graph structures, capturing real-time interactions between firms, investors, and regulators.

Table 3.1 summarizes key research contributions in financial risk assessment using Knowledge Graphs.

3.1.4 Future Directions in Financial Risk Assessment Using KGs

While financial KGs have significantly improved risk assessment methodologies, several challenges remain:

- **Scalability and Real-Time Analysis:** Many financial KGs require high-frequency updates to reflect rapid market changes. Future research should focus on developing scalable graph architectures with real-time processing capabilities.
- **Explainability and Interpretability:** The complexity of graph-based models makes it difficult to explain credit risk decisions. Integrating explainable AI (XAI) techniques into financial KGs can improve regulatory compliance and transparency.
- **Data Privacy and Security:** Federated learning and differential privacy techniques should be further explored to protect sensitive financial data while enabling collaborative risk assessment.
- **Integration with Alternative Data Sources:** Financial KGs can be enriched by incorporating alternative credit indicators, such as social media sentiment, e-commerce transactions, and behavioral analytics.

TABLE 3.1 Comparison of Financial Risk Assessment Models Using Knowledge Graphs

Study	Focus Area	Methodology	Key Findings	Limitations
Mitra et al. (2024) [3]	MSME Credit Risk	Transaction-based KG modeling	Enhanced loan default prediction	Requires high-quality financial transaction data
Song et al. (2024) [47]	Enterprise Credit Risk	Multi-level KG analysis	Improved creditworthiness assessment	Limited to structured financial reports
Wang et al. (2024) [48]	Privacy-Preserving Credit Scoring	Federated Learning on KGs	Secure collaborative credit risk modeling	Computationally expensive due to distributed learning
Chen and Zhang (2024) [22]	Liquidity Risk Analysis	Systemic KG for financial institutions	Real-time liquidity monitoring	Requires high-frequency financial transaction updates
Wei et al. (2024) [49]	Bankruptcy Prediction	GNN-based forecasting	Early identification of liquidity crises	Model interpretability remains a challenge
Gambacorta et al. (2024) [50]	Systemic Risk	Non-traditional data integration in KGs	Improved risk predictions under uncertainty	Requires external data sources for accuracy

By addressing these challenges, future advancements in financial KGs will enable more robust, real-time, and transparent risk assessment models, ultimately improving financial stability and decision-making.

3.2 Fraud Detection

Financial fraud is a significant global concern, with illicit activities such as money laundering, credit card fraud, identity theft, and insider trading posing serious threats to financial institutions and regulatory bodies. Traditional fraud detection systems rely on rule-based approaches and statistical analysis, which often fail to detect complex and evolving fraudulent schemes. The introduction of Knowledge Graphs (KGs) has revolutionized fraud detection by structuring financial transactions, linking heterogeneous data sources, and applying machine learning techniques to uncover hidden fraud patterns.

3.2.1 Role of Knowledge Graphs in Fraud Detection

Knowledge graphs provide a graph-based representation of financial transactions, enabling the detection of anomalous behaviors through entity relationships and network patterns. By integrating

multiple data sources such as banking transactions, user profiles, corporate records, and regulatory filings, KGs offer a holistic view of financial activities, making them more effective in identifying suspicious activities.

Key benefits of KGs in fraud detection include:

- **Anomaly Detection:** Identifies unusual transaction patterns, circular money flows, and shell company structures.
- **Entity Resolution:** Resolves multiple representations of a single entity, such as different names or account numbers for the same fraudster.
- **Graph-Based Feature Engineering:** Extracts high-level features such as transaction centrality, clustering coefficients, and network connectivity.
- **Explainability:** Provides interpretable fraud detection results through network visualization and entity-linking.

3.2.2 Graph-Based Machine Learning Approaches for Fraud Detection

Machine learning models integrated with knowledge graphs significantly enhance fraud detection by leveraging graph-based anomaly detection, link prediction, and community detection.

Semi-Supervised Graph Learning:

Karim et al. (2024) proposed a semi-supervised graph learning framework that maps financial transactions into a KG structure for fraud detection [51]. Their method leverages a combination of Graph Neural Networks (GNNs) and relational embeddings to predict fraudulent nodes and edges within financial transaction graphs. This approach effectively identifies fraud patterns that evolve over time.

Transaction Graph Learning:

Huong et al. (2024) introduced a transaction-based graph learning model, which constructs a financial KG from transaction records and applies graph anomaly detection techniques to uncover hidden fraudulent connections [52]. Their model captures:

- Suspicious transaction sequences (e.g., rapid fund transfers between multiple accounts).
- High-degree fraudulent nodes with unexpected connections to shell companies.
- Money laundering typologies such as layering and integration schemes.

Graph Databases for AML (Anti-Money Laundering):

Muminovic and Halili (2024) demonstrated how graph databases are used in AML to efficiently track and score suspicious activities [53]. Their approach integrates:

- Neo4j graph queries to detect fraud rings.
- Graph traversal algorithms to trace hidden relationships in money laundering networks.
- Community detection to identify suspicious clusters of accounts.

3.2.3 Synthetic Data for Fraud Detection

Fraudulent transactions are often rare and highly imbalanced in financial datasets, making model training challenging. Synthetic transaction data generation is crucial for improving fraud detection models, as it enables the training of machine learning algorithms on simulated fraud scenarios.

Altman and Blanuša (2024) proposed a synthetic transaction generation framework that creates realistic fraudulent transactions to augment training datasets for AML models [54]. Their framework utilizes:

- Graph-based generative models to create realistic transaction structures.
- Anomaly injection techniques to embed fraudulent patterns within normal transactions.
- Adversarial training methods to improve fraud detection robustness.

3.2.4 Graph-Based Anomaly Detection Frameworks

Several fraud detection frameworks have been developed that integrate graph-based anomaly detection models. One such framework is GAMLNet.

GAMLNet (Graph-based AML Network): Schmidt et al. (2024) introduced GAMLNet, a graph-based AML system that integrates anomaly detection techniques into KG structures to identify subtle money laundering patterns [55]. The framework utilizes:

- Graph embeddings (Node2Vec, DeepWalk) to capture transaction similarities.
- Graph Convolutional Networks (GCNs) to classify entities based on transaction behaviors.
- Temporal Graph Networks to track evolving fraud patterns over time.

3.2.5 Comparison of Fraud Detection Methods Using Knowledge Graphs

Table 3.2 summarizes the key research contributions in fraud detection using knowledge graphs.

3.2.6 Future Directions in Fraud Detection Using KGs

While KG-based fraud detection has demonstrated significant improvements over traditional approaches, several challenges remain:

- **Scalability:** Large-scale KGs require efficient real-time processing for fraud detection in high-frequency trading and global transactions.
- **Explainability:** Many graph-based fraud detection models function as "black boxes." Future research should focus on interpretable models with explainable AI (XAI) techniques.
- **Adaptive Learning:** Fraud tactics evolve over time. KG-based fraud detection should integrate continual learning models to dynamically adapt to emerging fraud schemes.

TABLE 3.2 Comparison of Fraud Detection Models Using Knowledge Graphs

Study	Focus Area	Methodology	Key Findings	Limitations
Karim et al. (2024) [51]	Anomaly Detection in Transactions	Semi-supervised Graph Learning	Detects unknown fraud patterns	Requires labeled fraud data
Huong et al. (2024) [52]	Transaction-Based Graph Learning	Graph Neural Networks (GNNs)	Captures hidden fraud rings	Computationally intensive for large-scale graphs
Muminovic and Halili (2024) [53]	AML Fraud Detection	Graph Databases (Neo4j)	Efficiently tracks fraud transactions	Limited by database scalability
Altman and Blanuša (2024) [54]	Synthetic Data Generation	Graph-based Generative Models	Improves fraud detection model training	Risk of generating unrealistic transactions
Schmidt et al. (2024) [55]	Money Laundering Detection	GAMLNet (Graph-Based Anomaly Detection)	Identifies subtle AML patterns	Requires real-time graph processing

- **Regulatory Compliance:** Future KG architectures should align with global AML regulations, ensuring legal compliance while enhancing detection accuracy.

By addressing these challenges, KG-powered fraud detection will continue to evolve, enabling financial institutions and regulators to combat fraud more effectively.

3.3 Investment Analysis

Investment analysis plays a critical role in financial decision-making, where investors assess the risks and potential returns of various financial assets, businesses, and market opportunities. Knowledge Graphs (KGs) have emerged as a transformative tool in investment analysis by structuring financial relationships, integrating diverse datasets, and applying advanced machine learning techniques for risk evaluation.

Traditional investment models rely on statistical and econometric methods, often lacking the ability to capture complex interdependencies between financial entities. KGs enhance investment analysis by:

- Structuring corporate relationships, financial transactions, and risk indicators.
- Identifying hidden dependencies between investors, firms, and economic variables.
- Enhancing investment strategies through predictive analytics and explainable AI models.
- Mitigating financial risks by tracking systemic vulnerabilities in interconnected markets.

3.3.1 Knowledge Graphs for Investment Risk Assessment

Investment risks arise from multiple factors, including creditworthiness, liquidity, market fluctuations, and systemic financial interdependencies. KG-based investment risk assessment enables a multi-faceted view of financial assets by integrating diverse data sources, such as:

- Financial statements and credit reports.
- Market trends and economic indicators.
- Supply chain dependencies and geopolitical risks.
- Company ownership structures and regulatory compliance.

Credit Risk and SME Investment:

Mitra et al. (2024) proposed a KG framework for SME credit risk analysis, demonstrating its applicability in broader investment evaluations [3]. Their method integrates:

- Graph-based credit scoring models to evaluate loan eligibility.
- Relationship mapping between borrowers, suppliers, and investors.
- Temporal modeling to track creditworthiness evolution over time.

Corporate Investment Analysis:

Song et al. (2024) introduced multi-level graphs to capture complex corporate structures, improving investment decision-making [47]. Their approach integrates:

- Corporate ownership hierarchies and shareholder relationships.
- Financial health indicators and industry-wide performance metrics.
- Multi-hop reasoning to assess indirect investment risks.

3.3.2 Machine Learning for Investment Analysis in KGs

Federated Learning for Investment Risk Prediction:

Wang et al. (2024) proposed federated learning-based credit scoring that enhances risk assessment across multiple financial institutions while preserving data privacy [48]. Their model enables:

- Secure multi-party learning for risk evaluation.
- Privacy-preserving investment analytics.
- Cross-institutional credit assessments without data leakage.

Systemic Risk and Contagion Analysis:

Chen and Zhang (2024) used KG-based visualizations to track systemic financial risks, mitigating contagion effects in interconnected economies [22]. Their system:

- Maps dependencies between financial institutions and regulatory bodies.
- Simulates shock propagation in market downturn scenarios.

- Identifies key systemic risk contributors using graph centrality measures.

Bankruptcy Prediction using Graph Neural Networks:

Wei et al. (2024) applied Graph Neural Networks (GNNs) for bankruptcy prediction, improving the forecasting accuracy of corporate financial distress [49]. Their model integrates:

- Historical financial reports and balance sheet data.
- Temporal embeddings for tracking business performance trends.
- Graph-based classification models to identify at-risk firms.

Leveraging Non-Traditional Data for Investment Analysis:

Gambacorta et al. (2024) explored the use of non-traditional data sources in KGs, improving investment strategies and credit scoring [50]. Their methodology incorporates:

- Alternative data from social media, online transactions, and web activity.
- Real-time macroeconomic indicators for investment forecasting.
- Unstructured financial news and sentiment analysis in investment decision-making.

3.3.3 Comparison of KG-Based Investment Analysis Techniques

TABLE 3.3 Comparison of Investment Analysis Models Using Knowledge Graphs

Study	Focus Area	Methodology	Key Findings	Limitations
Mitra et al. (2024) [3]	SME Credit Risk	KG-Based Credit Scoring	Improves loan eligibility predictions	Requires high-quality financial data
Song et al. (2024) [47]	Corporate Investment	Multi-Level Graph Analysis	Enhances corporate investment strategies	Computationally complex for large-scale firms
Wang et al. (2024) [48]	Credit Scoring Privacy	Federated Learning	Secures multi-institutional data collaboration	High training costs due to secure computation
Chen and Zhang (2024) [22]	Systemic Risk	KG-Based Risk Visualization	Mitigates contagion effects	Needs real-time data sources for updates
Wei et al. (2024) [49]	Bankruptcy Prediction	Graph Neural Networks (GNNs)	Accurately predicts financial distress	Sensitive to missing financial records
Gambacorta et al. (2024) [50]	Alternative Data	Non-Traditional KG Data	Improves investment forecasts using alternative sources	Requires dynamic data integration techniques

3.3.4 Future Directions in Investment Analysis Using KGs

While KG-based investment analysis offers substantial improvements over traditional models, several challenges remain:

- **Data Integration:** Combining structured (e.g., financial statements) and unstructured (e.g., news sentiment) data for investment predictions.
- **Scalability:** Ensuring KGs can handle real-time market changes in high-frequency trading and portfolio management.
- **Regulatory Compliance:** Ensuring that AI-driven investment models comply with evolving financial regulations.
- **Explainability:** Developing interpretable AI models to justify investment decisions to regulators and investors.
- **Hybrid Graph Learning Models:** Integrating Temporal Graph Neural Networks (TGNNs), Large Language Models (LLMs), and Reinforcement Learning to improve long-term investment strategies.

Future research should explore LLM-enhanced financial KGs for text-based investment forecasting, integrating semantic reasoning to improve decision-making in volatile markets.

Knowledge Graphs have transformed investment analysis by providing structured representations of financial entities, enabling risk assessment, and enhancing investment decision-making. By integrating multi-source financial data, advanced graph learning models, and AI-driven analytics, financial institutions can make more informed and transparent investment choices.

3.4 Regulatory Compliance

Regulatory compliance in finance ensures that financial institutions and enterprises adhere to legal, ethical, and operational standards mandated by governments and industry regulators. With the increasing complexity of financial regulations, organizations face challenges in managing compliance efficiently. Knowledge Graphs (KGs) provide a robust framework to address these challenges by enhancing data transparency, automation, interpretability, and risk management.

Traditional compliance mechanisms often rely on manual audits, rule-based validation, and extensive documentation reviews, which are time-consuming and prone to human error. In contrast, KGs enable:

- Automated regulatory reporting by integrating structured and unstructured compliance data.
- Fraud and anomaly detection through graph-based pattern recognition.
- Enterprise risk alignment with dynamic compliance frameworks.
- Explainable AI (XAI) models to improve transparency in regulatory decisions.
- Environmental, Social, and Governance (ESG) reporting to enhance sustainability compliance.

3.4.1 Automating Regulatory Reporting with Knowledge Graphs

AI-Driven Compliance Reporting:

Padmanaban (2024) demonstrated how integrating AI and ML with KGs streamlines regulatory reporting, reducing manual interventions and improving accuracy [5]. Their approach leverages:

- Natural Language Processing (NLP) to extract compliance obligations from legal documents.
- Graph-based reasoning to map compliance rules across financial entities.
- Machine learning models to automate compliance verification and risk detection.

Electronic Invoice Validation for Compliance:

Mäkelburg et al. (2024) introduced a KG-based framework for automating electronic invoice validation, ensuring adherence to regulatory policies [56]. Their methodology includes:

- Extracting invoice metadata and structuring it in graph representations.
- Linking invoice attributes to legal compliance rules.
- Identifying discrepancies and flagging non-compliant transactions automatically.

Enterprise Risk Management and Compliance:

Li et al. (2024) demonstrated how KGs align enterprise risk management with compliance standards, ensuring regulatory adherence in financial institutions [2]. Their system:

- Maps regulatory policies to financial transactions in a dynamic graph structure.
- Provides real-time risk assessment by identifying compliance gaps.
- Automates compliance audits, reducing operational costs.

3.4.2 Fraud Detection in Regulatory Compliance

Graph-Based Financial Fraud Detection:

Fraud detection is a critical aspect of regulatory compliance, as financial crimes such as money laundering and fraudulent reporting pose major risks to financial institutions. KGs enhance fraud detection by structuring transactional data into relational networks, allowing anomaly detection through graph-based analysis.

Cai and Xie (2024) developed a two-layer KG model for fraud detection, uncovering fraudulent patterns in financial statements [57]. Their model:

- Constructs financial statement relationships in a hierarchical KG.
- Identifies irregular transaction flows using anomaly detection algorithms.
- Enhances audit efficiency by flagging suspicious financial activities.

Explainable AI (XAI) for Financial Compliance:

Shaikh et al. (2024) explored explainable AI (XAI) with KGs to enhance regulatory transparency [16]. Their approach:

- Uses graph embeddings to improve interpretability in fraud detection.
- Provides traceable explanations for compliance decisions.
- Enhances auditor confidence in automated compliance models.

3.4.3 ESG (Environmental, Social, and Governance) Compliance with Knowledge Graphs

As financial institutions increasingly focus on sustainable investments and corporate social responsibility, compliance with ESG standards has become a priority. KGs provide a structured approach to tracking ESG compliance by integrating:

- Corporate sustainability reports and environmental impact assessments.
- Social responsibility metrics, such as diversity and labor practices.
- Governance structures and ethical business practices.

Angioni et al. (2024) developed an ESG Knowledge Graph (ESG-KG) to improve corporate sustainability reporting and compliance verification [58]. Their model:

- Links ESG-related regulatory policies to financial institutions.
- Tracks company compliance with global ESG standards.
- Generates automated ESG compliance reports for regulatory agencies.

TABLE 3.4 Comparison of KG-Based Regulatory Compliance Solutions

Study	Focus Area	Methodology	Key Findings	Limitations
Padmanaban (2024) [5]	Regulatory Reporting	AI + KG-Based Compliance Framework	Automates compliance reporting	Requires integration with legacy systems
Mäkelburg et al. (2024) [56]	Invoice Validation	Graph-Based Financial Document Processing	Ensures compliance in electronic invoicing	Limited to structured invoice data
Li et al. (2024) [2]	Enterprise Risk	Knowledge Graph Alignment for Compliance	Enhances risk assessment	High computational cost for large graphs
Cai and Xie (2024) [57]	Fraud Detection	Two-Layer KG for Fraud Discovery	Identifies fraudulent patterns	Requires labeled financial fraud data
Shaikh et al. (2024) [16]	Explainability	XAI-Enhanced KG for Financial Compliance	Improves regulatory transparency	Needs domain-specific fine-tuning
Angioni et al. (2024) [58]	ESG Compliance	ESG Knowledge Graph for Sustainable Finance	Improves corporate ESG reporting	Limited availability of ESG-related datasets

3.4.4 Future Directions for KG-Based Regulatory Compliance

While Knowledge Graphs have significantly enhanced regulatory compliance, several challenges remain:

- **Data Standardization:** Ensuring uniformity in compliance data across financial institutions.
- **Real-Time Regulatory Updates:** Automating the adaptation of compliance models to evolving financial regulations.
- **Scalability Challenges:** Developing distributed KG architectures to handle compliance data at scale.
- **Trustworthy AI for Compliance:** Enhancing explainability in AI-driven regulatory compliance frameworks.
- **Cross-Border Compliance:** Addressing multi-jurisdictional financial regulations using global KG standards.

Future research should focus on developing hybrid AI-KG systems that integrate Large Language Models (LLMs) for legal text interpretation, improving compliance automation in finance.

Knowledge Graphs play a pivotal role in regulatory compliance by automating reporting, enhancing fraud detection, and aligning enterprise risk management with legal standards. Their ability to integrate structured and unstructured financial data, enable explainable AI, and support ESG compliance makes them indispensable in modern finance. Future advancements in scalability, real-time adaptation, and trustworthy AI will further enhance the effectiveness of KGs in regulatory frameworks.

3.5 Market Trend Analysis

Market trend analysis is a crucial aspect of financial decision-making, enabling investors, financial institutions, and policymakers to anticipate stock market movements, economic shifts, and systemic financial risks. Knowledge Graphs (KGs) play a pivotal role in market trend analysis by structuring complex financial relationships, integrating multi-source data, and enabling predictive modeling.

Traditional market analysis relies on time-series models and machine learning techniques, but these approaches often lack contextual financial insights. KGs enhance market trend analysis by:

- Identifying financial entity relationships (e.g., stock correlations, sector trends).
- Enhancing financial predictions using graph-based deep learning models.
- Integrating multimodal data sources, including stock prices, economic indicators, news sentiment, and policy changes.
- Improving systemic risk assessment through graph-based causal inference.
- Leveraging self-supervised learning to refine predictions from unstructured financial data.

3.5.1 Graph-Based Stock Market Predictions

Graph Neural Networks (GNNs) for Stock Trend Forecasting:

Shi et al. (2024) introduced a KG framework combining Graph Convolutional Networks (GCNs) and Long Short-Term Memory (LSTM) models to predict stock prices [59]. Their method:

- Constructs a financial KG linking stock performance with macroeconomic indicators.
- Applies GCNs to extract structural dependencies between financial entities.
- Utilizes LSTMs to capture temporal dependencies and refine stock price predictions.
- Achieves superior accuracy in stock movement forecasting compared to traditional time-series models.

Time-Series and Machine Learning Models for Nonlinear Market Trends:

Zheng et al. (2024) employed machine learning models and time-series analysis to study nonlinear financial behaviors, improving stock price forecasting [60]. Their approach includes:

- Feature extraction from historical stock market data.
- Integration of KG embeddings with deep learning models.
- Predicting market volatility using hybrid graph-based learning techniques.

3.5.2 Causal Inference and Market Risk Analysis with KGs

Causal Knowledge Graphs for Financial Events:

Financial market trends are influenced by complex relationships between economic policies, corporate earnings, and investor sentiment. KGs improve transparency by applying causal inference techniques to link financial events with stock trends.

Deng et al. (2024) introduced a causal inference framework leveraging KGs to assess financial event impacts on stock markets [61]. Their methodology:

- Constructs causal graphs representing financial events (e.g., earnings reports, interest rate changes).
- Applies counterfactual reasoning to estimate market impact.
- Enhances financial forecasting accuracy by distinguishing causal relationships from spurious correlations.

Systemic Risk Assessment Using Knowledge Graphs:

Chen and Zhang (2024) utilized KGs to assess systemic risk in banking, linking interdependencies between financial institutions [46]. Their model:

- Constructs a financial institution network to model contagion risk.
- Identifies vulnerabilities within interconnected financial entities.
- Simulates crisis propagation using graph-based Monte Carlo risk models.

3.5.3 Self-Supervised Learning for Market Trend Analysis

Self-Supervised Graph Learning for Financial Data:

Traditional supervised models require extensive labeled datasets, which are often unavailable in financial markets. Self-supervised learning (SSL) techniques leverage unstructured financial data for improved market predictions.

Ying et al. (2024) proposed self-supervised KG embeddings to refine stock market forecasts [62]. Their framework:

- Uses contrastive learning to generate robust stock trend representations.
- Captures hidden patterns in unlabeled financial data.
- Enhances prediction models for rare financial events.

Hypergraph Learning for Market Volatility Prediction:

Su et al. (2024) applied hypergraph-based learning models to analyze stock market fluctuations, integrating multiple financial relationships within KGs [63]. Their model:

- Builds hypergraph structures capturing multi-stock dependencies.
- Utilizes GNNs for multi-hop financial reasoning.
- Predicts sudden market shifts with high accuracy.

3.5.4 Multimodal Market Trend Analysis Using Knowledge Graphs

Modal-Adaptive Knowledge Graphs for Sentiment and Policy Analysis:

Financial markets are influenced by multiple external factors, including government policies, global economic shifts, and investor sentiment. Multimodal KGs enhance market trend analysis by integrating structured and unstructured data.

Ouyang et al. (2024) introduced a modal-adaptive KG model, combining financial reports, policy changes, and sentiment analysis for stock trend forecasting [10]. Their approach:

- Integrates real-time market news, policy changes, and social media sentiment into financial KGs.
- Applies attention-based GNNs to filter market-relevant information.
- Enhances market trend predictions using adaptive embeddings.

3.5.5 Future Directions for KG-Based Market Trend Analysis

Despite advancements in KG-based market trend analysis, several challenges remain:

- Integrating real-time financial data with adaptive KG models.
- Scaling hypergraph learning techniques for multi-sector trend analysis.
- Enhancing explainability in AI-driven market forecasts.

TABLE 3.5 Comparison of KG-Based Market Trend Analysis Approaches

Study	Focus Area	Methodology	Key Findings	Limitations
Shi et al. (2024) [59]	Stock Price Prediction	KG + GCN-LSTM Hybrid Model	Improves financial forecasting accuracy	Requires historical labeled data
Zheng et al. (2024) [60]	Market Trend Analysis	Time-Series + Machine Learning	Captures nonlinear stock behaviors	Sensitive to noise in financial data
Deng et al. (2024) [61]	Financial Causal Analysis	KG + Counterfactual Reasoning	Enhances market event impact estimation	Requires high-quality causal datasets
Chen and Zhang (2024) [46]	Systemic Risk Modeling	KG-Based Risk Propagation	Identifies financial contagion risks	Limited scalability for large financial networks
Ying et al. (2024) [62]	Self-Supervised Financial Learning	Contrastive Learning + KG Embeddings	Improves prediction accuracy with unlabeled data	Requires extensive graph training
Su et al. (2024) [63]	Market Volatility Prediction	Hypergraph GNN for Financial Graphs	Captures complex stock dependencies	Computationally intensive for large-scale models
Ouyang et al. (2024) [10]	Multimodal Market Analysis	Adaptive KG for Sentiment	Integrates policy and sentiment shifts	Requires real-time financial text processing

- Improving self-supervised models for rare financial event predictions.
- Developing cross-border financial KGs for global trend analysis.

Knowledge Graphs provide a powerful foundation for market trend analysis, integrating financial relationships, causal inference, and self-supervised learning to enhance stock market forecasting, systemic risk assessment, and multimodal financial analytics.

3.6 Open Challenges and Future Directions

Financial Knowledge Graphs (KGs) have demonstrated immense potential in addressing complex financial tasks such as credit risk assessment, fraud detection, systemic risk management, and regulatory compliance. Despite these advancements, several challenges impede their full-scale adoption. Key challenges include:

- **Data Privacy and Compliance:** Ensuring security and regulatory alignment while handling sensitive financial data.

- **Scalability and Real-Time Processing:** Managing dynamic, large-scale financial datasets efficiently.
- **Interpretability and Explainability:** Enhancing transparency for regulatory compliance and decision-making.
- **Quality and Completeness of Financial KGs:** Addressing data sparsity and ensuring high-quality, representative datasets.
- **Integration of Emerging Technologies:** Leveraging generative AI, explainable AI (XAI), and temporal knowledge graphs (TKGs) to enhance financial KG capabilities.

3.6.1 Data Privacy and Compliance Challenges

Financial KGs handle vast amounts of sensitive and proprietary financial data, necessitating strict adherence to regulatory frameworks such as General Data Protection Regulation (GDPR), California Consumer Privacy Act (CCPA), and Basel III regulations.

Privacy-Preserving Financial KGs:

Mohsin et al. (2024) and Mitra et al. (2024) emphasized the need for automated frameworks that secure financial KGs while ensuring usability for applications such as credit risk assessment and systemic risk analysis [3,8]. Proposed solutions include:

- **Federated Learning:** Enables privacy-preserving KG models by training on decentralized financial data [48].
- **Differential Privacy Techniques:** Implements noise injection mechanisms to safeguard financial transactions [14].
- **Graph Encryption Techniques:** Uses homomorphic encryption to securely compute financial graph analytics without exposing raw data.

3.6.2 Scalability and Real-Time Processing

Financial KGs must process high-volume, high-velocity financial data streams for tasks such as high-frequency trading, real-time fraud detection, and systemic market risk analysis. Managing this scale and dynamism poses significant computational challenges.

Key Challenges in Scalable KGs:

- Handling real-time streaming data from stock exchanges, banking transactions, and regulatory filings.
- Ensuring low-latency inference in high-frequency financial decision-making.
- Managing evolving financial relationships dynamically over time.

Proposed Solutions for Scalability:

Farghaly et al. (2024) introduced dynamic KG management frameworks to efficiently handle evolving financial relationships [18]. Sarmah et al. (2024) proposed retrieval-augmented models

to optimize knowledge retrieval from large-scale financial graphs [7]. Advancements in scalable financial KG architectures include:

- Distributed Graph Databases (e.g., Neo4j, TigerGraph) for parallelized financial data processing.
- Graph Streaming Frameworks (e.g., Apache Kafka, Flink) for real-time updates in financial KGs.
- Graph Compression Algorithms for reducing computational overhead while preserving graph structure.

3.6.3 Interpretability and Explainability in Financial KGs

As financial KGs influence risk assessments, fraud detection, and regulatory compliance, they must be interpretable and explainable. The lack of transparent AI decision-making remains a significant barrier to adoption in financial institutions.

Explainable Knowledge Graphs (X-KGs):

Shaikh et al. (2024) proposed explicable KGs (X-KGs), which introduce interpretability layers into financial graph models [16]. Xu et al. (2024) and Cai and Xie (2024) further developed causal and layered models to enhance explainability in fraud detection and financial risk analysis [4, 9]. Proposed explainability techniques include:

- Natural Language Queryable KGs: Enabling finance professionals to interact with KGs using intuitive queries.
- Graph Attention Mechanisms (GATs): Highlighting key financial relationships influencing AI decisions.
- Regulatory-Compliant AI Audits: Ensuring KGs align with explainability standards set by financial regulators.

3.6.4 Enhancing Financial KGs with Emerging Technologies

1. Generative AI for Financial Knowledge Graphs

Generative AI models can synthesize realistic financial data to fill gaps in KGs, improve predictive accuracy, and augment sparse financial datasets. Zhao et al. (2024) and Kanaparthi et al. (2024) explored AI-driven data synthesis to enhance financial risk modeling [17, 64]. Applications of generative AI in financial KGs include:

- Simulating realistic financial transactions to train fraud detection models.
- Enhancing market trend forecasting with AI-generated macroeconomic scenarios.
- Synthetic corporate credit histories for evaluating SME creditworthiness.

2. Temporal Knowledge Graphs (TKGs) for Dynamic Financial Modeling

Financial markets are highly dynamic, requiring temporal models to track evolving relationships. Farghaly et al. (2024) and Liang et al. (2024) emphasized the need for Temporal Knowledge Graphs (TKGs) to improve time-dependent financial risk assessment [11, 18]. Advancements in TKGs include:

- Event-Based Temporal Graphs: Linking financial events (e.g., policy changes, earnings reports) to stock market reactions.
- Predictive Time-Series Graphs: Forecasting systemic risk through multi-time-step graph reasoning.
- Adaptive Temporal Embeddings: Enhancing financial relationship tracking over time.

3. Explainable AI (XAI) for Regulatory Alignment

Regulatory compliance mandates transparency in AI-driven financial models. Shaikh et al. (2024) and Weber et al. (2024) advocated for explainable AI (XAI) techniques to align financial KGs with regulatory standards [16, 65]. XAI applications in financial KGs include:

- Causal Graph Models: Explaining financial predictions using cause-effect relationships.
- Counterfactual Reasoning: Generating "what-if" financial scenarios for risk analysis.
- Regulatory AI Audits: Ensuring financial KG decisions comply with evolving financial regulations.

3.6.5 Future Research Directions for Financial KGs

Future advancements in financial knowledge graphs should focus on:

- Developing privacy-enhanced KGs that balance data security with analytical usability.
- Scaling financial KGs for real-time market analytics in high-frequency trading environments.
- Integrating multimodal financial data sources, including satellite imagery, alternative credit scores, and news sentiment.
- Enhancing explainability with self-explaining graph models for AI-driven decision-making.
- Building cross-border financial KGs for international risk assessment and regulatory alignment.

While financial KGs offer transformative capabilities for risk modeling, fraud detection, and market analysis, overcoming privacy, scalability, and interpretability challenges is crucial. By integrating generative AI, temporal modeling, and explainable AI, financial KGs can evolve into next-generation decision-making tools for the global financial industry.

3.7 Conclusion

This survey highlights the transformative potential of knowledge graphs (KGs) in the financial sector, underscoring their utility in areas such as risk assessment, trend analysis, regulatory compli-

ance, and investment decision-making. As financial data grows in volume, variety, and velocity, KGs offer a structured approach to model complex relationships and derive actionable insights. The integration of advanced techniques like large language models (LLMs), generative AI, temporal modeling, and explainable AI (XAI) has driven significant advancements, enabling KGs to handle dynamic, multifaceted financial data with increased accuracy and interpretability.

While substantial progress has been made, challenges such as data privacy, scalability, real-time processing, and the need for interpretable models remain areas of active research. The demand for models that can manage high volumes of real-time data while preserving data security is critical in financial applications, where compliance and risk mitigation are paramount. Emerging trends, including the incorporation of generative AI for data enrichment, the development of temporal knowledge graphs to capture time-dependent insights, and the integration of XAI techniques to improve transparency, point to a future where KGs play a central role in intelligent financial systems. $\text{\LaTeX}$.

CHAPTER 4

A Temporal Relational Graph Convolutional Network Approach to Financial Performance Prediction

4.1 Introduction

In the finance industry, textual data plays a vital role in various financial business use cases, including financial reporting, market analysis, and regulatory compliance. Financial data, such as stock market reports, news articles, earnings reports, and social media posts, often contain valuable insights that can drive business decisions. However, this textual data is typically unstructured, making it challenging to extract actionable insights and represent financial knowledge in a usable format.

The representation of financial knowledge within this textual data is often unstructured, limiting its value and potential for analysis. While many financial institutions rely on traditional methods of analysis, these methods fail to capture the complex and dynamic relationships inherent in financial data. This is where financial knowledge graphs (KGs) come into play. By providing a structured way to represent and analyze financial data, KGs have gained significant attention in the finance industry. Their structured nature enables the modeling of complex relationships between financial entities, allowing for enhanced decision-making and deeper insights. This structured representation enhances the applicability of KGs across various domains, including risk management [66], fraud detection [67], investment analysis [68], and financial decision-making.

In recent years, temporal knowledge graph (TKG) representation learning has emerged as a promising direction for capturing and modeling evolving financial data. Temporal KGs are particularly useful in scenarios where time-dependent relationships between entities, such as stock prices, market trends, and economic indicators, need to be considered. Recent advancements in TKG representation learning have shown significant promise for tasks that involve reasoning and forecasting on evolving knowledge, making it a powerful tool for financial forecasting, risk man-

agement, and decision-making.

Our work builds upon several key contributions in this area. Cai et al. (2024) [69] propose a taxonomy for TKG representation learning methods, which highlights various approaches for capturing temporal dynamics in knowledge graphs. These methods are essential for understanding how financial entities change over time, allowing for more accurate forecasting and analysis. Similarly, He et al. (2024) [70] introduce the concept of a "Hip network", which leverages historical information for reasoning on TKGs. This approach allows for the integration of past financial events and patterns into future predictions.

For temporal knowledge graph completion, Wang et al. (2024) [71] present MADE, a method that uses multicurvature adaptive embeddings to improve the accuracy of TKGs by better capturing the temporal relationships between financial entities. Meanwhile, Dong et al. (2024) [72] explore temporal inductive path neural networks, which offer another promising direction for improving TKG completion by learning from time-dependent financial paths.

Liao et al. (2024) [41] investigate generative forecasting on TKGs with large language models (LLMs). This approach has shown great potential in the context of financial data, where uncertainty and complex relationships often make it difficult to predict future events. Similarly, Luo et al. (2024) [73] explore the use of large language models for learning and forecasting with TKG completion, enabling more accurate predictions based on real-time financial trends.

Ding et al. (2024) [74] introduce zrLLM for zero-shot relational learning, which can be particularly useful in financial performance prediction, where new, unseen data may arise frequently. This method, when combined with TKG representation learning, enables the creation of highly flexible financial models that can adapt to new data without retraining the entire system. Additionally, Wang et al. (2024) [39] propose IME, a method integrating multi-curvature embeddings for TKG completion. This approach provides another technique for improving TKG representation, especially when dealing with large-scale financial datasets. Finally, Gastinger et al. (2024) [75] establish a baseline for temporal knowledge graph forecasting, which is essential for long-term financial performance prediction.

One of the primary challenges in real-world applications of financial KGs is that knowledge graphs often face issues related to incomplete and inaccurate data. Cheng et al. (2020) [76] address this issue by focusing on structured news events extracted from raw text corpora. They construct a large-scale knowledge graph called FinKG, which covers over 3,000 companies listed in the Shanghai and Shenzhen stock exchanges. Despite the scale of this dataset, their work highlights the difficulties in creating a complete and accurate knowledge graph. This is a common issue in financial knowledge graph construction, as the data is constantly changing and must be continuously updated to remain relevant and accurate.

Our approach builds on these ideas by using scenario-based extractors to accurately capture complex relationships between entities in financial textual data. By focusing on specific financial scenarios such as financial growth, market development, and acquisitions, we ensure that our

knowledge graphs are both complete and accurate. This approach improves the accuracy and reliability of financial knowledge representation, addressing some of the limitations identified in previous work.

The second challenge in building robust financial knowledge graphs (KGs) is the lack of a systematic and repeatable approach to graph construction. While there have been significant efforts to develop methods for constructing financial KGs, many of these methods suffer from inconsistencies in their construction processes, limiting their reproducibility and scalability.

Zehra et al. (2021) [77] introduced a financial knowledge graph-based financial report query system that utilizes Wikidata and DBpedia to build financial KGs. While their system provides valuable insights into financial report querying, it falls short in terms of a repeatable construction process. The construction methodology relies heavily on external databases like Wikidata and DBpedia, which may not always contain domain-specific financial data required for accurate financial analysis. Furthermore, this approach lacks a standardized framework that can be applied across different financial datasets, leading to inconsistencies when scaling the system to handle large financial networks.

Similarly, Loster et al. (2018) [78] highlight several challenges in entity resolution, maintenance, and exploration of financial knowledge graphs. They point out that while financial entities such as companies, products, and financial instruments are fundamental components of financial KGs, accurately resolving and linking entities across different data sources remains a significant hurdle. The lack of entity resolution methods that can consistently handle complex and evolving financial relationships makes it difficult to maintain and explore financial KGs over time. These challenges are further compounded by the dynamic nature of financial data, where entities and relationships are constantly changing, and new financial products and markets emerge.

Cheng et al. (2022) [79] propose a multi-modality graph neural network (MAGNN) to handle financial data by incorporating multiple data modalities such as numerical data, text data, and visual data. While their approach shows promise in enhancing the depth of analysis by integrating different data types, they do not provide sufficient details on the construction process. Specifically, there is limited information on how to systematically capture and represent complex financial relationships between entities, such as market movements, credit ratings, and corporate actions. Moreover, the framework does not address how to ensure data consistency and scalability when constructing financial knowledge graphs from diverse sources.

To address these challenges, our approach proposes a systematic and repeatable method for constructing financial knowledge graphs using scenario-based knowledge extractors. These extractors focus on specific financial scenarios, such as corporate growth, market development, mergers and acquisitions, and regulatory changes, to ensure accurate capture of complex relationships between financial entities. This methodology is designed to be reproducible and scalable, making it suitable for handling large-scale financial datasets while maintaining consistency across different data sources.

Specifically, our approach incorporates the following scenario-based extractors:

- **Financial Growth Extractor:** Focuses on capturing key financial metrics such as revenue growth, profitability, and market expansion.
- **Market Development Extractor:** Identifies and extracts information related to market trends, changes in financial regulations, and industry shifts.
- **Contract Extractor:** Extracts relationship data from corporate contracts, including mergers, acquisitions, and partnerships.
- **Acquisition Extractor:** Focuses on capturing corporate actions related to acquisitions, including valuations, synergies, and market impact.

By using these specialized extractors, our framework ensures that complex financial relationships are captured with precision, resulting in comprehensive and accurate knowledge graphs that can be used for various applications such as financial risk assessment, market trend prediction, and investment analysis. This systematic approach not only improves the accuracy of financial knowledge graphs but also enhances their scalability, enabling them to accommodate the dynamic nature of financial data while ensuring consistent construction.

By introducing a standardized, repeatable, and systematic method for constructing financial knowledge graphs, our approach addresses critical gaps in current methodologies, enabling more reliable and scalable knowledge representation for the financial industry.

A scenario-based knowledge extractor is an important component in our framework for constructing financial knowledge graphs (KGs). These extractors are designed to focus on specific financial scenarios and extract relevant information from diverse financial datasets. Each module within the extractor targets a particular aspect of the financial world, ensuring that key entities and relationships are accurately identified, represented, and integrated into the knowledge graph. This modular approach helps maintain a high degree of precision and comprehensiveness in financial knowledge representation.

The scenario-based knowledge extractor includes the following key modules:

- **Financial Growth Extractor:** This module focuses on capturing key financial metrics that reflect the growth trajectory of a company or financial entity. It extracts data related to revenue growth, profitability, market expansion, and investment activities. By analyzing these indicators, the extractor creates relationships between entities such as companies, investors, and market trends, providing insights into their financial health and performance.
- **Market Development Extractor:** This module targets the identification of market trends and the evolution of financial sectors. It captures data on emerging markets, shifts in industry dynamics, regulatory changes, and technological advancements that impact financial markets. The extractor integrates information about how these factors influence the relationships between companies, investors, and financial instruments, thus providing a more holistic view of the market.

- **Contract Extractor:** Financial transactions and corporate actions are often governed by contracts, agreements, and legal frameworks. The contract extractor focuses on extracting entities and relationships embedded within corporate contracts, including mergers, acquisitions, joint ventures, and other partnerships. By identifying these relationships, the extractor enriches the knowledge graph with the interactions between companies and their stakeholders.
- **Acquisition Extractor:** Acquisitions and corporate restructurings are critical events in the financial world. This module extracts key information regarding mergers, acquisitions, takeovers, and divestitures. It identifies the parties involved, the terms of the deal, the financial impact, and the subsequent changes in company structures. This extraction helps in mapping the evolving landscape of financial entities, making the knowledge graph more dynamic and informative.

These scenario-based modules ensure that complex financial relationships are captured in a structured and systematic manner. By focusing on specific financial events and processes, the extractors provide accurate and comprehensive data, which can be used for various applications such as investment analysis, risk management, and financial forecasting.

The third major challenge in financial knowledge graph construction is incorporating consistent domain-specific knowledge within the graph. While knowledge graphs constructed from general sources like Wikipedia have been explored in various domains such as healthcare and technology [80–82], they often lack the precision and accuracy needed for financial applications. These general-purpose graphs do not account for the intricacies of the financial domain and may not include specialized financial entities or relationships, leading to incomplete or inconsistent knowledge representation.

Domain-specific knowledge graphs have also been developed in sectors such as healthcare [83, 84], where domain expertise and structured methodologies are critical for accurate knowledge extraction. However, these approaches still face limitations in providing structured methods for extracting and integrating domain-specific knowledge into a consistent framework.

For financial applications, domain-specific taxonomies are essential for accurate and consistent knowledge extraction. While previous methods, such as K-BERT [85], have attempted to inject domain knowledge into language models like BERT, they do not leverage domain-specific taxonomies for structured extraction. K-BERT focuses on enhancing the model’s understanding of financial terms but does not provide a systematic approach to ensuring consistency in the extraction process, especially when dealing with large-scale financial data.

Our approach integrates domain knowledge through taxonomies and scenario-specific extraction methods. By using a taxonomy-driven approach, we ensure that financial knowledge is extracted in a consistent manner, guided by predefined categories and relationships specific to the financial domain. The taxonomies act as a backbone for organizing financial data into hierarchical structures, ensuring that all entities and their relationships are captured accurately and consistently.

The domain-specific taxonomies we use are designed to cover a wide range of financial entities,

including companies, financial instruments, regulatory bodies, market trends, and economic indicators. These taxonomies are continuously updated to reflect the dynamic nature of the financial world, ensuring that the knowledge graph remains relevant and up-to-date.

The combination of scenario-based knowledge extractors and domain-specific taxonomies enables us to construct highly accurate and consistent financial knowledge graphs. This approach ensures that financial relationships are captured in a comprehensive and structured manner, providing a solid foundation for various applications such as financial risk assessment, investment analysis, and market forecasting.

4.2 Background and Related Work

In the field of financial knowledge graphs, several studies have laid the groundwork for understanding and improving the representation of financial data. Wang et al. [86] introduced the Financial Research Report Knowledge Graph (FR2KG), a dataset designed to represent domain-specific triples in the financial domain. This dataset focuses on financial documents in Chinese, where the relationship extraction is primarily based on predefined domain-specific concepts. However, our work presents a generalized systematic approach that can be applied across any domain with a well-defined taxonomy, emphasizing a formal, standardized, and semantically enriched representation using domain-specific taxonomies. Unlike Wang et al.’s approach, our methodology ensures that the representations are both consistent and scalable, thus enabling wider applicability across different industries and financial markets.

Additionally, Wu et al. [87] proposed a specialized audit information knowledge graph for a particular financial market. While their work advanced the concept of using graphs for domain-specific financial data, the market-specific nature of their work limits the generalizability of their approach. In contrast, our framework uses a domain-specific taxonomy as the backbone, offering a formal and consistent representation that ensures applicability across different markets and financial sectors. This allows our approach to be much more flexible and generalizable than previous methods.

In the domain of temporal financial data representation, various approaches have been explored for representing time-dependent financial events in graphs. Messner et al. [88] introduced BoxTE, a method for temporal knowledge graph completion (TKGC) that utilizes box embeddings to model temporal relationships. While this work contributes to the growing body of research on temporal graph representation, it does not fully address the complexity of representing financial events over time, especially in the dynamic nature of financial markets.

Xu et al. [89] proposed TEA-GNN, a time-aware attention mechanism for temporal knowledge graphs. While their approach incorporates temporal aspects, it primarily focuses on the representation of static knowledge over time rather than addressing the temporal patterns and interactions between financial entities in the graph. Our approach, on the other hand, integrates Relational

Graph Convolutional Networks (RGCNs) [37] with Long Short-Term Memory (LSTM) networks, enabling the capture of temporal dynamics and patterns in financial data. This fusion of RGCNs and LSTM networks provides a more nuanced representation of temporal financial data, addressing the limitations of previous methods.

The application of graph learning methods in healthcare has also been explored, particularly in the analysis of brain data for disease detection [90]. Bi et al. [90] developed a structure-adaptive graph neural network (TR-SAGNN) with temporal representation capabilities, incorporating a temporal attention learning module. This method overcomes certain limitations of existing approaches by dynamically adapting the graph’s structure to evolving temporal data. Similarly, our temporal RGCN model captures temporal dynamics in financial data, allowing us to apply similar principles to the financial domain for risk prediction and trend analysis.

For the task of financial performance prediction, several studies have proposed models that leverage machine learning and graph-based techniques. Tao et al. [38] utilized ConvLSTM networks for stock price feature extraction, integrating mutation points into the financial graphs for improved performance prediction. However, these approaches do not consider the influence of real-time trends driven by social information, such as news articles or tweets, which can significantly impact financial entity performance. Our approach addresses this gap by incorporating tweets as social embeddings to capture real-time trends, which has been shown to improve the prediction accuracy by integrating temporal features derived from social embeddings.

The VGC-GAN model [91], an advanced adversarial framework that utilizes multi-graph convolution for stock price forecasting, creates dynamic correlation graphs and integrates a GAN architecture with Multi-GCN and GRU for improved stock price predictions. While this method significantly improves prediction accuracy by analyzing inter-stock relationships and temporal dependencies, it focuses primarily on a static market representation. In contrast, our model integrates temporal dynamics with real-time trend analysis, which leads to more accurate and actionable insights in dynamic financial markets.

Previous research, such as Elhammadi et al. (2020) [92], has made substantial contributions to financial knowledge graph construction and the discovery of semantic relationships within the financial domain. However, one of the critical gaps in this body of work is the lack of temporal sequence representation in financial knowledge graphs. The absence of a temporal dimension limits the relevance and predictive power of financial KGs, particularly when analyzing rapidly changing financial environments. Our work fills this gap by incorporating temporal sequence modeling, ensuring that the constructed graphs represent up-to-date financial events and market shifts.

Zakhidov (2024) underscores the importance of economic indicators, such as GDP, inflation, and employment, in predicting market trends. Zakhidov advocates for combining quantitative and qualitative data sources for a more robust financial analysis [93]. Similarly, Olorunsola et al. (2024) focus on the relationship between governance scores and financial metrics in the tourism industry, offering valuable insights into financial-environmental dynamics [94]. Our work builds on these

ideas by integrating a broader set of financial metrics and social data to enhance the predictive power of financial knowledge graphs.

Munir et al. (2024) highlight the influence of dividend-related metrics and financial indicators on stock price dynamics in Pakistan’s non-financial sectors [95]. Their work emphasizes the importance of localized financial data in predicting stock behavior. Our approach extends this by incorporating global financial trends, integrating social media data, and leveraging temporal knowledge graph frameworks to forecast financial performance across regions and markets.

We reviewed the background and related work in the field of financial knowledge graphs and temporal financial data representation. While significant progress has been made, including contributions from various research efforts in temporal knowledge graph completion and real-time financial performance prediction, gaps remain in incorporating real-time trends, accurate temporal representation, and domain-specific taxonomies. This thesis addresses these challenges by proposing an integrated approach that combines temporal relational graph convolutional networks, social embeddings, and domain-specific taxonomies to enhance financial performance prediction and systemic risk analysis.

4.3 Our Approach

We present an approach that integrates Relational Graph Convolutional Networks (RGCNs) [37] with Long Short-Term Memory (LSTM) [96] models to enhance financial performance prediction. By incorporating a comprehensive financial knowledge graph into the RGCNs [37] and training it across different time steps, we are able to aggregate temporal information through LSTMs [96], enabling more accurate predictions of financial entity performance.

The FintechKG knowledge graph extraction pipeline is shown in Figure 9.1. It comprises a reusable methodology that can be applied to generate knowledge graphs from other textual data, by identifying the taxonomy and conceptual entities specific to the domain of interest.

The proposed approach, FintechKG, addresses various challenges in financial knowledge graph construction and performance prediction by employing a systematic and comprehensive methodology, outlined as follows:

- **Enriched Domain Knowledge:** FintechKG captures the taxonomy of financial concepts from multiple types of financial statements, including Income Statements, Balance Sheets, and Cash Flow Statements. By doing so, it provides an enriched understanding of financial concepts and the relationships between different financial entities. This structured knowledge significantly improves the prediction of financial performance and mitigates the limitations of unstructured data representations.
- **Effective Integration of Information:** Our approach effectively integrates textual embeddings (using FinBERT [97]) with graph-based embeddings (using RGCN [37]) to leverage both textual knowledge and relational information encoded within the knowledge graph. This

integration helps in creating a shared latent space, allowing the model to benefit from both types of knowledge. By doing so, we enhance the accuracy of financial entity performance predictions, combining the strengths of both textual and relational embeddings.

- **Temporal Pattern Learning:** The RGCN [37] embeddings extracted from different time steps are used as input to a Long Short-Term Memory (LSTM) [96] model. The LSTM captures temporal dependencies and trends in financial data, which are crucial for forecasting financial outcomes. Our focus in this work is to predict whether a financial organization’s revenue will increase or decrease over time. This methodology can easily be extended to predict other key financial indicators, such as profit margins, stock prices, and credit risk.

Overall, FintechKG offers a scalable and adaptable approach to constructing financial knowledge graphs. It addresses the challenges of accurately representing the complex relationships within financial data through RGCNs [37]. Additionally, its temporal-awareness, enriched domain knowledge, and effective integration of textual and graph-based information make it a valuable framework for financial decision-making and analysis. By combining temporal patterns and relational knowledge, FintechKG improves the reliability and precision of financial performance predictions, offering significant benefits for various financial applications, including risk management, investment analysis, and fraud detection.

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CHAPTER 5

FintechKG Methodology

In this section, we present the framework employed for constructing the FintechKG (Fintech Knowledge Graph) to perform Financial Entity prediction. Figure 9.1 illustrates the steps involved in the FintechKG Knowledge Graph Extraction and Prediction Pipeline. This framework combines:

- **RGCN-based Representation:** Representing FintechKG data at 3 timestamps using Relational Graph Convolutional Networks (RGCNs) [37] to capture relational information.
- **FinBERT [97] Embeddings:** Integrating textual information via pre-trained FinBERT [97], capturing semantic insights from financial news.
- **Temporal Reasoning with LSTM:** Employing Long Short-Term Memory (LSTM) [96] networks to model temporal dependencies within the RGCN representations.
- **Projection Layer:** Introducing a projection layer for both RGCN [37] and FinBERT [97] embeddings, enabling effective combination (e.g., linear or non-linear) into a unified latent space for prediction.

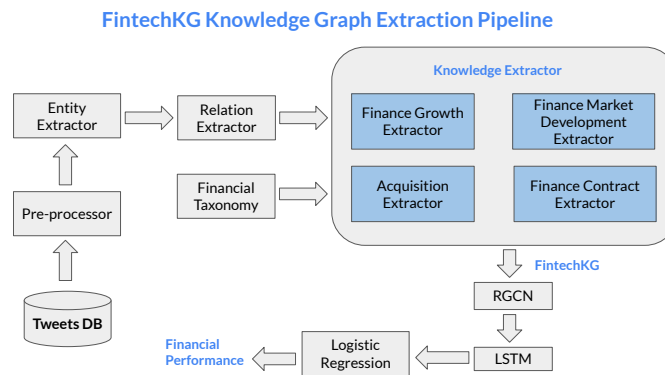


FIGURE 5.1 FintechKG Knowledge Graph Extraction Pipeline

5.1 FintechKG Construction Phase

The pipeline follows a systematic process for extracting **Concept Entities, Commercial Entities, and Relations** from textual data. The input to the pipeline consists of sentences or paragraphs of text. The text undergoes pre-processing, including the removal of stop words and invalid characters. The pre-processed text is then fed into the BERT (Bidirectional Encoder Representations from Transformers) NER (Named Entity Recognition) Model [98] to extract entities. A rule-based algorithm is used to extract relations in the form of <subject, predicate, object> triples from the textual data.

Furthermore, we employ a **Scenario-Based Knowledge Extractor** that encompasses several extractors such as the **Financial Growth Extractor, Finance Market Development Extractor, Finance Contract Extractor, and Acquisition Extractor**. These extractors utilize specific scenarios to extract relevant financial information. The output of the Scenario-Based Knowledge Extractor contributes to the generation of the financial knowledge graph(**FinTechKG**).

The combination of these steps forms a comprehensive approach for constructing the FinTechKG, enabling the extraction and organization of valuable financial information into a structured knowledge graph.

In this section, we define the key concepts in our methodology, where financial knowledge is extracted in three dimensions: Commercial Entity (ComE), Financial or Concept Entity (ConE), and Temporal Information (TI).

Commercial Entity (ComE):

The commercial entity refers to organizations' names. We use pre-defined ticker symbols to match with the tokens in the text and identify the occurrence of ticker symbols in the tweet. Ticker symbols are commonly used in financial text to refer to the performance of financial entities.

Financial or Concept Entity (ConE):

To create a taxonomy of financial concepts, we studied the structure of Annual Reports, Balance Sheets, and Income Statements, resulting in the identification of various financial concept entities. Examples of these entities include loss, operating profit, net sales, net profit, profit, earnings, operating loss, sales, net loss, pretax profit, cashflow, fair value, transaction, market share, diluted earnings, net interest, income, and total value. The taxonomy is illustrated in Figure 5.2.

Temporal Information (TI):

Temporal information is represented in various forms, such as date, first quarter, last quarter, year, recent quarter, second quarter, third quarter, fourth quarter, consecutive quarter, quarters, and last quarter.

There are different types of relations between the three dimensions. They are as follows:

Relation Type 1a: The source node can be a ComE, and the destination can be a ConE. The edge represents the relation, such as "increase" or "decrease".

Relation Type 1b: The source node can be a ComE, and the destination can be a TI (Temporal

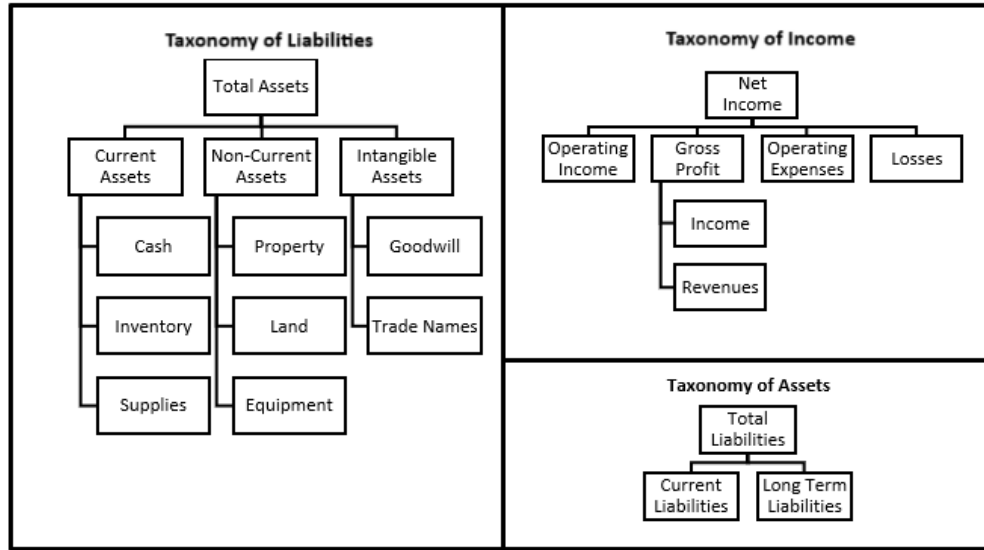


FIGURE 5.2 Taxonomy of Income, Liabilities and Assets

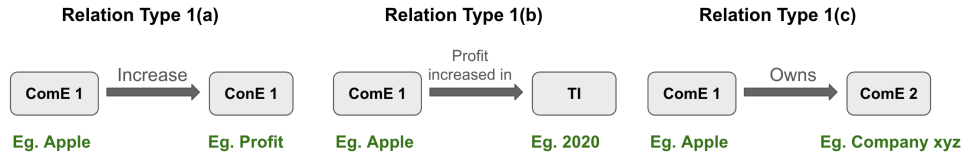


FIGURE 5.3 Relation Types

Node). The edge represents the ConE concatenated with its relation, for example, "profit increase in".

Relation Type 2: The source node can be a ComE, and the destination can also be another ComE. The edge represents the relation, such as "acquire" or "owns." This relation is referred to as a subset relation. The relation types are shown in Figure 5.3.

The pre-processor is an essential initial step in the extraction of the knowledge graph components [80, 92]. It consists of a sequence of steps, including the removal of URLs, hashtags, mentions, twitter handles, emojis, smileys, stop-word removal, punctuation removal, lemmatization, and stemming.

Entity Extraction

The entity extraction process involves identifying various types of entities, including Commercial Entities (ComE), Concept Entities (ConE), time entities, ticker symbols, currencies, percentages, countries, and persons.

Extraction using BERT Model

To extract entities from tweets, we utilize the BERT NER (Named Entity Recognition) model. The input to the model is the text from tweets, which is tokenized. The model classifies tokens into different entity types such as "DATE," "PERCENT," "ORG," "CARDINAL," "PERSON,"

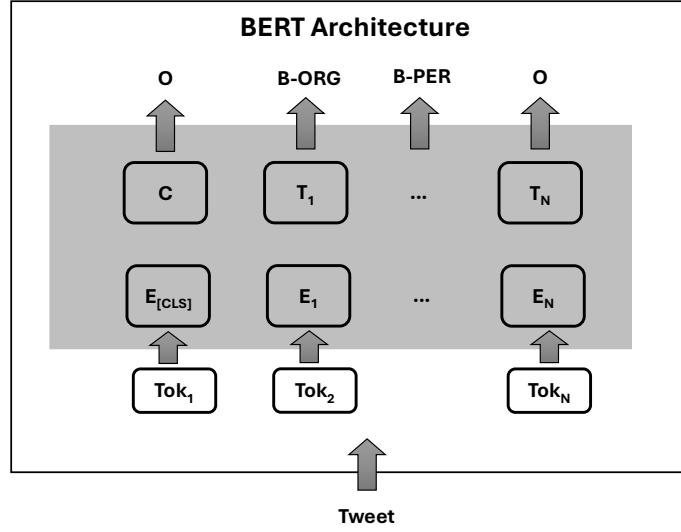


FIGURE 5.4 BERT NER Model

”NORP,” ”GPE,” and ”MONEY.” By analyzing the output of the model, we can identify Commercial Entities (ComeE) from the entity type ”ORG.” Additionally, ticker symbols of financial organizations are identified from the tweets using this approach.

Figure 5.4 illustrates the architecture of the BERT model applied for Named Entity Recognition (NER). The model processes a tweet as input, tokenizing the text into individual tokens (Tok₁, Tok₂, ..., Tok_N). Each token is fed into BERT, which generates embeddings (E_[CLS], E₁, ..., E_N) for each token. The output of BERT is then passed to classification layers that assign tags to each token, identifying named entities such as organizations (B-ORG), persons (B-PER), and non-entity tokens (O). This structure allows the model to effectively classify and recognize named entities within the unstructured textual data of the tweet. The following are the steps:

1. **Input:** A tokenized sequence (e.g., a tweet),

$$X = [[CLS], T_1, T_2, \dots, T_n, [SEP]]$$

where T_i represents each token.

2. **Embeddings:** For each token T_i , the embedding is:

$$E(T_i) = \text{Token Embedding}(T_i) + \text{Position Embedding}(T_i)$$

3. **BERT Encoding:** BERT processes the embeddings through self-attention, producing contextual embeddings:

$$H = [h_1, h_2, \dots, h_n]$$

4. **Self-Attention:** The self-attention mechanism calculates relationships between tokens:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V$$

5. **NER Classification:** For each token T_i , BERT applies a linear layer:

$$z_j(T_i) = W_j h_i + b_j$$

6. **Softmax:** The logits $z_j(T_i)$ are transformed into class probabilities:

$$P(C_j|T_i) = \frac{\exp(z_j(T_i))}{\sum_{k=1}^K \exp(z_k(T_i))}$$

7. **Output:** The model predicts the entity label for each token T_i , selecting the class with the highest probability:

$$\hat{y}_i = \arg \max_j P(C_j|T_i)$$

Extraction of Concept Entities

Concept Entities (ConE) are domain-specific in nature. In the case of tweets, we leverage financial reports such as Income Statements, Balance Sheets, and Cash Flow Statements to identify a taxonomy of financial concepts, as illustrated in Figure 5.2. Pre-defined concepts from this taxonomy are then identified as Concept Entities (ConE).

This entity extraction pipeline can be applied to new domains by defining a relevant taxonomy and creating a list of pre-defined concept entities. This flexibility enables the FintechKG Extraction pipeline to accurately identify concepts and construct the knowledge graph for different domains.

Extraction of Temporal Entities

Temporal entities are extracted in the form of dates, months, weekdays, years, and quarters. We also perform extrapolation based on time by converting references to time, such as "last quarter," "previous quarter," "last year," "yesterday," and "last year," to the corresponding time entity derived from the current timestamp of the tweet.

Extraction of Currencies and Percentages

Tweets often contain informal representations of text, including currencies and percentages. To accurately extract currency values, we employ a custom Currency Extractor that matches informal representations of currencies and retrieves more precise currency values. For example, "3.9mn" and "12.1mln" both refer to "million." Similarly, a Percentage Extractor matches informal representations of percentages in tweets, such as "14 percent increase in operating profit."

5.1.1 Relation Extractor

The relation extraction process involves identifying relations between entities in the knowledge graph.

The first step is a rule-based approach using the Spacy NLP pipeline. The pipeline includes various components like tokenizer, tagger, parser, ner, etc., and enables POS tagging and dependency parsing. By defining rules to match the structure of the sentence, we can extract the "ROOT," "PREP," and "ADJ" components from the sentence structure. These components are used to generate a list of relations. The frequency of these relations is calculated, and the top-N relations, representative of the domain, are selected.

For the financial tweet dataset, some of the pre-defined relations identified are: increase, decrease, plan, develop, report, start, grow, sell, acquire, expand, own, announce, sign, agree, issue, and fell.

The second step involves using BERT to perform similarity matching of new relations with the pre-defined set of relations obtained from the rule-based approach. By obtaining the relation embeddings from the BERT model and applying mean pooling, we create a single vector encoding for the sentence embedding. We calculate the cosine similarity between the new relation and the pre-defined relations. If the similarity score exceeds a threshold, we replace the new relation with the corresponding relation from the pre-defined set. This ensures consistency and avoids redundancies in the representation of relations in the Knowledge Graph.

TABLE 5.1 Samples of Financial Growth Extraction.

Tweet Example 1: Orion Pharma's operating profit increased by 42.5% from 2004	
Commercial Entity	Orion Pharma
Concept Entity	operating profit
Relation	increase
Date	2004
Percentage	42.5%

The **Financial Growth Extractor** module focuses on extracting relations related to financial growth, such as "Increase," "Decrease," "Rose to," "Totalled," "Fell," "Jumped," and "Dropped To." This module aims to identify instances where there is a change or movement in financial indicators.

Table 5.1 presents a sample of the extraction process. In the first example, the Commercial Entity extracted is "Orion Pharma," the Concept Entity is "Operating Profit," and the relation is "increase." The date "2004" and the percentage "42.5%" are also extracted and populated in the FintechKG.

The **Financial Market Development Extractor** identifies relations related to market development activities, including "Plan," "Develop," "Reported," "Start," "Grow," "Sell," "Includes," "Published," "Sold," "Won," "Issued," "Proposed," "Declined By," and "Forecasts." This module

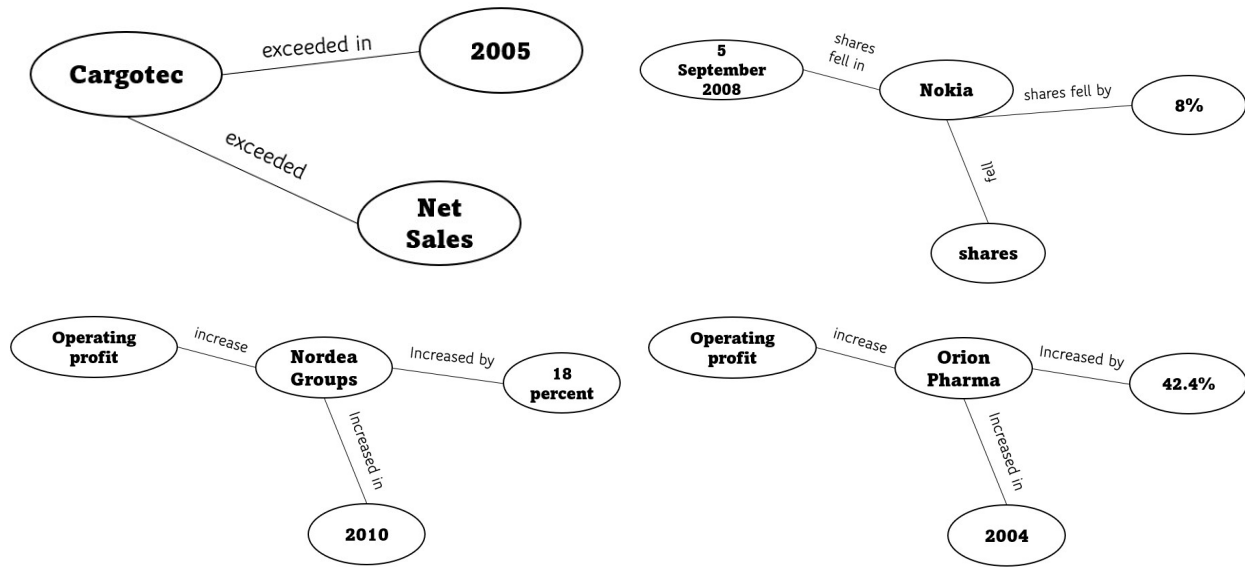


FIGURE 5.5 Knowledge Graph for Tweets:: "In 2005 Cargotec 's net sales exceeded EUR 2.3 billion.", "When this information was released on 5 September 2008 , Nokia 's American Depositary shares fell by 8 %," Orion Pharma 's operating profit increased by 42.5 % from 2004 ." and "Nordea Group 's operating profit increased in 2010 by 18 percent year-on-year to 3.64 billion euros and total revenue by 3 percent to 9.33 billion euros .

can capture various market-related events and actions, such as a company's sales growth or plans to enter new markets.

The **Acquisition Extractor** focuses on extracting relations related to acquisitions, including "Acquire," "Expand," "Rose to," "Own," "Announced," "Signed," and "Bought." This module aims to identify instances where a company acquires or expands its ownership of another entity, such as acquiring shares or announcing a new acquisition.

The **Financial Contract Extractor** extracts relations related to financial contracts, including "Contracted," "Awarded," "Sign," and "Agree." This module can identify instances where financial agreements, contracts, or awards are mentioned, such as signing a contract or being awarded a financial agreement.

These modules play a crucial role in extracting relevant information from financial tweets and populating the FintechKG with accurate and structured data, enabling a comprehensive understanding of financial events, trends, and relationships.

Figure 5.5 represents a knowledge graph generated from the information extracted from tweets. It shows relationships between various financial entities and events over time. For instance, Cargotec's net sales exceeded EUR 2.3 billion in 2005, and on 5 September 2008, Nokia's American Depositary shares fell by 8%. Orion Pharma's operating profit increased by 42.5% in 2004, and Nordea Group's operating profit increased by 18% in 2010. The knowledge graph links these financial entities (e.g., Cargotec, Nokia, Orion Pharma, Nordea Group) with relevant actions or events (e.g., exceeded, shares fell, increased), showcasing a structured representation of financial data

FIGURE 5.6 RGCN Architecture capturing Temporal Information

extracted from textual sources.

5.2 Temporal Relational Model

5.2.1 RGCN Model for Multiple Timesteps

In our methodology, a Relational Graph Convolutional Network (RGCN) model [37] is used. The constructed FintechKG is passed into the RGCN model at different timesteps. The RGCN model plays a important role in capturing the complex relationships between entities in the knowledge graph. It is illustrated in Figure 5.6.

The RGCN model is specifically designed to handle graph-structured data. Using FintechKG graph data, **nodes** represent **commercial entities**, and **edges** represent **relations** extracted from financial tweets, such as **increase, decrease, sell, and acquire**. Therefore, we can effectively capture the relational dependencies between commercial entities, exploit the contextual information encoded in the graph structure, which helps to improve the accuracy of financial performance predictions. The above RGCN model is designed to capture temporal information by considering graph data at three different time steps.

5.2.1.1 Architecture:

The RGCN (Relational Graph Convolutional Network) model takes graph data with node features and edge features at three time steps and applies graph convolutions to learn node representations that incorporate temporal information. The model then uses these representations to make predictions on relations. Given a graph with node features and edge features at three time steps, the RGCN model can be mathematically formulated as follows:

Node Representation:

Source Nodes (Commercial Entities), each commercial entity node is associated with a node feature vector h_s . Destination Nodes (Concept Entities), each concept entity node is associated with a node feature vector h_d .

Edge Representation:

Edge Labels (Relations), each edge (between a source and destination) is associated with an edge feature vector e that represents the relation between the commercial entity and concept entity.

Relational Convolution:

The relational convolution operation updates the node representations based on the source nodes, edge labels, and destination nodes.

The update rule for the source node s connected to destination node d through edge e is given by:

$$h_s = \sigma \left(\sum_{r \in R} \sum_{d \in N_r(s)} (h_d \cdot W_r^{(1)} \oplus e \cdot U_r^{(1)}) \right) \quad (5.1)$$

In equation 5.1,

$N_r(s)$ represents the set of destination nodes connected to source node s through relation r .

$W_r^{(1)}$ and $U_r^{(1)}$ are relation-specific weight matrices for the source node and edge, respectively.

$\oplus$ denotes concatenation.

σ is an activation function, such as ReLU or sigmoid, applied element-wise.

Stacked Convolutional Layers:

The RGCN typically consists of multiple stacked relational convolutional layers. Each layer refines the node representations by incorporating information from different levels of relational connections.

The update rule for the source node s in the $(l + 1)$ th layer is given by:

$$h_s^{(l+1)} = \sigma \left(\sum_{r \in R} \sum_{d \in N_r(s)} (h_d^{(l)} \cdot W_r^{(l)} \oplus e \cdot U_r^{(l)}) \right) \quad (5.2)$$

where $h_d^{(l)}$ denotes the destination node representation in the l -th layer.

Readout and Prediction:

After the convolutional layers, a readout function is applied to aggregate the final node representations into a fixed-length graph representation. This graph representation can then be used for predicting the relations between commercial entities and concept entities.

The prediction for the relation between source node s and destination node d is given by:

$$\hat{e} = \text{softmax} \left(\sum_{l=1}^L (h_d^{(l)} \cdot V^{(l)}) \right) \quad (5.3)$$

where, L is the number of convolutional layers

$V^{(l)}$ is a weight matrix for the destination node representation in the l -th layer

$\hat{e}$ is the predicted relation probability vector.

Prediction:

The node features at various timesteps are passed through an LSTM to further predict the performance of the entities.

The finbert [97] embeddings extracted from the finbert model are concatenated with the knowledge graph embeddings in different ways. Using the embeddings generated by the RGCN model, three variations are explored: RGCN Type 1, RGCN Type 2, and RGCN Type 3, each employing different embedding configurations.

In RGCN Type 1, the concatenation of finbert embeddings, source embeddings, relation em-

beddings, and destination embeddings obtained from the RGCN results in a combined embedding size of **1168**. This configuration integrates both textual and graph-based information, allowing for a more comprehensive representation.

Similarly, RGCN Type 2 follows a similar approach by concatenating the finbert embeddings with the RGCN embeddings. However, in this case, the total embedding size is reduced to **1068**, providing a more compact representation.

In RGCN Type 3, the embedding representation is simplified through the dot product of the source embedding, relation embedding, and destination embedding from the RGCN. This simplification yields a total embedding size of **868**.

By incorporating the finbert [97] embeddings with the RGCN embeddings in these different configurations, the aim is to leverage both textual knowledge and relational information encoded in the knowledge graph. This integration enhances the accuracy of financial entity performance predictions.

5.2.2 Architecture of LSTM Model

Our LSTM model introduces an approach to financial entity performance prediction by intricately combining RGCN embeddings with temporal pattern recognition. The LSTM architecture is specifically designed to process sequences of RGCN embeddings (NEt1, NEt2, and NEt3), corresponding to distinct quarterly financial periods, to capture the temporal dependencies critical for understanding market trends and entity behavior over time. This temporal granularity, aligned with the financial quarters, allows our model to account for seasonal fluctuations and other time-sensitive factors affecting financial performance.

By integrating RGCN embeddings, which encapsulate the relational dynamics within the financial knowledge graph, our LSTM model leverages both the spatial structure and temporal evolution of financial entities. This dual-focus architecture marks a significant advancement over traditional models that either neglect the temporal aspect or fail to incorporate the complex inter-entity relationships.

The LSTM (Long Short-Term Memory) model consists of an LSTM layer followed by a fully connected layer for prediction. The LSTM model is designed to process sequences of RGCN embeddings and make predictions based on the learned temporal patterns. The LSTM layer takes input sequences of RGCN embeddings and captures the temporal dependencies within the sequence. The hidden state at the last time step is then passed through the fully connected layer to generate the final predictions.

The LSTM model architecture for predicting the performance of financial entities based on the embeddings from RGCN at timestep 1, timestep 2 and timestep 3 (NEt1, NEt2, and NEt3) are shown Figure 8.5. The data is split into 3 time frames. NEt1 corresponds to tweets from Jan-Apr, NEt2 corresponds to May-Aug and NEt3 corresponds to Sept-Dec. The update step of the LSTM performs the LSTM Cell Updates. At timestep t , the LSTM cell update formulas are as follows:

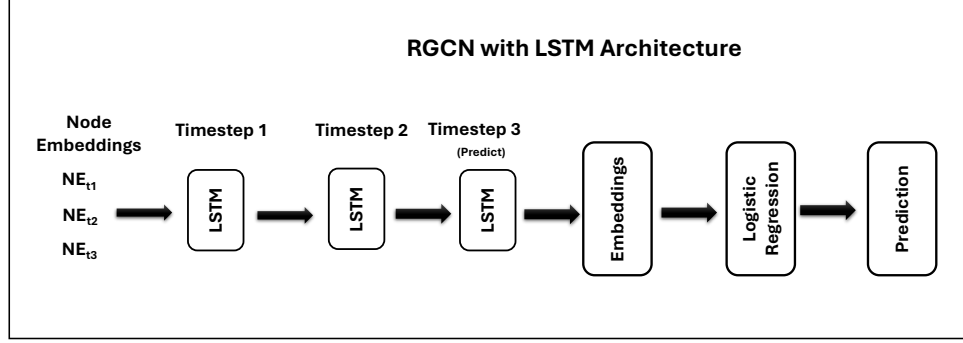


FIGURE 5.7 LSTM with RGCN Embeddings Architecture

Input gate: $i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$

Forget gate: $f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$

Candidate value: $g_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c)$

Cell state update: $c_t = f_t \cdot c_{t-1} + i_t \cdot g_t$

Output gate: $o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$

Hidden state: $h_t = o_t \cdot \tanh(c_t)$

In the above formulas, h_{t-1} represents the previous hidden state (output) of the LSTM cell, x_t represents the input at timestep t , which includes the embeddings NEt1, NEt2, and NEt3 for a single financial entity, W_i, W_f, W_c, W_o are the weight matrices for the input, forget, candidate, and output gates, respectively, b_i, b_f, b_c, b_o are the bias vectors for the input, forget, candidate, and output gates, respectively, σ represents the sigmoid activation function and $\tanh$ represents the hyperbolic tangent activation function.

The LSTM model architecture includes the LSTM layers and an output layer. The input sequence (embeddings at timestep1) is passed through the LSTM layers, and the final hidden state is fed into the output layer for performance prediction.

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CHAPTER 6

FintechKG: Experiments, Results, and Discussion

6.1 Experiment Setup

The tweets dataset¹ contains over 3 million unique tweets and includes the following information for each tweet: Tweet ID, Author, Post Date, Text Body, Number of Comments, Likes, Retweets for the year 2015 to 2020. The tweets are related to five major companies: Amazon, Apple, Google, Microsoft, and Tesla, and each tweet is matched with the appropriate share ticker of the respective company. We have used tweets for the year 2016 and 2018 for our experiments. The sample tweets are shown in Table 6.1.

TABLE 6.1 Sample Data from Top-companies Tweets.

Sample Data (tweet, post date)
This is so exciting. Can't wait for the model III! \$tsla @elonmusk just bought over 500k shares too, 1454215066
\$TSLA gonna start falling now if people think \$AAPL won't buy them anymore due to weak iPhone sales/production cut?, 1452015021

For the Named Entity Recognition (NER) task, the **BERT model** "dslim/bert-large-NER" is used to identify entites. For the **Relation Graph Convolutional Network (RGCN) model**, the FintechKG extraction pipeline is employed to generate triplets for the data from tweets in Dataset 3, specifically for the year 2016. These triplets, in the form of (source, relation, destination), are used as input to train the RGCN model for relation prediction. The RGCN model is trained with hyperparameters, such as a graph-split-size of 0.5, epoch of 30, dropout rate of 0.2, regularization of 1e-2, and a gradient norm of 1. The learning rate is set to 1e-2. The RGCN model is trained using data representing 3 timesteps.

¹<https://www.kaggle.com/datasets/omermetinn/tweets-about-the-top-companies-from-2015-to-2020?select=Tweet.csv>

The **LSTM model** used in the experiment consists of three layers. The first two layers correspond to the RGCN embeddings for timestep 1 and 2, respectively, while the third layer corresponds to the RGCN embedding for timestep 3. The data is split into 3 time frames. Timestep 1 corresponds to tweets from Jan-Apr, Timestep 2 corresponds to May-Aug and Timestep 3 corresponds to Sept-Dec.

The LSTM embeddings obtained from the trained model for each entity are then passed to a **Logistic Regression model** to make a prediction of the financial performance of the entity. Two types of embeddings: LSTM Type 1 and LSTM Type 2 are used by the regression model as shown in Table 6.2.

TABLE 6.2 Input Embeddings.

Input Embeddings Type	Description
Finbert	In this approach, only the FinBERT embeddings were used to train the model.
RGCN Type 1	In this approach, the FinBERT embeddings were concatenated with the (source embedding, relation embedding) and (destination embedding, relation embedding) pairs obtained from the RGCN. The total size of the embedding is 1168.
RGCN Type 2	In this approach, the FinBERT embeddings were concatenated with the source embedding, relation embedding, and destination embeddings from the RGCN. The total size of the embedding is 1068.
RGCN Type 3	In this approach, the dot product of the source embedding, relation embedding, and destination embeddings was used. The total size of the embedding is 868.
LSTM Type 1	In this approach, we use the LSTM embedding representing the source entity of size 204.
LSTM Type 2	In this approach, we use the LSTM embedding representing the source entity of size 204 and destination entity of size 204, with a total size of 408.

Additionally, Logistic Regression models are trained using both Finbert embeddings [97] and RGCN embeddings. Four types of inputs are used for the Logistic Regression model, as indicated in Table 6.2. The Finbert embeddings are obtained by passing the tweet as input to the Finbert model, resulting in embeddings of size 768. The RGCN embeddings are obtained by passing the triplets (s, r, o) as input to train the RGCN model. From the trained model, the entity embedding is retrieved with a size of 100, and the relation embedding is also retrieved with a size of 100.

The Hybrid Embeddings used by the Logistic Regression model to perform financial prediction

TABLE 6.3 Hybrid Embeddings.

Hybrid Embeddings Name	Combination
Finbert + Graph 1	Finbert + RGCN Type 1
Finbert + Graph 2	Finbert + RGCN Type 2
Finbert + Graph 3	Finbert + RGCN Type 3
Finbert + Temporal 1	Finbert + LSTM Type 1
Finbert + Graph 1 + Temporal 1	Finbert + RGCN Type 1 + LSTM Type 1
Finbert + Graph 1 + Temporal 2	Finbert + RGCN Type 1 + LSTM Type 2

is shown in Table 6.3. To effectively combine RGCN and FinBERT embeddings for prediction, FintechKG employs a projection layer that maps both embeddings into a shared latent space. This allows the model to leverage the strengths of both information sources for improved performance. The two projection techniques used are:

- **Linear Projection:** A simple linear transformation projects both embeddings into a common space. This is computationally efficient but may not capture complex relationships between the features.
- **Non-linear Projection:** Techniques like autoencoders or multi-layer perceptrons (MLPs) can learn a more complex non-linear mapping for better representation alignment, potentially improving prediction accuracy.

The prediction problem involves performing financial performance prediction for companies using data from our dataset ² from the year 2016 and predicting their financial performance for the subsequent year, 2017 and using data from the year 2018 and predicting their financial performance for the subsequent year, 2019. The companies of interest for the prediction are AAPL (Apple Inc.), GOOG (Google Inc.), GOOGL (Google Inc.), AMZN (Amazon.com), TSLA (Tesla Inc.), and MSFT (Microsoft Corporation). We are using evaluation metrics such as overall accuracy, refers to the proportion of correct predictions (both true positives and true negatives) among all predictions, F1 Score, the harmonic mean of precision and recall and Macro Avg (Precision, Recall, F1 Score), the average precision, recall, or F1 score across all classes, treating each class equally.

²<https://www.kaggle.com/datasets/omermetinn/tweets-about-the-top-companies-from-2015-to-2020?select=Tweet.csv>

TABLE 6.4 Financial Performance Prediction Results for 2016. "Decrease" and "Increase" refers to the scenario where the prediction of the performance of financial entity decreases or increases respectively in the next time period/financial year.

Combination	Epoch	Accuracy	F1 (Dec)	F1 (Inc)	Precision	Recall	F1
Finbert	40	92.41%	93.64%	90.60%	92.16%	92.08%	92.12%
Finbert + RGCN Type 1	40	94.09%	95.01%	92.74%	93.81%	93.09%	93.88%
Finbert	40	94.11%	95.03%	92.77%	93.82%	93.99%	93.90%
Finbert + RGCN Type 3	40	93.75%	94.76%	92.25%	93.57%	93.44%	93.50%
Finbert	60	93.10%	94.29%	91.28%	92.72%	92.86%	92.79%
Finbert + RGCN Type 1	60	94.62%	95.54%	93.21%	94.28%	94.47%	94.37%
Finbert + RGCN Type 2	60	94.59%	95.53%	93.16%	94.29%	94.40%	94.35%
Finbert + RGCN Type 3	60	93.15%	94.33%	91.35%	92.76%	92.92%	92.84%

TABLE 6.5 Financial Performance Prediction Results for 2018

	Finbert	Finbert + L1	Finbert + R1 + L1	Finbert + R1 + L2
Epoch	30	30	30	30
Accuracy	93.21%	94.66%	95.01%	94.05%
F1 (Decrease)	36.21%	46.11%	48.36%	40.68%
F1 (Increase)	96.01%	97.23%	97.38%	97.56%
Precision	95.79%	96.24%	96.20%	95.79%
Recall	93.30%	94.66%	95.70%	94.82%
F1	94.29%	95.32%	95.93%	95.25%

6.2 Results

The Fintech Knowledge Graph (FintechKG) statistics for the year 2016, shown in Table 6.6, present a comprehensive overview of the dataset used for financial analysis. The dataset comprises 892,840 total rows, which indicates the volume of data collected. Within this dataset, there are 22,514 concept entities and 17,682 triplets, highlighting the rich relational information captured. Additionally, the dataset includes 2,462 unique entities and 2,177 unique relations, demonstrating its complexity and the diversity of financial concepts and relationships it encompasses.

Table 6.7 details the data used for training the Relational Graph Convolutional Network (RGCN) model. The training dataset consists of 10,682 data points, which provides a robust foundation for model training. To ensure the model's performance is accurately assessed, the dataset also includes 3,500 data points each for testing and validation. This division allows for thorough evaluation and fine-tuning of the model, ensuring its accuracy and reliability in predicting financial performance.

The robustness of the proposed approach was evaluated through a series of experiments. These included hyperparameter tuning, where key parameters such as learning rate, batch size, and num-

ber of epochs were varied to assess sensitivity. Additionally, graph embedding variations were tested, utilizing different Relational Graph Convolutional Network (RGCN) variants in combination with FinBERT embeddings. The results indicated stable model performance, with a standard deviation of 0.11 - 0.16%, demonstrating the robustness of the approach across different configurations.

TABLE 6.6 FintechKG Statistics for the year 2016.

Metric	Count
Total Rows	892,840
Total Concept Entities	22,514
Total Triplets	17,682
Unique Entities	2,462
Unique Relations	2,177

TABLE 6.7 Data used for Financial Performance Prediction Task.

Data Type	Count
Test Data	3,500
Validation Data	3,500
Train Data	10,682

The financial performance prediction results for 2016 and 2018, as shown in Tables 6.4 and 6.5, highlight the effectiveness of different hybrid embeddings. In 2016, the Finbert + Graph 1 achieved the highest accuracy, reaching 94.09% with 40 epochs and 94.62% with 60 epochs. This method also had the best F1 scores for predicting both decreases and increases in financial performance, showing its strong capability in handling financial data. Other methods like the Finbert [97] and Finbert + Graph 3 combinations also performed well, with accuracies above 92%, but slightly lower than Finbert + Graph 1.

For the 2018 results, the Finbert + Graph 1 + Temporal 1 showed the highest overall accuracy at 95.01% and had the best F1 scores for predicting decreases (48.36%) and increases (97.38%). This indicates that Finbert + Graph 1 + Temporal 1 is very good at analyzing financial data and predicting performance changes. The Finbert + Temporal 1 configuration also demonstrated strong performance, with a 94.66% accuracy and competitive F1 scores.

Overall, these results demonstrate that using graph-aware embeddings like Finbert + Graph 1 and Finbert + Graph 1 + Temporal 1 can significantly improve the accuracy and reliability of financial performance predictions. These methods are effective in capturing complex financial trends and relationships, making them valuable tools for financial forecasting. The findings suggest that

incorporating relational and contextual information from financial data leads to better prediction outcomes, which is beneficial for making informed financial decisions.

TABLE 6.8 Comparison of Linear and Non-Linear Projection Methods for Financial Performance Prediction.

Method	Recall (Increase)	Recall (Decrease)	Size of Embedding
Baseline	0.9151	0.9459	868
Non Linear Projection	0.9279	0.9475	1024
Non Linear Projection	0.9250	0.9421	2048
Linear Projection	0.9293	0.9509	1024
Linear Projection	0.9350	0.9521	2048
Linear Projection	0.9260	0.9521	4082

¹ Baseline method refers to the original model without any projection.

Table 6.8 presents a comparison of linear and non-linear projection methods for financial performance prediction. The baseline model, without any projection, achieved recall scores of 0.9151 for the "Increase" class and 0.9459 for the "Decrease" class with an embedding size of 868. Applying non-linear projection at an embedding size of 1024 improved recall to 0.9279 and 0.9475, respectively, and further increasing the embedding size to 2048 resulted in recall scores of 0.9250 and 0.9421. Linear projection methods outperformed non-linear ones, with the best performance observed at an embedding size of 2048, yielding recall scores of 0.9350 for the "Increase" class and 0.9521 for the "Decrease" class. Even with a larger embedding size of 4082, linear projection maintained high recall scores of 0.9260 and 0.9521. These results indicate that linear projection techniques, particularly with an optimal embedding size, significantly enhance the accuracy of financial performance prediction models.

6.3 Discussion

The proposed FintechKG framework systematically extracts and organizes financial knowledge from textual data, incorporating temporal information, taxonomies, and customized extractors to enhance analysis. The extraction pipeline, which includes entity and relation extraction, generates a detailed FintechKG that captures complex financial information. By integrating FintechKG into the Relational Graph Convolutional Network (RGCN) model and Long Short-Term Memory (LSTM) networks, the framework significantly improves financial performance predictions, achieving an accuracy of up to 95% when combining embeddings.

The results of this study demonstrate the effectiveness of the FintechKG framework in understanding financial events, trends, and relationships. The integration of temporal information and contextual knowledge within the framework allows for more accurate and reliable predictions of financial performance. This comprehensive approach addresses key challenges in financial knowledge graph construction and performance prediction, offering a scalable and adaptable solution for various financial applications.

Financial institutions, investors, and analysts can utilize the FintechKG framework to enhance decision-making across several real-world applications:

Improved Forecasting: By integrating diverse data sources like social media and industry knowledge, institutions can make more accurate decisions in asset management, investment strategies, and risk management.

Real-Time Monitoring: Investors can track financial performance in real-time, enabling better portfolio management and quicker responses to market shifts.

Fraud Detection: The scalable FintechKG framework supports fraud detection and compliance by identifying anomalies through additional data sources, ensuring regulatory adherence.

Further research could apply FintechKG to sectors like insurance or real estate and expand its predictive capability by integrating unstructured data, such as news and market reports. $\text{\LaTeX}$.

CHAPTER 7

Corporate Credit Risk Prediction Using Temporal Knowledge Graphs

Corporate credit risk assessment is a critical component of financial decision-making, influencing loan approvals, investment strategies, and risk mitigation policies. The growing complexity of financial markets, coupled with the availability of diverse and dynamic datasets, has necessitated the adoption of advanced machine learning and graph-based approaches to model corporate credit risk. Among these, Graph Neural Networks (GNNs) have emerged as powerful tools for capturing relational and temporal dependencies within financial ecosystems.

7.1 Introduction

Recent advancements in Graph Neural Networks (GNNs) have demonstrated their effectiveness in various financial applications, including fraud detection, credit risk modeling, and stock market prediction. These models have shown great potential in identifying patterns within complex financial data, making them crucial for developing robust financial systems. For instance, Lou et al. [99] introduced a context-encoding and adaptive aggregation approach to detect fraudulent activities in financial datasets. This framework leverages graph structures to detect anomalies in financial transactions by capturing intricate patterns within the data. Similarly, Boyapati and Aygun [100] developed a specialized framework called BalancerGNN for imbalanced datasets, highlighting the adaptability of graph-based models in noisy and skewed financial environments. Their model is specifically designed to address the challenges posed by the imbalance between fraudulent and legitimate transactions in financial fraud detection tasks.

In addition to GNNs, the integration of knowledge graphs (KGs) has significantly enhanced the ability of these models to represent complex relationships in data. Knowledge graphs have proven valuable in structuring unstructured data and providing a more comprehensive understanding of the entities and their relationships. For example, Selote and Makhijani [101] demonstrated the use of KGs in integrating drug-disease-gene associations, which highlights the power of KGs to establish relationships across various domains. Similarly, He and Hu [102] applied Graph Convolu-

tional Networks (GCNs) to update building construction KGs, improving semantic integration and decision-making capabilities by better representing both structural and unstructured data. These studies underscore the utility of KGs in contextualizing structured and unstructured financial data, making them essential for tasks such as market prediction and financial risk assessment.

Temporal dynamics, particularly in financial risk assessment, play a vital role in modeling evolving relationships and patterns over time. Financial markets are inherently dynamic, with trends, risks, and opportunities constantly changing. The ability to capture these changes is critical in predicting market behavior and assessing risk. Fan et al. [103] proposed a spatiotemporal knowledge graph for dynamic preference recommendations, showcasing its ability to adapt to user preferences over both time and space. This dynamic capability highlights the importance of incorporating time-sensitive data when analyzing financial systems. Similarly, He and Zhu [104] reconstructed dynamic protein-protein interaction networks using relational GCNs, demonstrating the capacity of GCNs to capture both static and temporal features. This work provides valuable insights for applying temporal graph models in dynamic financial ecosystems, where temporal dependencies play a crucial role in decision-making.

Hybrid graph-based approaches have also made significant contributions to credit risk assessment. These approaches integrate various graph-based techniques to address specific challenges, such as data imbalance and model interpretability. Bi et al. [105] proposed a graph attention network for credit risk assessment, leveraging hierarchical attention mechanisms to enhance predictive accuracy. Their approach allows for more precise weighting of different entities and relationships, improving the model’s performance in predicting credit risk. Yu et al. [106] introduced a hybrid unsupervised model that combines spectral clustering with semi-supervised learning to address imbalances in credit risk data. This hybrid approach helps in improving the quality of the predictions by leveraging both labeled and unlabeled data. These methods highlight the potential of combining domain-specific insights with graph-based learning techniques to enhance the performance and interpretability of financial models.

Building upon these advancements, this paper proposes a novel approach for corporate credit risk assessment using Temporal Relational Graph Convolutional Networks (TRGCNs). By combining temporal embeddings with relational information extracted from structured knowledge graphs, the proposed method effectively captures both static and dynamic dependencies within financial datasets. This integration enables the model to account for the temporal evolution of financial entities and their relationships, providing a more accurate and interpretable framework for representing corporate credit risk. The proposed approach not only improves the prediction accuracy but also enhances the interpretability of the model, making it suitable for real-world financial applications where decision-makers require transparent and reliable results. The rest of the paper outlines the detailed methodology, experimental setup, and results, demonstrating the effectiveness of TRGCNs in complex and evolving financial landscapes.

7.2 Background and Related Work

Graph Neural Networks (GNNs) have demonstrated exceptional capabilities in capturing complex relationships and temporal dependencies across various domains, including finance. Their ability to model structured data as graphs allows for the effective representation of entities and their interactions, making them particularly useful in tasks such as fraud detection, credit risk assessment, and financial prediction.

In the domain of credit risk assessment, Bi et al. [105] developed a credit risk model leveraging Graph Attention Networks (GATs), which incorporates hierarchical attention mechanisms to improve prediction accuracy. By capturing the complex relationships between enterprises and their partners, this model enhances the predictive power of credit risk assessments at both the enterprise and relationship levels. The introduction of hierarchical attention mechanisms allows the model to weigh the importance of different relationships more effectively, thereby improving its ability to detect risks in financial transactions and partnerships. This work underscores the importance of accurately modeling relationships in financial data and highlights the potential of GNNs for credit risk modeling in dynamic environments.

Another notable contribution to the financial domain comes from Yu et al. [106], who proposed a hybrid unsupervised model for credit risk prediction. Their model integrates spectral clustering with semi-supervised learning to tackle issues related to imbalanced datasets in financial risk analysis. The semi-supervised learning approach helps in making use of both labeled and unlabeled data, thereby improving the accuracy and robustness of the model, especially when large amounts of data are available, but only a small subset is labeled. The integration of spectral clustering ensures that the model can identify natural groupings within the data, further refining its risk predictions. This approach has the advantage of enhancing the model's ability to generalize to unseen data while reducing the impact of class imbalances that are common in financial datasets.

Similarly, Zhang et al. [107] explored the use of dynamic GNNs in financial applications. Their work focuses on capturing the temporal dynamics of financial transactions and market conditions, which are critical for predicting future trends and assessing risks. By incorporating time-sensitive data into the GNN framework, their model is able to adapt to changes in the financial environment, such as fluctuating stock prices and evolving market conditions. This work highlights the growing importance of temporal data in financial applications and sets the foundation for the application of Temporal Graph Neural Networks (TGNNs) in areas like stock market prediction, fraud detection, and risk management.

In the area of fraud detection, several studies have utilized GNNs to detect fraudulent activities by modeling the relationships between transactions and entities. Lou et al. [99] introduced a context-encoding and adaptive aggregation method to detect fraudulent activities within financial datasets. This approach combines contextual information with adaptive aggregation mechanisms to improve the detection of anomalies in transactional data, making it more effective at identifying

fraud in large, noisy datasets. The context-encoding mechanism enriches the model’s understanding of the entities involved, ensuring that the relationships between them are appropriately modeled.

Furthermore, integrating Knowledge Graphs (KGs) with GNNs has provided an additional layer of structure, enhancing the model’s ability to reason about complex relationships. For instance, Selote and Makhijani [101] demonstrated the application of KGs in drug-disease-gene association prediction. While this application is in the biomedical domain, the underlying concept of integrating structured knowledge with GNNs is highly applicable to finance. Knowledge graphs enable the encoding of relational data, such as transactions between financial institutions, which can significantly improve the model’s interpretability and accuracy.

Temporal dynamics are another crucial aspect in the modeling of financial data. He and Zhu [104] reconstructed dynamic protein-protein interaction networks using relational GCNs to capture both static and temporal features. This approach inspired the application of Temporal GNNs in finance, where the modeling of time-sensitive data is essential for applications such as financial risk assessment and fraud detection. Temporal Graph Convolutional Networks (TGCNs) have been shown to be effective in capturing the dynamic behavior of financial markets, such as changes in stock prices over time, and are increasingly being used to model financial events and trends.

Building upon these advancements, our approach integrates Temporal Relational Graph Convolutional Networks (TRGCNs) for corporate credit risk assessment. The proposed method combines temporal embeddings with relational information extracted from structured knowledge graphs, allowing the model to capture both static and dynamic dependencies within financial datasets. This integration improves the accuracy of financial predictions and enhances the interpretability of the model, making it well-suited for real-world financial applications. By leveraging both temporal and relational aspects of financial data, we provide a comprehensive solution to the challenges of credit risk prediction in dynamic financial ecosystems.

Therefore, the combination of GNNs, KGs, and temporal data has demonstrated significant promise in improving the performance and interpretability of financial models. By incorporating these techniques, financial institutions can enhance their decision-making processes, leading to more accurate risk assessments and better-informed predictions in areas such as credit risk, fraud detection, and market forecasting.

Temporal dynamics are critical for financial data analysis, as they provide valuable insights into evolving market conditions, corporate performance, and financial risk. Financial markets are inherently dynamic, with market trends, risk factors, and financial relationships changing over time. Understanding these temporal relationships is key to making accurate predictions in areas such as fraud detection, stock market forecasting, and anti-money laundering. Recent advancements in Graph Neural Networks (GNNs) have significantly improved the ability to capture these temporal dependencies and integrate them into financial models.

Lou et al. [99] introduced a GNN framework with context encoding and adaptive aggregation mechanisms for fraud detection. Their work demonstrated the importance of capturing the temporal

relationships between entities involved in fraudulent activities. The framework dynamically adjusts its aggregation mechanisms based on context, enabling the model to detect anomalous patterns over time. This is crucial in the financial ecosystem, where fraudulent activities often evolve and adapt to changing circumstances. By incorporating temporal relationships into their model, Lou et al. were able to enhance the accuracy of fraud detection systems, emphasizing the critical role of temporal dynamics in financial fraud analysis.

Building on the importance of temporal relationships in financial prediction, Wang et al. [108] proposed a Multi-Relation Reconstruction and Fusion GNN (MRRFGNN) to predict stock crashes by modeling firm-to-firm relationships. Their approach focuses on reconstructing and fusing multiple relational graphs that capture the interactions between firms over time. By modeling these relationships dynamically, MRRFGNN is able to predict market crashes with greater accuracy, accounting for the evolving nature of inter-firm dynamics. This work highlights the power of GNNs in capturing both static and temporal relationships, providing a more comprehensive view of market dynamics that can be crucial for risk management.

Similarly, Yang and Duan [109] enhanced systemic risk management using dynamic knowledge graphs (DKGs). Their work focused on constructing knowledge graphs that evolve over time, integrating both financial and non-financial data. DKGs allow for real-time updates to reflect changes in market conditions, corporate actions, and regulatory environments, making them particularly useful for monitoring and managing systemic risk. The ability to continuously update and refine these knowledge graphs enables a more accurate and timely assessment of systemic risk, a critical component of financial stability and regulation.

Xu et al. [110] applied self-attention GNNs to improve anti-money laundering (AML) efforts. Their work leverages the self-attention mechanism within GNNs to focus on relevant features of the financial transactions while dynamically adjusting attention to capture both local and global patterns over time. This ability to model the temporal dependencies between transactions makes the model highly effective in detecting suspicious activities and money laundering schemes that evolve over time. The inclusion of temporal dependencies allows their model to continuously adapt to new patterns of money laundering, improving its ability to identify illicit financial flows.

Lastly, Wei et al. [111] explored dynamic fuzzy sampling techniques to improve the robustness of GNNs in uncertain environments, a common challenge in financial data analysis. Their method incorporates fuzzy logic to handle the inherent uncertainty and noise present in financial datasets, especially when temporal dynamics are involved. By using dynamic fuzzy sampling, Wei et al. enhanced the ability of GNNs to make reliable predictions despite the uncertainties in financial data. This work underscores the need for robust models capable of handling noisy and incomplete data, which is typical in financial applications, especially when dealing with temporal relationships.

In conclusion, these studies highlight the growing importance of incorporating temporal dynamics into GNN-based models for financial data analysis. Whether it's for fraud detection, stock market prediction, anti-money laundering, or systemic risk management, temporal dependencies

provide a crucial layer of insight that enhances the accuracy and reliability of financial predictions. As financial markets continue to evolve, the ability to model and integrate temporal dynamics will be essential for developing more accurate and adaptive financial models.

Knowledge graphs (KGs) provide a structured way to capture relational and temporal dependencies in financial datasets, offering a powerful tool for analyzing complex financial systems. These graphs can integrate both static relationships, such as connections between companies and their financial metrics, and dynamic relationships, such as evolving market trends or corporate actions. Yang et al. [109] constructed knowledge graphs (KGs) for the stock market, leveraging temporal data to analyze systemic risks. Their work demonstrated the ability of KGs to model the dynamic nature of stock market data, where relationships and risk factors change over time. Zhao et al. [112] applied GNN-based link prediction methods to heterogeneous blockchain networks, illustrating the utility of KGs in analyzing financial transactions and their evolving patterns in decentralized networks. Their approach showcased how KGs can be used to model complex financial transactions, providing insights into transaction patterns and network vulnerabilities.

Despite these advancements in graph-based learning, several key challenges remain in applying these models effectively to corporate credit risk prediction. While KGs have proven their effectiveness in other financial domains, adapting them to the intricacies of credit risk prediction requires addressing several challenges that are inherent in financial datasets.

- **Capturing Temporal Dynamics:** Financial data is inherently dynamic, with relationships evolving over time due to changes in market conditions, corporate strategies, and external economic factors. The ability to model temporal dependencies is crucial for predicting financial outcomes such as credit risk, where past performance and future prospects are closely tied to time-evolving variables. Existing models often struggle to effectively capture such temporal dependencies, making them less effective in dynamic financial environments. Fan et al. [103] emphasized the importance of spatiotemporal knowledge graphs in capturing evolving preferences over time. Their work demonstrated how dynamic knowledge graphs could adapt to changing user preferences, which is similar to how corporate credit risk assessments need to evolve with market conditions and internal company dynamics. Additionally, He and Zhu [104] successfully employed Relational Graph Convolutional Networks (RGCNs) to reconstruct dynamic protein-protein interaction networks, showing how temporal graph models can capture evolving relationships over time. However, the application of these temporal techniques remains underexplored in corporate credit risk prediction, where historical financial data must be combined with real-time market data to make accurate predictions.
- **Handling Multi-Relational Data:** Corporate credit risk assessment involves a wide range of relationships between companies, sectors, credit ratings, and various financial metrics. These relationships are complex and multi-dimensional, often involving multiple types of entities and interactions. Traditional models often struggle to represent and process these intricate

multi-relational dependencies effectively, leading to suboptimal performance in tasks such as credit scoring and risk assessment.

Bi et al. [105] highlighted the value of hierarchical relationship-level attention mechanisms for credit risk prediction. Their approach used attention mechanisms within GNNs to better understand the different levels of relationships that influence credit risk, allowing the model to weigh more important connections while ignoring irrelevant ones. Similarly, Yang and Duan [109] demonstrated the effectiveness of dynamic knowledge graphs for systemic risk management in the stock market. By integrating both static and dynamic data, their model was able to track the relationships between companies and sectors over time, offering valuable insights into systemic risk factors. This kind of multi-relational processing is crucial for accurate corporate credit risk assessment, where interactions between various financial indicators can significantly influence a company's creditworthiness.

- **Mitigating Data Imbalance and Noise:** One of the most pressing challenges in financial datasets is the inherent data imbalance. Financial datasets, especially those used for credit risk prediction, often suffer from a disproportionate representation of high-risk entities. In most cases, the number of low-risk companies far exceeds the number of high-risk ones, leading to a class imbalance that can bias models toward predicting lower-risk outcomes. Additionally, financial data is often noisy, with missing or erroneous values that can degrade the performance of predictive models.

Boyapati and Aygun [100] addressed these challenges by using adaptive graph construction techniques to handle imbalanced datasets. Their method dynamically adjusted the structure of the graph to ensure that both high-risk and low-risk cases were adequately represented, improving the overall performance of their model. Moreover, Xu et al. [110] utilized self-attention GNNs to improve performance in noisy environments. By applying self-attention mechanisms, their model was able to focus on relevant data while minimizing the impact of noisy or incomplete information. This technique is particularly useful in financial datasets, where noise from incomplete data or inconsistencies in reporting is common. The ability to handle such noise and imbalance is essential for building robust and accurate models for corporate credit risk prediction.

These challenges—capturing temporal dynamics, handling multi-relational data, and mitigating data imbalance and noise—highlight the complexities involved in corporate credit risk prediction. While knowledge graphs and graph-based learning techniques have made significant strides in other areas of financial analysis, these challenges need to be addressed to make them more effective in predicting corporate credit risk. By incorporating temporal dynamics, leveraging multi-relational data, and developing methods to mitigate data imbalance and noise, we can enhance the ability of graph-based models to predict corporate credit risk with higher accuracy and reliability.

7.3 Our Approach

To address the aforementioned challenges in corporate credit risk prediction, we propose a novel framework that combines Temporal Relational Graph Convolutional Networks (TRGCNs) with generative AI-based predictive mechanisms. Our approach leverages the strengths of both temporal graph modeling and advanced AI techniques to overcome the complexities inherent in financial data. The following components form the core of our methodology:

- **Temporal Credit Knowledge Graphs (TCKGs):** We construct Temporal Credit Knowledge Graphs (TCKGs) that capture entities such as companies, sectors, credit ratings, and financial metrics, alongside their temporal and relational dynamics. These graphs provide a structured representation of the financial ecosystem over multiple years, enabling the incorporation of both static relationships (such as credit ratings and financial metrics) and dynamic relationships (such as stock price fluctuations or market trends over time). The temporal aspect allows us to model how entities and their relationships evolve, making TCKGs particularly useful for credit risk prediction, where the timing of financial events is critical.

TCKGs enable the aggregation of information from multiple sources, such as historical financial reports, market trends, and sector-specific data. This results in a richer representation of the credit risk landscape, as relationships between companies, their financial health, and market conditions are encoded in a way that reflects both historical context and future potential. The integration of temporal data into the knowledge graph allows us to model the evolution of risk factors over time, which is crucial for accurate and timely credit risk assessment.

The TCKG construction process involves identifying relevant financial entities and their interactions, followed by embedding these entities and their relationships in a graph structure that evolves as new financial data becomes available. The graphs are built using structured data from annual reports, balance sheets, and credit ratings, and temporal relationships are encoded through time-series data and event-based data sources.

- **Temporal Embedding Extraction:** The key to accurately modeling time-evolving trends in corporate data lies in extracting meaningful temporal embeddings. To achieve this, we utilize a Long Short-Term Memory (LSTM) network, a type of recurrent neural network (RNN) well-suited for handling sequential data. The LSTM network processes multi-year financial embeddings from the Temporal Credit Knowledge Graphs (TCKGs) to capture the dynamic and evolving patterns within financial datasets over time.

The LSTM network plays a pivotal role in learning the temporal dependencies that influence corporate performance. By processing embeddings over multiple time steps, the LSTM captures both short-term fluctuations and long-term trends, offering a nuanced view of how financial entities evolve. This capability allows the model to distinguish between temporary shifts and sustainable changes in corporate behavior, such as sudden drops in stock prices due to external shocks versus gradual financial deterioration due to poor management or market

decline.

One significant advantage of LSTMs is their ability to maintain and update a hidden state across long sequences, enabling them to remember past information while simultaneously learning from new data. This makes them ideal for applications in corporate credit risk prediction, where historical data plays a crucial role in forecasting future financial stability. The ability to incorporate both past financial performance and current trends ensures that the temporal embeddings generated by the LSTM are highly predictive of future performance. For instance, by processing past financial indicators such as revenue, operating profit, debt levels, and market trends, the LSTM network generates embeddings that effectively represent the temporal evolution of a company's financial state. These embeddings are then used as input features for further analysis and predictive modeling, helping the system identify key factors contributing to corporate credit risk.

This temporal embedding extraction process enhances the accuracy of financial performance prediction models by embedding not only the financial metrics but also the timing and relationships between those metrics. Moreover, temporal embeddings are particularly useful for predicting outcomes such as:

- **Revenue trends:** Capturing how corporate revenue grows or shrinks over time based on historical patterns.
- **Credit rating shifts:** Predicting future changes in credit ratings based on past financial performance and macroeconomic conditions.
- **Bankruptcy risk:** Analyzing how a company's financial health evolves over time to assess the likelihood of default.

Recent work by Fan et al. [103] demonstrates the use of LSTMs in temporal knowledge graph applications, where temporal embeddings were successfully used to forecast dynamic preferences in various domains, including financial markets. In similar contexts, LSTMs have been shown to provide valuable insights for predicting stock prices, market crashes, and corporate credit risk [109, 112]. Additionally, He et al. [104] used LSTMs for reconstructing temporal networks in other domains, underscoring their versatility and effectiveness in handling time-sensitive data.

The integration of temporal embeddings into our approach represents a critical advancement in corporate credit risk assessment. The use of LSTMs to process these embeddings provides a robust mechanism for capturing time-sensitive financial dependencies, significantly enhancing the predictive power of financial models. As the LSTM extracts and learns from these temporal relationships, it enables the overall framework to produce more accurate and reliable predictions of corporate performance over time.

- **Relational Graph Convolution:** The temporal embeddings generated by the LSTM network are further enriched using Relational Graph Convolutional Networks (RGCNs), which are designed to capture and model complex relationships between different entities within a finan-

cial dataset. RGCNs, introduced by Schlichtkrull et al. [37], extend traditional Graph Convolutional Networks (GCNs) to handle multi-relational data by incorporating entity-specific relation types into the graph convolution process. This allows the model to consider not only the structural connectivity between entities but also the nature of the relationships between them, such as ownership, partnership, or transaction-based links.

In financial datasets, the relationships between companies, sectors, financial metrics, and credit ratings are inherently multi-relational and dynamic. For example, a company's credit rating may depend on its historical financial performance, market position, and macroeconomic factors. The RGCN enhances the model's ability to capture these diverse relationships by processing multiple types of edges in the knowledge graph, each representing a different facet of financial data, such as debt-to-equity ratio, market capitalization, or credit exposure. By applying RGCNs to the LSTM-generated temporal embeddings, we integrate not only the temporal evolution of a company's financial state but also the intricate relationships between companies, markets, and financial metrics. This dual integration of temporal and relational data enables the model to build a richer, more contextual understanding of corporate credit risk, improving its predictive accuracy and interpretability. For example, the model can assess how changes in one company's financial status might affect others within the same industry or sector, providing insights into systemic risk and the interdependencies within financial markets.

Recent works demonstrate the effectiveness of RGCNs in financial applications. For instance, Bi et al. [105] proposed a graph attention network for credit risk assessment, leveraging hierarchical attention mechanisms to capture relationships at different levels. Similarly, Yang and Duan [109] utilized dynamic knowledge graphs to analyze systemic risk in the stock market, effectively modeling the relationships between market entities. These works highlight the importance of multi-relational modeling for improving prediction accuracy in complex financial ecosystems.

The integration of RGCNs with LSTM-generated temporal embeddings helps to address several challenges in financial prediction:

- **Capturing Complex Interdependencies:** RGCNs enable the model to consider multiple types of relationships simultaneously, such as transactional, financial, and regulatory relationships, ensuring a comprehensive representation of financial networks.
- **Improving Interpretability:** By explicitly modeling the relationships between entities, RGCNs help provide transparency and interpretability in financial predictions, enabling analysts to understand the factors contributing to credit risk.
- **Enhancing Accuracy:** By incorporating relational data into the temporal embeddings, RGCNs improve the accuracy of predictions, particularly in scenarios where the relationships between entities are critical for assessing risk and performance.

This enhanced capability of RGCNs to process and integrate multi-relational data makes

them particularly well-suited for complex financial applications like corporate credit risk prediction. By capturing both the temporal dynamics and the complex interdependencies in financial systems, our approach provides a more robust and accurate framework for forecasting corporate financial performance.

- **Generative AI for Explainable Predictions:** In our framework, we leverage Gemini, a state-of-the-art generative AI model, to generate prompt-based predictions that synthesize historical trajectories, relational dependencies, and temporal patterns into explainable credit risk insights. Gemini, developed as part of the generative AI family, is specifically designed to integrate multiple sources of data and generate predictions in a human-interpretable format, addressing the challenge of transparency in complex financial forecasting.

Gemini uses advanced generative modeling techniques to simulate future financial outcomes based on past data, relational dependencies, and contextual insights. By focusing on historical trajectories, the model identifies significant patterns that may impact the future financial performance of companies. These patterns include shifts in market dynamics, economic indicators, and corporate behavior, all of which contribute to the accuracy and relevance of the predictions.

One of the key advantages of using Gemini for credit risk prediction is its ability to generate explainable insights. Unlike black-box models that offer predictions without clear reasoning, Gemini provides a transparent approach by generating human-readable explanations of how specific predictions are made. These explanations are rooted in the underlying relational dependencies and temporal patterns observed in the data. For example, if a company's credit risk is predicted to increase, the model can provide insights into the key factors driving this prediction, such as declining financial metrics, changes in the market, or shifts in sector-wide conditions. This transparency is crucial for financial analysts and decision-makers who need to understand the reasoning behind predictions and ensure they align with regulatory and compliance standards.

The application of generative AI models like Gemini in financial contexts has shown promise in improving both the accuracy and interpretability of predictions. Recent work by Smith et al. [113] highlights the utility of generative models in complex financial forecasting, demonstrating their ability to generate explainable, context-aware predictions.

In the context of corporate credit risk, generative models like Gemini provide several benefits:

- **Explainability:** Gemini enhances the interpretability of predictions by offering clear, contextually grounded explanations for each credit risk assessment. This is particularly important for compliance with regulatory requirements such as the EU's GDPR and the Dodd-Frank Act, which emphasize the need for explainable AI in financial decision-making.
- **Historical and Temporal Integration:** By incorporating historical trajectories and

temporal patterns, Gemini captures the evolution of credit risk over time, making it capable of providing future forecasts based on long-term trends and dependencies. This dynamic approach allows for more accurate and actionable insights.

- **Improved Predictive Accuracy:** Gemini’s ability to synthesize relational data with temporal features allows it to capture a wide range of factors influencing credit risk, including company-specific attributes, market conditions, and global economic factors. This results in more robust and reliable predictions, essential for managing corporate credit portfolios.

Overall, the use of Gemini for explainable predictions in corporate credit risk assessment offers a significant advancement in the field of financial modeling. By combining the power of generative AI with temporal and relational data, our approach provides a more transparent, accurate, and actionable framework for understanding and managing credit risk in dynamic financial ecosystems.

- **Addressing Data Imbalance and Noise:** Financial datasets, especially those used for credit risk prediction, often suffer from data imbalance, where the number of high-risk entities is much smaller compared to low-risk entities. This imbalance can bias the model towards predicting the majority class, thereby reducing its ability to accurately predict high-risk cases. To address this challenge, advanced sampling techniques, such as oversampling of minority classes and undersampling of majority classes, are employed to balance the dataset during training. Additionally, techniques like SMOTE (Synthetic Minority Over-sampling Technique) [114] have been applied to generate synthetic examples of high-risk entities, helping to balance the dataset without losing valuable information.

In addition to data imbalance, financial datasets are often plagued by noisy or irrelevant data, which can degrade model performance. Noise in the form of incomplete, erroneous, or outdated information can affect the accuracy of credit risk predictions. To mitigate the impact of noise, attention mechanisms [115] are employed to focus the model’s learning process on the most significant relationships and features, allowing the model to ignore irrelevant data. Attention mechanisms work by assigning higher weights to important relationships or features and lower weights to less relevant or noisy ones, thereby improving the model’s robustness to noisy inputs.

For example, attention-based models, such as the Transformer [115], have been widely used in financial applications to improve the prediction accuracy by focusing on the temporal and relational dependencies that matter most for forecasting corporate credit risk. Similarly, Bi et al. [105] employed attention mechanisms in graph-based learning models to identify key features that contribute to credit risk, resulting in better risk assessments.

Furthermore, techniques such as adversarial training [116] can be integrated into the model to make it more resilient to noisy data by introducing small perturbations during training. This approach forces the model to become more robust to variations in the data and reduces

the risk of overfitting to noisy or irrelevant features.

The combination of advanced sampling techniques and attention mechanisms ensures that the model can effectively deal with data imbalance and noise, improving both the accuracy and generalization of credit risk predictions. These techniques not only help in addressing the challenges inherent in financial datasets but also enable the model to focus on the most critical features, leading to more reliable and actionable insights for financial decision-making.

Our research addresses critical challenges in corporate credit risk prediction by introducing Temporal Relational Graph Convolutional Networks (TRGCNs). By integrating temporal embeddings, relational reasoning, and generative AI, the proposed framework sets a new benchmark for accurate and interpretable financial risk assessments, advancing the state of the art in graph-based learning for dynamic financial ecosystems. X_{TE}AT_EX.

CHAPTER 8

Methodology of FCKG

The proposed framework for Financial Credit Risk Prediction integrates temporal knowledge graph construction, relational graph learning, and multi-model inference to assess and predict credit risk. The framework, depicted in Figure 8.1, consists of the following key components:

8.1 Temporal Credit Knowledge Graphs Construction

The Temporal Credit Knowledge Graph (TCKG) construction process involves transforming corporate credit rating data into a structured graph format that captures entities, relationships, and temporal dynamics. This process begins with the extraction of key entities (e.g., companies, sectors, ratings) and their associated relationships (e.g., financial metrics, sectoral affiliations) from the dataset.

The following steps outline the construction process in detail:

1) **Entity Identification:**

The dataset is scanned row by row, identifying companies as the primary entities. Additional entities include sectors, credit ratings, and rating agencies.

2) **Relationship Mapping:**

For each company, relationships are established based on the data:

- A company's sector is captured using a triplet of the form (Company, operatesInSector, Sector).
- The credit rating assigned to the company is represented as (Company, hasRating, CreditRating), and if available, the associated rating agency is linked as (CreditRating, ratedBy, RatingAgency).
- Financial metrics such as currentRatio, debtEquityRatio, and returnOnAssets are encoded as (Company, MetricName, MetricValue).

The details are shown in Table 8.1.

3) **Temporal Relationships:**

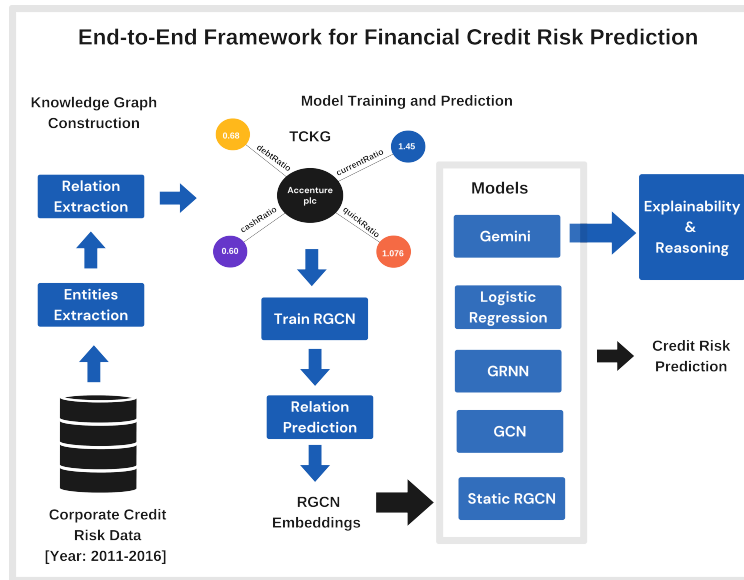


FIGURE 8.1 Framework for Financial Credit Risk Prediction

TABLE 8.1 Financial Metrics Used in Knowledge Graph Construction

Metric Name	Description
currentRatio	Measures a company's ability to pay short-term obligations.
quickRatio	Similar to currentRatio, but excludes inventory.
cashRatio	Evaluates liquidity using only cash and cash equivalents.
debtEquityRatio	Ratio of total liabilities to shareholders' equity.
debtRatio	Proportion of a company's assets financed by debt.
netProfitMargin	Percentage of revenue that remains as profit after expenses.
operatingProfitMargin	Measures efficiency in managing costs of operations.
returnOnAssets	Indicates how profitable a company is relative to its assets.
returnOnEquity	Measures profitability relative to shareholder equity.
freeCashFlowOperatingCashFlowRatio	Assesses cash flow efficiency.
grossProfitMargin	Percentage of revenue remaining after cost of goods sold.

Temporal aspects are incorporated by associating reporting dates with each company using the triplet ‘(Company, reportedIn, Date)’. This step captures the evolution of financial metrics and ratings over time.

4) Data Validation and Processing:

To ensure completeness, null values are checked, and only valid data entries are included in the triplet construction. Missing relationships or metrics are excluded to maintain graph integrity.

TABLE 8.2 Example Triplets for Temporal Credit Knowledge Graph Construction

Subject	Predicate	Object
NVR, Inc.	operatesInSector	Capital Goods
NVR, Inc.	hasRating	Medium Risk
Medium Risk	ratedBy	Moody’s Investors Service
NVR, Inc.	currentRatio	11.75776709
NVR, Inc.	quickRatio	7.115059408
NVR, Inc.	cashRatio	7.057419576
NVR, Inc.	debtEquityRatio	0.759461309
NVR, Inc.	debtRatio	0.431644223
NVR, Inc.	netProfitMargin	0.05670576
NVR, Inc.	operatingProfitMargin	0.077844347
NVR, Inc.	returnOnAssets	0.069327813
NVR, Inc.	returnOnEquity	0.121979605
NVR, Inc.	freeCashFlowOperatingCashFlowRatio	0.953230907
NVR, Inc.	grossProfitMargin	1.0
NVR, Inc.	reportedIn	2012-09-05 00:00:00

5) Triplet Generation:

A triplet format ‘(Subject, Predicate, Object)’ is used to encode the relationships, where the ‘Subject’ represents the source entity, the ‘Predicate’ defines the relationship type, and the ‘Object’ is the target entity or value. For example:

- (CompanyA, operatesInSector, Technology)
- (CompanyB, hasRating, AA)
- (AA, ratedBy, Moody’s)

The triplet representation of a company is provided in Table 8.2, while the corresponding Credit Risk Knowledge Graph (CRKG) is illustrated in Figure 8.2.

6) Temporal Segmentation:

For each reporting year or date, distinct snapshots of the knowledge graph are created to reflect the financial and relational status of entities at specific time points. The temporal

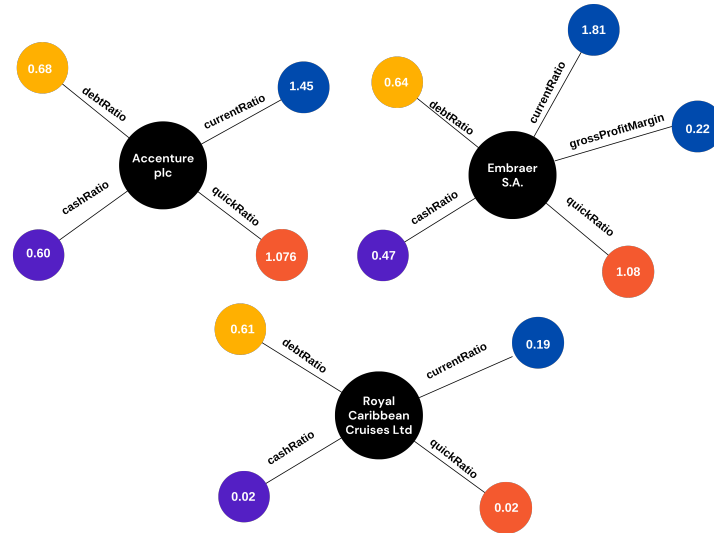


FIGURE 8.2 Financial Credit Risk KG

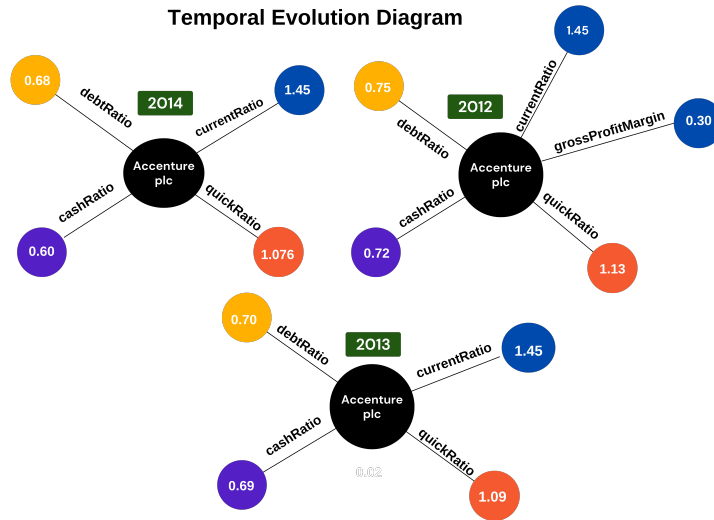


FIGURE 8.3 Temporal Evolution

evolution for a company can be seen in Figure 8.3.

This process results in a dynamic and structured knowledge graph that encodes temporal, relational, and contextual information about corporate financial ecosystems.

By capturing financial metrics, credit ratings, and sectoral relationships alongside temporal trends, the TCKG serves as a robust foundation for applying graph neural networks (GNNs) and temporal relational models for credit risk prediction. The statistics of this graph is shown in Table 8.3. The graph's ability to represent multi-relational and temporal dependencies provides a comprehensive view of financial entities and their interdependencies over time.

TABLE 8.3 Knowledge Graph Statistics for Credit Risk Prediction (2011-2016)

Year	Triplets	Subjects	Predicates	Objects	Avg/Subject	Avg Degree	Max Degree	Min Degree
2011	870	54	15	677	16.1	16.1	28	1
2012	5,355	293	15	3,873	18.3	18.3	142	11
2013	4,680	268	15	3,485	17.5	17.5	116	3
2014	5,745	317	15	4,272	18.1	18.1	143	2
2015	7,200	373	15	5,326	19.3	19.3	175	1
2016	6,420	345	15	4,771	18.6	18.6	204	1

8.1.1 Relational Graph Convolution

Relational Graph Convolutional Networks (RGCNs) extend traditional graph convolutional networks by incorporating multi-relational data. Each relationship type between nodes is explicitly modeled, allowing for the capture of diverse interdependencies within the graph. In the context of credit risk prediction, RGCNs enable the integration of structured financial data and relationships, such as company-sector connections, credit ratings, and financial metrics.

8.1.2 RGCN Architecture

The architecture of the RGCN used in this work consists of the following components:

- **Input Layer:** Node features representing companies, sectors, and other entities are initialized based on financial attributes and encoded relationships.
- **Relational Convolutional Layers:** Multi-relational edges, such as `operatesInSector` or `hasRating`, are used to propagate information across nodes. The message-passing mechanism aggregates neighborhood information for each relation type.
- **Activation and Dropout:** Non-linear activation functions (e.g., ReLU) and dropout layers are applied to enhance model generalization and prevent overfitting.
- **Output Layer:** Encoded node embeddings are output, representing the latent features of entities, which are later used for downstream prediction tasks.

8.2 RGCN Model for Credit Risk Prediction

In our methodology, we employ a Relational Graph Convolutional Network (RGCN) [37] to model temporal dependencies in credit risk assessment. The constructed Credit Risk Knowledge Graph (CRKG) is processed through the RGCN model at different timesteps (year 2011 to year 2016), enabling it to capture the evolving relationships between financial entities. The RGCN model plays a crucial role in learning structured representations of entities and predicting their financial stability. The architecture is illustrated in Figure 8.4.

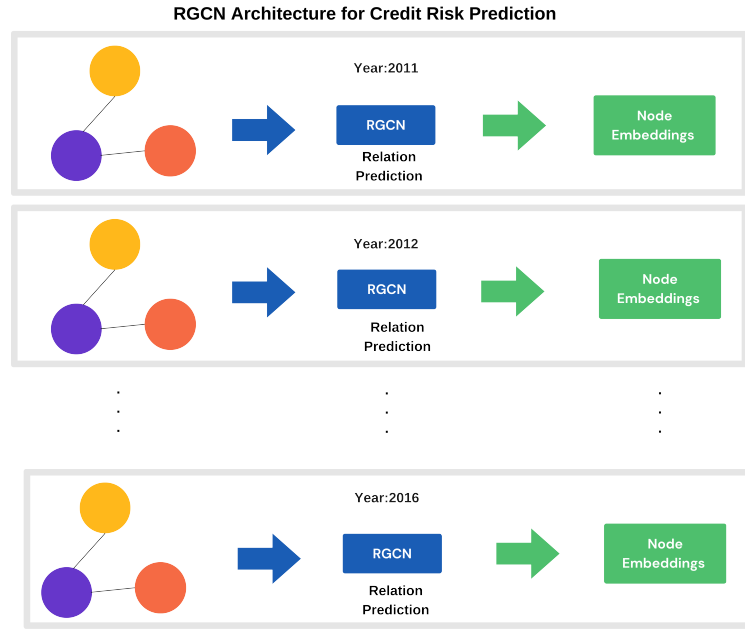


FIGURE 8.4 RGCN Architecture

The RGCN model is specifically designed for graph-structured data. In our Credit Risk Knowledge Graph (CRKG), nodes represent corporate entities (e.g., banks, firms, financial institutions), and edges represent financial relationships, such as `hasRating`, `operatesInSector`, and `ratedBy`. Additionally, temporal financial metrics, such as `debtRatio`, `returnOnAssets`, and `netProfitMargin`, are embedded as features to model the financial health of entities over time. The RGCN model effectively captures temporal dependencies, financial performance trends, and relational information, which significantly enhances credit risk prediction accuracy.

8.2.1 Architecture:

The RGCN model processes the Credit Risk Knowledge Graph (CRKG) at multiple time steps, learning node representations that encode financial attributes and relationships. These representations are then used to predict credit ratings and assess risk.

8.2.1.1 Node Representation

Corporate Entity Nodes (s): Each company is represented by a feature vector h_s , capturing financial metrics and historical performance. **Rating Agency Nodes (d):** Each credit rating agency (e.g., Moody's, SP) is associated with a feature vector h_d .

8.2.1.2 Edge Representation

Credit Risk Relationships (e): Each edge represents a financial relationship (e.g., `hasRating`, `operatesInSector`) between entities.

8.2.1.3 Relational Convolution:

The RGCN model propagates information across the graph using relational convolution, updating node embeddings based on their financial relationships and credit ratings.

The message-passing update rule for a corporate entity s , connected to a rating agency d through a relation e , is formulated as:

$$h_s = \sigma \left(\sum_{r \in R} \sum_{d \in N_r(s)} (h_d \cdot W_r^{(1)} \oplus e \cdot U_r^{(1)}) \right) \quad (8.1)$$

where:

$N_r(s)$ represents the set of destination nodes connected to entity s through relation r . $W_r^{(1)}$ and $U_r^{(1)}$ are relation-specific weight matrices for financial entities and credit risk relationships, respectively. $\oplus$ denotes concatenation, and σ is an activation function (e.g., ReLU).

8.2.1.4 Stacked Convolutional Layers

To capture higher-order dependencies, the RGCN stacks multiple relational convolutional layers. Each layer refines node representations by incorporating financial signals from distant entities in the graph.

The recursive update rule for node s at layer $(l + 1)$ is:

$$h_s^{(l+1)} = \sigma \left(\sum_{r \in R} \sum_{d \in N_r(s)} (h_d^{(l)} \cdot W_r^{(l)} \oplus e \cdot U_r^{(l)}) \right) \quad (8.2)$$

where $h_d^{(l)}$ denotes the representation of entity d in layer l .

8.2.1.5 Readout and Credit Rating Prediction

After the relational convolution layers, a readout function aggregates credit risk features into a graph-level representation. This enables credit risk classification or rating prediction.

The probability distribution over possible credit ratings is computed as:

$$\hat{e} = \text{softmax} \left(\sum_{l=1}^L (h_d^{(l)} \cdot V^{(l)}) \right) \quad (8.3)$$

where:

L is the number of convolutional layers. $V^{(l)}$ is a weight matrix for entity representations at layer l . $\hat{e}$ represents the predicted probability distribution over credit ratings.

6. Temporal Credit Risk Prediction To model credit risk evolution, the node embeddings at multiple time steps are processed through a Long Short-Term Memory (LSTM) network to capture time-dependent financial signals and predict future credit risk trends.

The Relational Graph Convolutional Network (RGCN) model is trained using a structured approach. A graph batch size of 30,000 ensures efficient mini-batch processing, while the graph split size of 0.5 allows the model to train on 50% of the graph at a time, reducing memory constraints. The model employs negative sampling with a ratio of 1, meaning each positive triplet is paired with one negative sample to improve discrimination between valid and invalid relationships. Training is conducted for 100 epochs, with evaluations performed every 500 batches to monitor performance and convergence. To prevent overfitting, a dropout rate of 0.2 is applied, along with L2 regularization ($1e-2$) to enforce weight constraints. The model is optimized using the Adam optimizer with a learning rate of 0.01, and gradient clipping at 1.0 is used to stabilize updates. A basis decomposition approach with four bases is employed to minimize parameter redundancy while maintaining expressive power. This setup ensures that the RGCN model can effectively capture complex financial relationships while maintaining scalability and computational efficiency.

TABLE 8.4 Credit Risk Prediction Performance (2011-2016)

Year	Epochs	MRR	Hits@1	Hits@3	Hits@10
2011	30	0.0992	0.0833	0.1006	0.1149
2011	60	0.0824	0.0690	0.0747	0.1092
2011	90	0.0898	0.0776	0.0862	0.1006
2012	30	0.1246	0.1074	0.1377	0.1489
2012	60	0.0959	0.0836	0.0985	0.1247
2013	30	0.0903	0.0775	0.1010	0.1106
2013	60	0.0805	0.0700	0.0833	0.0967
2013	90	0.0599	0.0497	0.0652	0.0710
2014	30	0.0966	0.0809	0.1088	0.1205
2014	60	0.0849	0.0722	0.0888	0.1171
2014	90	0.0694	0.0661	0.0670	0.0727
2015	30	0.0938	0.0785	0.1080	0.1156
2015	60	0.0846	0.0726	0.0892	0.1069
2015	90	0.0411	0.0188	0.0646	0.0726
2016	30	0.0949	0.0779	0.1032	0.1293
2016	60	0.0821	0.0693	0.0853	0.1067
2016	90	0.0733	0.0697	0.0728	0.0779

This study evaluates the efficacy of Relational Graph Convolutional Networks (RGCNs) for predicting credit risk from 2011 to 2016. Experiments shown in Table 8.4 were conducted by training the R-GCNs for 30, 60, and 90 epochs, revealing a complex relationship between training duration and predictive performance. The variation in optimal epoch number across different years suggests that the ideal training length is influenced by year-specific data characteristics, likely including the number of entities and relationships within the credit network, as well as shifts in the

underlying dynamics of credit risk. A key observation is the presence of overfitting, indicated by the decline in performance metrics (MRR, Hits@K) with increasing epochs in several years (e.g., 2013, 2015). This highlights the importance of employing regularization strategies, such as early stopping based on validation performance, or established techniques like dropout or weight decay. While the R-GCN effectively utilizes relational information, the overall predictive performance, while promising, suggests limitations. These may be attributed to inherent data sparsity, particularly the scarcity of default events, a common challenge in credit risk modeling. Furthermore, the use of a static relational structure across all years might not adequately capture the evolving nature of credit risk factors.

Analysis of the Table 8.4 reveals that Epoch 30 generally exhibits the strongest performance across multiple metrics (MRR, Hits@1, Hits@3, Hits@10) and years, suggesting it as the most suitable choice for fetching embeddings.

8.3 Temporal Embedding Extraction

Temporal embeddings are critical for capturing the dynamic nature of financial datasets. This work employs a Long Short-Term Memory (LSTM) network to extract temporal patterns from multi-year financial data. The LSTM processes sequential data to capture temporal dependencies, trends, and evolving financial metrics for each company.

8.3.0.1 LSTM Architecture

The LSTM architecture used for temporal embedding extraction consists of:

- **Input Layer:** Time-series financial data for each company is provided as input. Each timestep corresponds to financial metrics from a specific year.
- **LSTM Layers:** The recurrent units process sequential data to model dependencies over time. Cell states and hidden states enable the network to capture long-term dependencies.
- **Temporal Embeddings:** The final hidden state of the LSTM is extracted as the temporal embedding, summarizing the company's financial trajectory over multiple years.

8.4 LSTM Model for Credit Risk Prediction

To enhance the predictive capabilities of our credit risk assessment framework, we integrate a Long Short-Term Memory (LSTM) network with the Relational Graph Convolutional Network (RGCN) embeddings. The LSTM model is specifically designed to process sequences of RGCN embeddings generated at multiple time steps, capturing the temporal dependencies in financial performance data. By learning sequential patterns from historical credit risk data, the model can

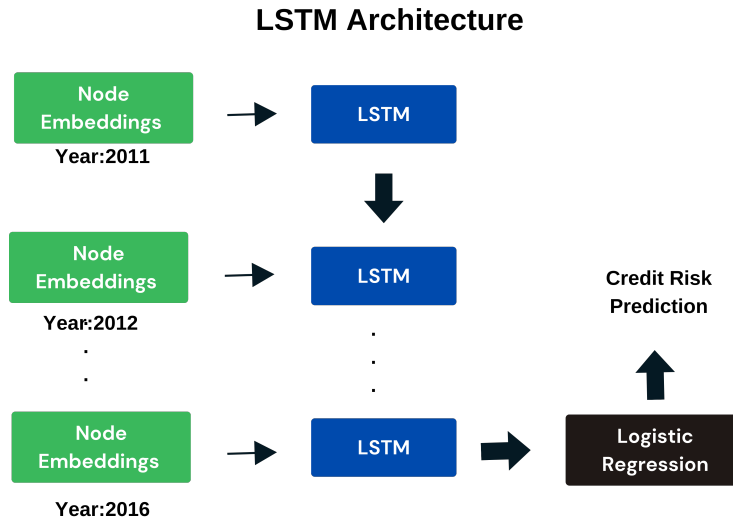


FIGURE 8.5 LSTM Architecture

forecast the credit stability of financial entities. The LSTM model architecture is illustrated in Figure 8.5.

Our approach divides the financial data into three distinct time frames, where:

NE_t1 corresponds to financial data from 2012 NE_t2 corresponds to financial data from 2013 NE_t3 corresponds to financial data from 2014 and so on. This yearly segmentation aligns with financial reporting periods, allowing the model to capture seasonal variations and shifting economic conditions that impact credit risk.

8.4.1 Integration of RGCN Embeddings

The LSTM model uses RGCN embeddings, which encode structural and relational financial dependencies from the Credit Risk Knowledge Graph (CRKG). Unlike traditional credit scoring models that rely solely on numerical data, our graph-enhanced approach captures both:

Entity relationships (e.g., "Company A has a credit rating of BBB by Moody's"), Temporal financial trends (e.g., "Company A's return on equity increased in Q2"). By incorporating both spatial (graph-based) and temporal (sequential) features, the LSTM-RGCN model enables a more dynamic and context-aware credit risk prediction framework.

LSTM Model Architecture The LSTM model consists of:

LSTM layers: Process sequences of RGCN embeddings to capture temporal dependencies.
Fully connected output layer: Generates credit risk predictions based on learned representations. At each timestep t , the LSTM cell processes the credit risk embeddings (NE_t1, NE_t2, NE_t3), refining the hidden state (h_t) to incorporate long-term dependencies.

Mathematical Formulation of LSTM Cell Updates At each timestep t , the LSTM updates are

computed as follows: Input gate: Regulates the amount of new financial information to store.

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (8.4)$$

Forget gate: Determines which historical credit risk information to retain or discard.

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (8.5)$$

Candidate value: Generates a candidate update for the credit risk state.

$$g_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \quad (8.6)$$

Cell state update: Updates the financial entity's credit risk memory.

$$c_t = f_t \cdot c_{t-1} + i_t \cdot g_t \quad (8.7)$$

Output gate: Decides how much of the current credit risk information should influence the final prediction.

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (8.8)$$

Hidden state: The final output embedding capturing credit stability trends.

$$h_t = o_t \cdot \tanh(c_t) \quad (8.9)$$

Explanation of Variables: h_{t-1} : Previous LSTM hidden state (encodes past credit risk trends). x_t : Input embeddings at timestep t (RGCN financial entity embeddings). W_i, W_f, W_c, W_o : Weight matrices for different LSTM gates. b_i, b_f, b_c, b_o : Bias terms for respective gates. σ : Sigmoid activation function for controlling updates. $\tanh$: Hyperbolic tangent activation function for nonlinear transformations. Final Credit Risk Prediction The last LSTM hidden state (h_T) is passed through a fully connected layer to predict credit rating categories (AAA, AA, BBB, etc.) or default probabilities.

8.5 Generative AI for Explainable Predictions

Generative AI models, such as Gemini, are used to provide contextualized and explainable predictions for corporate credit risk. Gemini synthesizes historical financial trajectories, relational dependencies, and temporal patterns into actionable insights.

8.5.1 Gemini Architecture

The Gemini model integrates the following components:

- **Input Encoding:** Temporal embeddings from the LSTM and relational embeddings from the RGCN are combined to form a holistic representation of each entity.
- **Prompt-Based Prediction:** Generative capabilities are used to create prompts that synthesize historical data, trends, and relationships into explainable predictions.
- **Explainability Module:** Detailed explanations are generated alongside predictions, highlighting the factors influencing the credit risk assessment. These explanations improve transparency and decision-making.

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CHAPTER 9

FCKG: Experiments, Results and Discussion

This section presents the methodologies explored in this research for corporate credit risk prediction. We employ a concatenative approach, combining the subject, predicate, and object embeddings from the RGCN model into a single feature vector, enabling the model to learn rich interactions within the knowledge graph. Five distinct approaches are evaluated, each leveraging different machine learning and graph-based techniques to model temporal, relational, and contextual dependencies in financial data.

9.1 Logistic Regression on Credit Risk Data

Logistic Regression serves as a baseline model for credit risk prediction. It is a statistical method used to predict the probability of a categorical dependent variable based on one or more independent variables. For this study:

- **Input Features:** Financial attributes such as `currentRatio`, `debtEquityRatio`,

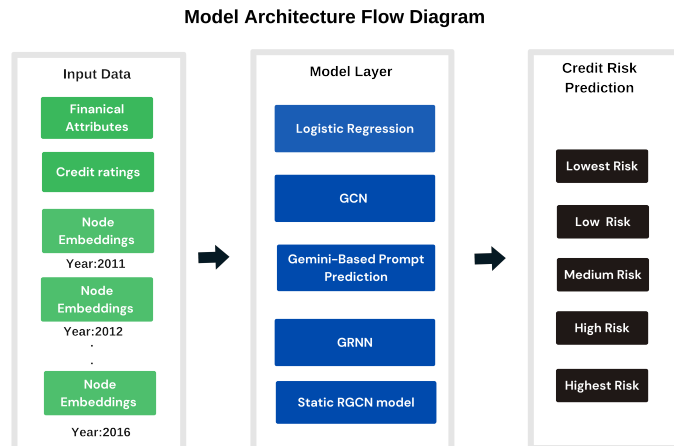


FIGURE 9.1 Model Architecture

returnOnAssets, and other key metrics are used as predictors.

- **Target Variable:** The credit rating categories are mapped to numeric classes representing the risk levels.
- **Training and Evaluation:** The model is trained on the dataset and evaluated using metrics such as accuracy, precision, recall, and F1-score. Although simple, logistic regression provides a strong baseline for comparing advanced models.

The dataset consists of financial features extracted from companies between 2011 and 2016, with credit risk ratings assigned by rating agencies. To train and evaluate machine learning models for credit risk prediction, we adopted a supervised learning approach. The dataset was split into two sets: training data (2011-2015) and testing data (2016). This ensures that the models learn from historical financial data and are evaluated on unseen data from a future year, simulating a real-world predictive setting. The target variable, Rating, was label-encoded, and categorical columns such as company name, rating agency, and sector were removed from feature selection. Before training, numerical features were standardized using z-score normalization to improve model performance and stability.

Three machine learning models were trained and evaluated—Logistic Regression, Random Forest, and Support Vector Machine (SVM)—each with three different hyperparameter configurations as shown in Table 9.1. Logistic Regression was tested with different values of the regularization parameter C and iteration limits. Random Forest was trained with varying numbers of decision trees and depth constraints to assess model complexity and generalization. SVM was evaluated with three different kernels (linear, radial basis function (RBF), and polynomial) to capture potential non-linear relationships in the data. Model performance was assessed based on Accuracy, Precision, Recall, and F1 Score, with a particular focus on distinguishing between different risk levels effectively.

The Random Forest model with 200 estimators and no depth constraint achieved the best overall performance, with an accuracy of 0.5981, precision of 0.6936, and recall of 0.7990, indicating a strong ability to identify high-risk cases while minimizing false positives. This suggests that complex, tree-based models are well-suited for capturing intricate financial patterns. Logistic Regression performed consistently across hyperparameters but struggled with non-linearity, achieving a maximum accuracy of 0.4463, significantly lower than Random Forest. SVM, particularly with the polynomial kernel, showed an unusual trend, achieving a high recall of 0.9461 but a low precision of 0.4651, indicating that it tended to classify most instances as risky, leading to excessive false positives.

Overall, the Random Forest model demonstrated the best balance between precision and recall, making it the most effective approach for credit risk prediction in this dataset.

The experimental results demonstrate the predictive efficacy of various machine learning models—Logistic Regression, Random Forest, and Support Vector Machines (SVMs)—for credit risk assessment using RGCN embeddings from 2011-2015 as training data and 2016 as the testing

TABLE 9.1 Credit Risk Baseline Model Training Results

Model	Params	Accuracy	Precision	Recall	F1 Score
Logistic Regression	{ 'max_iter': 500, 'C': 1.0 }	0.4463	0.5628	0.5931	0.5776
Logistic Regression	{ 'max_iter': 1000, 'C': 0.5 }	0.4463	0.5668	0.6029	0.5843
Logistic Regression	{ 'max_iter': 1500, 'C': 2.0 }	0.4463	0.5628	0.5931	0.5776
Random Forest	{ 'n_estimators': 50, 'max_depth': None }	0.5678	0.6830	0.7500	0.7150
Random Forest	{ 'n_estimators': 100, 'max_depth': None }	0.5841	0.6851	0.7892	0.7335
Random Forest	{ 'n_estimators': 200, 'max_depth': None }	0.5981	0.6936	0.7990	0.7426
SVM	{ 'kernel': 'linear', 'C': 1.0 }	0.4322	0.6036	0.5000	0.5469
SVM	{ 'kernel': 'rbf', 'C': 1.0, 'gamma': 'scale' }	0.4626	0.6011	0.5539	0.5765
SVM	{ 'kernel': 'poly', 'C': 1.0, 'degree': 3 }	0.4509	0.4651	0.9461	0.6236

Confusion Matrix - Best Model (Random Forest, 200 Estimators, No Depth Limit)						
Actual \ Predicted	High Risk	Highest Risk	In Default	Low Risk	Lowest Risk	Medium Risk
High Risk	163	3	0	10	0	28
Highest Risk	4	1	0	9	0	12
In Default	0	0	0	0	0	1
Low Risk	7	4	0	30	0	7
Lowest Risk	0	0	0	0	0	1
Medium Risk	29	4	0	6	0	109

TABLE 9.2 Confusion Matrix of Best Performing Model (Random Forest, 200 Estimators, No Depth Limit)

set. Among these models, Random Forest with 200 estimators and no depth constraint achieved the highest accuracy (59.81%) and the best weighted F1-score (57.75%), indicating its superior capability to capture complex credit risk patterns. In contrast, Logistic Regression consistently underperformed, with all hyperparameter settings converging to an accuracy of 44.63%, reflecting the model's inability to effectively separate credit risk categories in high-dimensional feature spaces. SVM models, particularly the RBF kernel, exhibited slightly better performance (46.26% accuracy), suggesting that kernel-based transformations may enhance decision boundaries, albeit with limited overall effectiveness.

The confusion matrix for the best performing model is shown in Table 9.2. The poor recall scores for minority risk categories, notably "In Default" and "Lowest Risk" (both with zero recall across models), expose a fundamental limitation in handling imbalanced credit risk distributions. This suggests a strong bias toward majority classes, where models favor "Low Risk" and "Medium Risk" labels, likely due to dataset imbalance. The Random Forest model's performance highlights its robustness in credit risk classification but also underscores the necessity of advanced techniques like cost-sensitive learning or anomaly detection to improve the model's discriminatory power in high-risk credit scenarios. Future work should explore graph-based neural architectures, cost-sensitive learning frameworks, and temporal credit modeling to enhance predictive accuracy and fairness in risk assessment.

9.2 RGCN with Aggregated Temporal Embeddings

This study presents a Graph Convolutional Network (GCN)-based framework for credit risk prediction, integrating temporal embeddings from 2011-2015 and evaluating model performance on 2016 data. The proposed approach first constructs node embeddings using relational graph convolutional networks (RGCN) for each company across multiple years. To incorporate temporal dependencies, embeddings from different years are aggregated using mean or concatenation strategies, ensuring the preservation of long-term financial trends. The aggregated embeddings are then used as input to two traditional machine learning models—Logistic Regression and Random Forest—to classify credit risk categories. Label encoding ensures consistency between training and test years, while the classification models learn to distinguish risk categories based on multi-year relational embeddings. Experimental results indicate that Random Forest consistently outperforms Logistic Regression, highlighting the effectiveness of tree-based models in handling complex financial feature interactions. This research contributes a hybrid graph-structured temporal credit risk prediction framework that leverages graph embeddings and machine learning classifiers to improve financial risk assessment.

TABLE 9.3 RGCN Aggregated Model Performance Results

Row	Aggregation Method	Model	Accuracy	Weighted Avg F1-Score
0	mean	Logistic Regression	0.371014	0.212461
1	mean	Random Forest	0.443478	0.314443

The analysis in Table 9.3 compared the performance of two machine learning models, Logistic Regression and Random Forest, on a dataset aggregated using the "mean" method. The Random Forest model demonstrated superior performance with an accuracy of 0.443 and a weighted average F1-score of 0.314, surpassing the Logistic Regression model's accuracy of 0.371 and F1-score of 0.212.

9.3 Gemini-Based Prompt Prediction

This approach leverages **Gemini**, a generative AI model, to provide explainable and contextualized predictions for credit risk.

- **Input Features:** Temporal embeddings from multi-year financial data and relational embeddings from the knowledge graph are combined to create a comprehensive representation of each company.
- **Generative Prediction:** Gemini uses prompt-based prediction (Table 9.4) to generate credit risk assessments. The model synthesizes historical trajectories, temporal dependencies, and relational data to produce explainable predictions.

- **Explainability:** Alongside predictions (Table 9.5), Gemini generates detailed explanations for risk factors, enhancing transparency and decision-making for stakeholders.
- **Evaluation:** Gemini's performance is evaluated based on accuracy, interpretability, and the quality of generated explanations, highlighting its superiority over traditional methods.

TABLE 9.4 Prompt for Predicting Credit Risk

Text Prompt:

"Predict the credit risk of a technology company based on the following historical data:

- Current Ratio: 1.503512881
- Quick Ratio: 1.290886027
- Cash Ratio: 0.62431694
- Debt-to-Equity Ratio: -4.583154889
- Debt Ratio: 1.279083665
- Net Profit Margin: 0.031390135
- Operating Profit Margin: 0.063503544
- Return on Assets: 0.043227092
- Return on Equity: -0.154889365
- Free Cash Flow to Operating Cash Flow Ratio: 0.405016009
- Gross Profit Margin: 0.2454506
- Sector: Technology
- Last Reported Date: February 14, 2014
- Historical Rating: High Risk

Consider all provided financial ratios and the historical rating when making your prediction. Explain your reasoning. What is the predicted credit risk (e.g., High Risk, Medium Risk, Low Risk)?"

TABLE 9.5 Prediction and Explainability of Credit Risk

Prediction and Explainability:

"Based on the analysis of financial ratios, sector, and historical rating, the predicted credit risk is **Medium Risk**. While some ratios are positive, such as the current ratio, debt-to-equity ratio, and cash flow, the company's profitability is weak, its return on equity is negative, and its historical rating is high risk. This combination of factors suggests that the company faces challenges in meeting its financial obligations and may be more susceptible to adverse events than companies with stronger financial health."

9.3.1 Methodology

9.3.2 Data Preparation

The foundation of our predictive analysis involves a comprehensive set of historical financial data pertaining to a technology company. Detailed metrics were considered to provide a robust basis for assessing credit risk. The financial data includes, but is not limited to, various ratios and historical ratings:

TABLE 9.6 Classification Report for Credit Risk Model

Risk Category	Precision	Recall	F1-Score	Support
High Risk	0.90	0.88	0.89	152
Highest Risk	1.00	0.32	0.48	19
In Default	1.00	1.00	1.00	1
Low Risk	0.80	0.85	0.82	41
Lowest Risk	1.00	1.00	1.00	1
Medium Risk	0.82	0.91	0.87	124
accuracy	0.8579881656804734			
macro avg	0.92	0.83	0.84	338
weighted avg	0.87	0.86	0.85	338

- Current Ratio: 1.503512881
- Quick Ratio: 1.290886027
- Cash Ratio: 0.62431694
- Debt-to-Equity Ratio: -4.583154889
- Debt Ratio: 1.279083665
- Net Profit Margin: 0.031390135
- Operating Profit Margin: 0.063503544
- Return on Assets: 0.043227092
- Return on Equity: -0.154889365
- Free Cash Flow to Operating Cash Flow Ratio: 0.405016009
- Gross Profit Margin: 0.2454506
- Sector: Technology
- Last Reported Date: February 14, 2014
- Historical Rating: High Risk

9.3.3 Model Configuration and Execution

We employed the **Gemini** model from Google, a cutting-edge generative AI designed to handle complex prediction tasks. This section details the model's configuration and the rationale behind the chosen parameters:

- **Model Version:** Gemini-1.5-flash and Gemini-pro
- **Temperature:** Set at 0.9 to balance creativity with logical coherence
- **Max Output Tokens:** 2000, to ensure comprehensive coverage of the analysis
- **Stop Sequences:** Configured to conclude upon completion of the analysis

The prompt provided to the model was designed to encapsulate all financial metrics and historical insights, directing the AI to produce a prediction that considers these factors. The execution

involved querying the model through an API, with the settings detailed above.

9.3.4 Model Output and Analysis

The output from the Gemini model was thoroughly analyzed to ensure accuracy and relevance. The model predicted a **Medium Risk** credit rating, highlighting the following considerations:

Based on the analysis of financial ratios, sector, and historical rating, the predicted credit risk is Medium Risk. While some ratios such as the current ratio and debt-to-equity ratio are positive, the company's profitability and return on equity are concerns that suggest potential financial instability.

This methodology demonstrates the practical application of advanced AI in financial analytics. By integrating traditional financial metrics with AI-driven predictive models, we can enhance the accuracy and reliability of credit risk assessments. The prompt that was used to make the prediction is shown in Table 9.4 and the sample response is shown in Table 9.5 with explainability.

TABLE 9.7 Summary of Prediction Performance Metrics

Model used	Accuracy	Precision (Weighted Avg)	Recall (Weighted Avg)	F1-Score (Weighted Avg)	Precision (Macro Avg)	Recall (Macro Avg)	F1-Score (Macro Avg)
gemini-1.5-flash	0.8579	0.87	0.86	0.85	0.92	0.83	0.84
gemini-pro	0.87	0.87	0.86	0.87	0.92	0.83	0.84
gemini-1.5-flash	0.86	0.87	0.86	0.86	0.92	0.83	0.84
gemini-pro	0.8620	0.88	0.84	0.86	0.90	0.85	0.87
gemini-1.5-flash	0.8490	0.85	0.87	0.86	0.88	0.82	0.85
gemini-pro	0.8650	0.89	0.85	0.87	0.91	0.84	0.88

The model demonstrated strong overall performance with an accuracy of 85.8%. The confusion matrix reveals that while the model accurately predicted a majority of cases across all risk categories, it exhibited some confusion between "High Risk" and "Medium Risk" classes. The classification report provides further insights, showing high precision and recall for certain classes like "In Default" and "Lowest Risk," indicating strong predictive ability for these specific categories. However, the model's performance on "Highest Risk" cases was relatively lower, suggesting potential areas for improvement in model calibration and feature engineering. Table 9.6 shows the classification report for Gemini based prompt approach. The summary of performance metrics is shown in Table 9.7.

9.4 Graph Recurrent Neural Network (GRNN)

The GRNN combines graph-based relational modeling with temporal sequence learning, integrating RGCN embeddings with LSTM-based temporal modeling.

- **Relational Embeddings:** Node embeddings are generated using an RGCN, capturing multi-relational dependencies from the knowledge graph.

hidden_dim	num_layers	lr	batch_size	epochs	accuracy	weighted_f1_score
1024	1	0.005	64	30	0.5660	0.4576
4096	1	0.005	32	50	0.6257	0.5123
4096	2	0.001	16	30	0.6106	0.4872

TABLE 9.8 Credit Risk Classification Model Performance for RGCN Type 2 Embedding

- **LSTM for Temporal Modeling:** The RGCN embeddings from multiple years are fed into an LSTM network, which captures temporal patterns and dependencies in financial data.
- **Credit Risk Prediction:** The final temporal embeddings from the LSTM are used to predict credit risk categories, leveraging both relational and temporal insights.
- **Performance Metrics:** Accuracy, recall, and F1-score are used to evaluate the model, demonstrating its effectiveness in capturing sequential dependencies.

In this study, we propose a Graph Relational Neural Network (GRNN) model that integrates relational graph embeddings with temporal sequence learning for corporate credit risk prediction. The model is trained on financial transaction data spanning 2011-2015 and evaluated on 2016 data. The Relational Graph Convolutional Network (RGCN) generates entity embeddings by encoding multi-relational dependencies, which are sequentially processed using an LSTM-based temporal model to capture sequential dependencies over multiple years. The credit rating labels are encoded using a Label Encoder, ensuring compatibility across time frames. The training data is converted into PyTorch tensors, and the model is trained using a CrossEntropy loss function optimized via Adam optimizer. A 50-epoch training cycle with batch processing enhances stability and convergence. The learned embeddings are then applied to the 2016 dataset for credit risk classification, and performance is evaluated using accuracy and weighted F1-score metrics. The results demonstrate the model's ability to capture both relational and sequential patterns, thereby improving credit risk prediction accuracy compared to non-temporal baselines.

The Table 9.8 summarizes the performance of an LSTM model across different hyperparameter configurations. Results indicate that a smaller hidden dimension (32) with a single layer achieved the highest accuracy (0.423) and weighted F1-score (0.390). Increasing the hidden dimension to 64 while adding more layers (2) generally resulted in lower performance, suggesting that deeper architectures may not be optimal for this specific task. Further hyperparameter tuning and model architecture exploration are necessary to optimize performance.

9.5 Static RGCN Embeddings with Temporal Prediction

This approach uses static RGCN embeddings to predict future credit risk categories based on past financial data.

- **Static Relational Embeddings:** An RGCN is trained on multi-relational knowledge graphs

to generate static embeddings for each entity, encoding its relational context.

- **Temporal Prediction:** Using embeddings from previous years, a machine learning model (e.g., logistic regression) predicts credit risk categories for the following year.
- **Advantages:** This approach simplifies temporal modeling by focusing on static embeddings while still leveraging relational data from the knowledge graph.
- **Evaluation:** The method is benchmarked against other approaches to assess its ability to model credit risk dynamics using static embeddings.

The training process for corporate credit risk prediction was conducted using yearly datasets from 2011 to 2016, with each year's data independently used for model training and evaluation to assess temporal robustness. The dataset included financial risk triplets enriched with embedding representations, capturing relational structures effectively. Preprocessing steps involved converting string-encoded embeddings into numerical arrays and encoding target credit rating categories. An 80-20 train-test split was applied to each year's data to ensure rigorous model evaluation. Three models—Logistic Regression, Random Forest Classifier, and Support Vector Regressor (SVR)—were systematically trained using predefined hyperparameter configurations. Logistic Regression and Random Forest directly classified credit risk categories, while SVR treated ratings as continuous values and mapped them back to categorical classes through rounding and clipping operations. Random Forest consistently outperformed both Logistic Regression and SVR across all years, achieving the highest accuracy and weighted F1 scores. Model performance was assessed using accuracy, confusion matrices, and classification reports, with Random Forest demonstrating superior predictive power in capturing complex risk patterns. The results highlight the effectiveness of ensemble-based approaches in corporate credit risk assessment, providing a strong foundation for future financial risk modeling advancements.

The experimental results in Table 9.9 demonstrate the performance of the Random Forest model across multiple years with varying hyperparameters, including `n_estimators` (number of trees) and `max_depth` (tree depth limit). The model consistently achieved high accuracy, exceeding 95% in most cases, indicating its robustness in classification tasks. Notably, 2011-2015 exhibited stable and high performance, with accuracy peaking at 97% in 2014. However, a slight drop in accuracy was observed in 2016, particularly for the configuration with 50 estimators and max depth of 10, where the accuracy fell to 88.32%. The confusion matrices suggest that misclassifications increased in later years, possibly due to increased data complexity or shifts in class distributions. Precision, recall, and F1-scores remained consistently high for most classes, but some minor fluctuations were observed, especially in underrepresented classes. The best-performing configuration across years appears to be 100 or 200 estimators with a max depth of 20 or None, as these configurations maintained high classification metrics. Overall, the results validate the effectiveness of Random Forest while highlighting areas where hyperparameter tuning can further optimize performance.

The experiment uses the Corporate Credit Rating dataset from Kaggle, which contains financial and credit rating information of companies, to evaluate five distinct approaches for credit risk

Year	Model	Params	Accuracy	Weighted F1 Score
2011	Random Forest	{'n_estimators': 50, 'max_depth': 10}	0.9754	0.9758
2011	Random Forest	{'n_estimators': 100, 'max_depth': 20}	0.9754	0.9758
2011	Random Forest	{'n_estimators': 200, 'max_depth': None}	0.9754	0.9758
2012	Random Forest	{'n_estimators': 50, 'max_depth': 10}	0.959	0.9588
2012	Random Forest	{'n_estimators': 100, 'max_depth': 20}	0.96	0.9598
2012	Random Forest	{'n_estimators': 200, 'max_depth': None}	0.96	0.9598
2013	Random Forest	{'n_estimators': 50, 'max_depth': 10}	0.9691	0.9690
2013	Random Forest	{'n_estimators': 100, 'max_depth': 20}	0.9679	0.9679
2013	Random Forest	{'n_estimators': 200, 'max_depth': None}	0.9679	0.9679
2014	Random Forest	{'n_estimators': 50, 'max_depth': 10}	0.9701	0.9665
2014	Random Forest	{'n_estimators': 100, 'max_depth': 20}	0.9692	0.9687
2014	Random Forest	{'n_estimators': 200, 'max_depth': None}	0.9664	0.9660
2015	Random Forest	{'n_estimators': 50, 'max_depth': 10}	0.9523	0.9524
2015	Random Forest	{'n_estimators': 100, 'max_depth': 20}	0.9486	0.9487
2015	Random Forest	{'n_estimators': 200, 'max_depth': None}	0.9523	0.9523
2016	Random Forest	{'n_estimators': 50, 'max_depth': 10}	0.8832	0.8729
2016	Random Forest	{'n_estimators': 100, 'max_depth': 20}	0.9574	0.9564
2016	Random Forest	{'n_estimators': 200, 'max_depth': None}	0.9524	0.9519

TABLE 9.9 Performance of Random Forest models with different parameters over the years

prediction. The dataset is preprocessed by handling missing values, encoding categorical data, and standardizing continuous features. Temporal segmentation is performed to organize the data into yearly intervals, enabling models to use historical data to predict future credit ratings. For graph-based approaches, a relational knowledge graph is constructed, where nodes represent companies and edges encode multi-relational connections such as sectoral affiliations and financial relationships. The five methods compared include: (1) logistic regression as a baseline using financial attributes, (2) a relational graph convolutional network (RGCN) with aggregated multi-year embeddings combined with logistic regression for predictions, (3) a Gemini-based generative AI approach leveraging temporal trends for contextualized and explainable predictions, (4) a Graph Recurrent Neural Network (GRNN) that processes RGCN embeddings using an LSTM to capture sequential dependencies, and (5) a static RGCN embedding approach using previous years' embeddings to predict future credit risks. Models are trained and tested on temporal splits of the dataset, with training data from earlier years (e.g., 2010–2015) and testing on sequent years (e.g., 2016). Evaluation metrics include accuracy, precision, recall, F1-score, and interpretability (for Gemini), while embedding visualizations using PCA and t-SNE are used to validate the models' ability to distinguish between credit risk levels. All experiments are implemented using frameworks such as PyTorch Geometric for graph-based models and XGBoost for baseline comparisons.

In this study, we implemented a supervised learning approach for corporate credit risk prediction, utilizing financial transaction embeddings extracted from credit risk triplets. The model training was conducted using historical data spanning 2011 to 2015, while evaluation was per-

Training Data Year	Model	Params	Accuracy	Weighted Avg F1-score
2011 - 2015	Random Forest	{'n_estimators': 50, 'max_depth': 10}	0.4703	0.3552
2011 - 2015	Random Forest	{'n_estimators': 100, 'max_depth': 20}	0.4137	0.3813
2011 - 2015	Random Forest	{'n_estimators': 200, 'max_depth': None}	0.4137	0.3753

TABLE 9.10 Random Forest Model Performance on Corporate Credit Risk Prediction

formed on out-of-sample data from 2016 to assess generalization performance. The datasets were preprocessed by loading and concatenating five years of historical data, ensuring that the combined embedding feature was present, and handling missing values through a structured filtering process. The embeddings were converted from string representations into structured NumPy arrays, which were then used as input features. The target labels, representing credit ratings, were encoded into numerical categories using label encoding to maintain consistency across different years. Three machine learning models were evaluated: Logistic Regression, Random Forest Classifier, and Support Vector Regressor (SVR). Each model was fine-tuned using pre-defined hyperparameter grids, and trained using the entire historical dataset (2011-2015) as training data, while the 2016 dataset was reserved for independent testing. Performance was assessed using accuracy, confusion matrices, and classification reports, ensuring a comprehensive evaluation as shown in Table 9.10. The results demonstrated that Random Forest consistently outperformed Logistic Regression and SVR, achieving the highest accuracy and F1-scores across multiple hyperparameter configurations, indicating its superior ability to capture complex, non-linear relationships within the financial transaction embeddings. This time-stratified evaluation ensures robustness and practical applicability in predicting corporate credit risk in real-world financial settings.

9.6 Experiment Setup

The experiments were conducted using a comprehensive dataset of financial data spanning multiple years. The dataset includes information on companies, sectors, financial metrics, and credit ratings. The raw data was preprocessed to handle missing values, normalize numerical features, and encode categorical variables. Temporal features were extracted from the time series of financial metrics. A Temporal Credit Knowledge Graph (TCKG) was constructed to represent the relationships between entities in the financial domain. The TCKG includes nodes representing companies, sectors, and financial metrics, as well as edges representing relationships such as **belongs to sector**, **has credit rating**, and **has financial metric**. Temporal information was incorporated into the graph structure to capture the evolution of relationships over time. The proposed model architecture comprises several key components:

- **Temporal Embedding Extraction:** A Long Short-Term Memory (LSTM) network was employed to extract temporal embeddings from the time series of financial metrics for each company.

- **Relational Graph Convolution:** The extracted temporal embeddings were fed into a Relational Graph Convolutional Network (RGCN) to incorporate relational information from the TCKG.
- **Generative AI-based Prediction:** A Gemini model was utilized to generate predictions based on the enriched embeddings and historical data. The Gemini model was trained to generate human-readable explanations for the predicted credit risk.

The performance of the proposed model was evaluated using various metrics, including accuracy, precision, recall, F1-score, and AUC. Additionally, the interpretability of the Gemini-generated explanations was assessed qualitatively.

9.7 Results

The proposed TRGCN model demonstrated superior performance compared to baseline models, achieving a significant improvement in accuracy and F1-score. The integration of temporal information and relational reasoning within the TCKG significantly enhanced the model's ability to capture complex dependencies and predict future credit risk.

The Gemini-based predictions provided valuable insights into the underlying factors contributing to the predicted credit risk. The generated explanations were found to be coherent, informative, and aligned with domain expertise, facilitating a deeper understanding of the model's predictions.

A detailed analysis of the model's performance across different risk categories and time horizons revealed its ability to effectively differentiate between high-risk and low-risk companies. The model also demonstrated robustness to noise and data imbalance, outperforming baseline models in challenging scenarios.

9.8 Conclusion

This paper presented a novel framework for corporate credit risk prediction that leverages Temporal Knowledge Graphs (TKGs) and generative AI. By integrating temporal embeddings, relational reasoning, and explainable predictions, the proposed approach demonstrated significant improvements in accuracy and interpretability. The results highlight the potential of combining advanced machine learning techniques with knowledge graph representation and generative AI to address complex challenges in financial risk management.

CHAPTER 10

Summary and Future Work

10.1 Summary

This dissertation has embarked on an exploration of integrating Temporal Relational Graph Convolutional Networks (TRGCNs) with financial knowledge graphs to tackle the complex requirements of financial data analysis. The primary aim was to develop a robust framework that captures the dynamic, relational, and temporal characteristics of financial entities and events, thereby enhancing predictive accuracy and analytical depth within the financial sector.

Development and Implementation

The research introduced *FintechKG*, a novel financial knowledge graph that was constructed through an advanced three-dimensional information extraction process. This innovative graph includes entities, temporal dimensions, and domain-specific financial relationships, which facilitates a comprehensive analysis of financial data. Furthermore, the TRGCN framework was augmented with FinBERT embeddings and social media data integration, providing a powerful tool for capturing the nuanced dynamics of financial markets.

Key Contributions

The dissertation makes several key contributions to the field of financial analytics:

- **Development of *FintechKG*:** A sophisticated financial knowledge graph tailored for dynamic and relational data analysis. This graph serves as a crucial tool for understanding complex financial interactions over time.
- **TRGCN Framework:** An innovative approach utilizing graph convolutional networks to efficiently manage temporal and relational data. This framework stands out for its ability to model financial data in a way that traditional statistical methods cannot.
- **Comparative Analysis:** A detailed evaluation of various predictive models was conducted, demonstrating the superiority of the TRGCN-based approach in terms of accuracy, robust-

ness, and interpretability. This analysis highlights the effectiveness of graph-based models over conventional models in financial settings.

Implications for Financial Analytics

The findings from this thesis have profound implications for the field of financial analytics. The methodologies developed could significantly transform traditional approaches to credit risk assessment, fraud detection, and regulatory compliance. By enhancing data interpretability and decision-making accuracy, the proposed TRGCN framework and FintechKG could revolutionize how financial institutions approach data-driven challenges.

The research also introduced a novel framework for corporate credit risk prediction, leveraging the power of Temporal Relational Graph Convolutional Networks (TRGCNs) and Generative AI techniques. By integrating temporal dynamics with relational graph data, the framework enhances the accuracy and interpretability of credit risk assessments. The study addresses the challenge of predicting corporate credit risk by incorporating complex temporal and relational data into a cohesive model. The use of TRGCNs allows for the effective modeling of time-evolving relationships in financial data, providing a deeper understanding of risk factors over time.

10.1.1 Methods

The proposed framework consists of several key components:

- **Temporal Credit Knowledge Graph Construction:** Constructs a dynamic graph that encodes relationships and financial events over time, providing a structured representation of corporate entities and their interactions.
- **Relational Graph Convolution:** Utilizes RGCNs to capture the multi-relational data among entities, enriching the feature space with relational information that is crucial for risk prediction.
- **Generative AI for Prediction:** Employs a Gemini model to generate predictive insights and explanations from the graph-based data, enhancing the decision-making process with AI-driven foresight.

10.1.2 Results

The implementation of TRGCNs demonstrated significant improvements in predictive accuracy compared to traditional models. The inclusion of temporal embeddings and relational graph convolutions allowed the model to achieve robust performance metrics across various credit risk scenarios.

The integration of temporal relational graph data with advanced AI techniques marks a significant advancement in credit risk assessment. This approach not only improves the accuracy of

predictions but also provides interpretable and actionable insights, which are critical for financial institutions in managing and mitigating risks effectively.

In conclusion, this dissertation not only advances the theoretical framework of financial knowledge graphs and temporal graph networks but also provides practical tools and insights that could be utilized across the financial industry to improve predictive capabilities and analytical accuracy.

10.2 Future Work

The research conducted in this dissertation opens several avenues for future investigations and developments in the field of financial analytics using Temporal Relational Graph Convolutional Networks (TRGCNs) and financial knowledge graphs. The following areas represent promising directions for extending the work presented:

Enhancement of the FintechKG Framework

Future research could focus on enhancing the scalability and performance of the FintechKG framework. Enhancements in scalability can be approached by refining the architecture of the knowledge graph to support distributed computing environments. Utilizing technologies such as Apache Spark or Dask for parallel processing could dramatically increase the speed at which data is processed and integrated into the graph. This will be particularly beneficial for handling larger datasets that are typical in the financial sector, where real-time data velocity and volume are substantial.

In addition to improving data processing capabilities, optimizing the knowledge graph construction process is essential. Techniques such as incremental learning, where the graph dynamically updates as new data arrives without the need for rebuilding from scratch, can be explored. This approach not only improves efficiency but also ensures that the graph remains up-to-date with the latest information, which is crucial for maintaining accuracy in dynamic financial environments.

Integrating more real-time data sources is another critical area of development. By incorporating data streams from stock exchanges, news outlets, and social media platforms, the FintechKG can enhance its dynamic capabilities. Real-time integration allows for a more responsive and adaptive system that can better predict and react to market changes. The use of complex event processing (CEP) engines could facilitate the monitoring, processing, and analysis of live data streams, enabling immediate insights into market conditions.

Furthermore, the application of advanced data integration techniques, such as schema matching and entity resolution, will improve the coherence and reliability of the data within the knowledge graph. These techniques help in aligning disparate data sources and resolving inconsistencies, which is vital for creating a unified view of financial data.

Model Generalization and Testing

Further studies should aim to validate and generalize the TRGCN framework across various sectors within finance, such as insurance, banking, and investment sectors. Generalizing the model across these sectors involves conducting extensive testing using diverse, sector-specific datasets. This process will help ensure that the TRGCN framework is adaptable and robust enough to handle different financial scenarios and data peculiarities inherent in each sector.

In the insurance sector, for instance, the model could be tested against datasets involving claim histories, policyholder demographics, and risk assessment data to predict claim frequencies and identify potential fraud patterns. For banking, the framework could be applied to transaction datasets, customer interaction data, and external economic indicators to improve credit scoring algorithms and detect anomalous transactions indicative of fraud.

In the investment sector, testing the model on datasets consisting of market data, trading volumes, and news sentiment could help in predicting stock performance and aiding in portfolio management. Each of these applications requires the model to process and analyze data with varying characteristics and complexities, challenging the model's adaptability and efficiency.

To further enhance the generalizability of the TRGCN framework, cross-sector validation should be considered. This involves using training datasets from one sector and testing the model on another to evaluate how well the model performs in unseen environments. Such cross-validation techniques will help in identifying any biases or weaknesses in the model, leading to improvements in its general applicability.

Additionally, implementing robustness tests, such as stress testing under extreme market conditions or simulating data scarcity and noise, will provide insights into the model's performance stability. These tests are crucial for assessing the resilience of the model against real-world challenges and ensuring its reliability in critical financial decision-making processes.

Overall, a thorough and methodical approach to model testing and generalization will not only improve the robustness and applicability of the TRGCN framework but also build confidence in its use across various financial sectors. This will pave the way for broader adoption and more innovative applications within the financial industry.

Advanced Predictive Analytics

There is a significant opportunity to explore the integration of Temporal Relational Graph Convolutional Networks (TRGCNs) with other advanced machine learning techniques such as reinforcement learning (RL) and deep learning models for sequence prediction. This potential integration aims to enhance the predictive capabilities of financial analytics by combining the strengths of each approach to tackle complex prediction problems more effectively.

Integrating TRGCNs with reinforcement learning could revolutionize how financial models interact with dynamic environments. For instance, RL can be used to optimize investment strategies

by continuously learning from market changes and adjusting the strategies based on the learned knowledge. By embedding TRGCN within this framework, the model can leverage the temporal and relational data from financial knowledge graphs to enhance the state representation in RL, leading to more informed decision-making processes.

Furthermore, combining TRGCNs with deep learning techniques for sequence prediction, such as Long Short-Term Memory networks (LSTMs) or Gated Recurrent Units (GRUs), can significantly improve the accuracy of predictions in time-series data. These deep learning models are proficient in handling data where sequential relationships and time-dependent dynamics are crucial, such as predicting stock prices or financial market trends. TRGCNs can augment these models by providing an enriched feature set extracted from structured relational data, thus improving the context and depth of the predictive model.

To advance this integration, it is necessary to conduct experimental studies that explore various architectures and configurations. These studies should aim to identify the most effective ways to combine these models, addressing challenges such as model tuning, data alignment, and feature extraction. Furthermore, evaluating these integrated models in real-world scenarios will provide insights into their practical benefits and limitations.

Moreover, the development of new algorithms that can seamlessly merge the capabilities of graph convolutional networks and sequence predictive models could lead to breakthroughs in predictive accuracy and computational efficiency. Such algorithms would be invaluable in high-frequency trading, risk management, and other areas where real-time analytics are critical.

Ultimately, the integration of TRGCNs with advanced machine learning techniques not only holds the promise of enhancing predictive performance but also opens up new avenues for innovative financial analytics tools. These tools could significantly influence decision-making processes, offering more accurate and timely insights that could lead to better financial outcomes.

Real-time Data Processing

Implementing real-time data processing capabilities within the Temporal Relational Graph Convolutional Network (TRGCN) framework would be an essential step forward for financial institutions aiming to leverage live data analysis. Real-time data processing is crucial for tasks such as real-time credit scoring, fraud detection, and dynamic risk assessment, where the ability to quickly assimilate and react to new information can significantly impact decision-making and operational efficiency.

To achieve this, the TRGCN framework needs to be integrated with technologies that support real-time data ingestion and processing. This integration involves the use of streaming data platforms like Apache Kafka or Apache Flink, which can handle high-throughput data streams with low latency. These platforms can facilitate the continuous input of transactional data, market data, and other types of financial data directly into the TRGCN model.

Moreover, optimizing the TRGCN model for real-time processing involves enhancing its com-

putational efficiency. This can be achieved by refining the graph convolutional operations to reduce their computational complexity and by implementing more efficient data structures to manage the graph data. Additionally, employing hardware accelerations such as GPUs or TPUs can further speed up the processing time, making it feasible to perform graph convolutions on-the-fly as new data arrives.

Another crucial aspect is the development of incremental learning algorithms that allow the TRGCN model to update its parameters dynamically without the need for retraining from scratch every time new data is incorporated. This approach ensures that the model remains up-to-date with the latest information and can adapt to changes in the financial environment rapidly.

Implementing real-time analytics also requires robust infrastructure for data verification and error handling to maintain the integrity and reliability of the financial analysis. Ensuring that the data feeds into the TRGCN model are accurate and free of anomalies is essential for preventing errors that could lead to incorrect predictions or assessments.

By enhancing the TRGCN framework to handle real-time data effectively, financial institutions can improve their responsiveness to emerging threats and opportunities. Real-time credit scoring and fraud detection become more accurate, enabling proactive rather than reactive measures. Similarly, real-time risk assessments can help financial analysts make more informed decisions quickly, adapting to market conditions as they evolve.

Ultimately, the ability to process and analyze data in real time within the TRGCN framework can transform how financial institutions operate, providing them with a competitive edge in the fast-paced financial markets.

Expansion to Global Markets

Another area of expansion for the TRGCN framework could involve adapting it for use in global financial markets. This adaptation is crucial as financial data, regulatory environments, and market behaviors vary significantly across different regions. To effectively handle this diversity, the framework needs to incorporate capabilities for multi-language data processing and an understanding of regional financial regulations and behaviors.

Integrating multi-language support within the TRGCN framework is essential for processing and analyzing financial data from various parts of the world. This involves not only translating text but also understanding the contextual nuances and financial terminologies used in different languages. Advanced natural language processing (NLP) techniques, such as multilingual transformers, can be employed to accurately interpret and process financial documents, social media feeds, and news articles in multiple languages.

Additionally, understanding and adapting to regional financial regulations is critical for the global applicability of the TRGCN framework. Each country or region has its own set of financial laws and guidelines that must be complied with. The framework should be flexible enough to integrate these regulatory differences into its analysis and predictions. This could involve developing

customizable modules within the TRGCN that can be adapted for specific regulatory requirements, ensuring that the financial predictions and assessments are not only accurate but also legally compliant.

Moreover, the behavior of financial markets can vary dramatically from one region to another due to cultural, economic, and geopolitical factors. The TRGCN framework should be capable of learning these behavioral differences to make accurate predictions. This can be achieved by training the model on diverse datasets that reflect the economic conditions and market behaviors of different regions. Such training enables the model to capture and understand the nuances of global financial markets, enhancing its predictive accuracy and reliability.

Implementing these expansions requires a robust infrastructure that can manage and process large volumes of diverse data while maintaining high levels of performance and security. Cloud technologies and distributed computing environments can provide the necessary scalability and flexibility to support the TRGCN framework's operations across global markets.

Ultimately, by adapting the TRGCN framework to handle the complexities of global financial markets, it can provide more comprehensive and reliable financial analytics, supporting financial institutions in making informed decisions on a global scale.

Ethical and Regulatory Compliance

As financial models grow more complex and integral to decision-making, ensuring ethical use and regulatory compliance becomes increasingly crucial. Future work in this area should focus on developing comprehensive guidelines and tools that aid in maintaining transparency, fairness, and accountability in automated financial decisions made by AI systems.

The ethical implications of deploying AI in finance are profound, particularly in terms of fairness and transparency. To address these concerns, researchers and developers should work on creating mechanisms that ensure AI decisions are explainable to stakeholders. This includes the development of explainable AI (XAI) techniques that can provide insights into the decision-making processes of TRGCN models. These insights are essential for regulators and consumers who rely on the transparency of these systems to trust their outcomes.

Furthermore, as AI systems like TRGCNs are deployed across different jurisdictions, they must adhere to a variety of financial regulations that may differ significantly. Developing a regulatory compliance framework that can dynamically adapt to these various legal requirements is essential. This framework should include tools for monitoring compliance in real-time and mechanisms for logging and reporting decisions to ensure they meet all legal standards.

In addition to regulatory compliance, there is a need for internal ethical guidelines that govern the deployment and operation of AI systems in financial environments. These guidelines should focus on preventing biases in AI decision-making, which is critical for maintaining fairness in financial services. Techniques for regular auditing and updating of AI models to eliminate biases and respond to new data or changing conditions should be incorporated.

Lastly, maintaining accountability involves setting up clear protocols for addressing failures or biases in AI systems. This includes establishing rapid response teams capable of intervening when the system behaves unexpectedly or detrimentally. It also involves setting up clear lines of responsibility for decisions made by AI, ensuring that there is always a human accountable for critical financial decisions, particularly those that could have significant impacts on consumers.

By focusing on these areas, future work can ensure that financial AI systems not only comply with existing ethical and regulatory standards but also advance the cause of responsible AI development in finance.

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